INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

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Abstract

A rollup holds many smart contracts, and wallet applications, explorers and auditors must choose which to examine first. PageRank methods for ledgers walk over transfers among all addresses, user accounts included, and do not read the ERC-20 allowance by which one contract lets another spend its tokens. This thesis asks whether one PageRank walk over allowance and transfer edges between contracts on Arbitrum One finds the contracts that users rely on sooner, and predicts their later use better, than single-layer walks and raw counts.

Its method, a coupled PageRank (C-PR), mixes the allowance and transfer rows of each contract in one walk; its references are Adaptive Weighted PageRank (AWP) on transfers and EndorseRank, the same weights on the latest allowances. After the registered results were known, the study defined this analysis: graphs of contracts, with no user account in any; each allowance counted only up to its owner's balance; amounts in US dollars on five tokens; scores from the twelve months before each freeze (31 March and 30 June 2026); and a discovery-order test, which crawls the contract graph in the order a score gives on the contracts read so far and measures how soon it reaches those that user accounts pay and authorize most. Out of window, it tests $7{,}864$ and $7{,}915$ spender contracts against the approval pairs and senders of the next three months, under the rules of its analysis plan and registered second window.

The discovery-order test answers no. In its deciding run the crawl ordered by C-PR reached the contracts with the most user-account inflow and user-authorized value sooner than a breadth-first crawl (difference in area under the discovery curve $+0.0520$ $[+0.0485, +0.0558]$ and $+0.0061$ $[+0.0014, +0.0111]$) but later than a crawl ordered by contract backlinks (-0.0051 $[-0.0087, -0.0010]$ and -0.0571 $[-0.0621, -0.0519]$). Rule A is not met: C-PR ranks approval pairs below its allowance layer ($\Delta\tau = -0.025$ $[-0.045, -0.004]$) and senders below its transfer layer (-0.014 $[-0.019, -0.008]$). Rule B is not met: the thesis applies two of its registered criteria,

F1 and F2, and under the registration the rule holds only if each of its criteria holds; C-PR gains on approval pairs over its transfer layer ($+0.074$ $[0.060, 0.087]$) but loses more than the margin on senders (-0.016 $[-0.022, -0.012]$). EndorseRank ranks approval pairs below the in-approve degree it smooths in both windows (-0.036 $[-0.046, -0.027]$, -0.034 $[-0.043, -0.025]$); AWP ranks senders slightly above the transfer in-degree ($+0.012$ $[0.003, 0.023]$, $+0.011$ $[0.001, 0.022]$) but below it among first-window spenders a contract had already paid (-0.035 $[-0.049, -0.022]$); and C-PR lies above the count of both edge types on approval pairs ($+0.016$ $[0.007, 0.024]$, $+0.011$ $[0.003, 0.019]$) and below it on senders (-0.053 $[-0.061, -0.043]$, -0.056 $[-0.065, -0.047]$). C-PR takes 1.09 times AWP's solve time, and EndorseRank, with 22.4 times fewer edges, 27.9 times less.

The thesis contributes the coupled walk, the balance-capped allowance between contracts as a ranking edge, a discovery-order test for ledger rankings, the walk's negative answers under that test and the rules of the plan and registration, and an evaluation of every walk against its count. The allowance layer between contracts is thin, and the labels count mostly user accounts. Every Sybil address must be a deployed, funded contract, yet a closed farm collects 6.16 times its population share of EndorseRank and about 374 farm contracts reach the cohort's top score, so no score should decide rewards without Sybil controls; none reads a contract's code or was tested for credit decisions about contracts.

Key terms: coupled PageRank; smart contract; crawl ordering; on-chain reputation; ERC-20 allowance; EndorseRank; Adaptive Weighted PageRank; decentralized finance; temporal holdout; pre-registration; Arbitrum One; Kendall's tau

Table of Contents

Declaration 1
Abstract 2
Abbreviations and Notation 11
1 **Introduction and Research Problem** 13
1.1 Background: contracts on a rollup and how trust in them is formed 13
1.2 Motivation for on-chain reputation 15
1.3 Research problem 16
1.4 Temporal holdout 19
1.5 Research objectives 21
1.6 Research questions 22
1.7 Summary of the literature review 23
1.8 Summary of the research methodology 24
1.9 Contributions 25
1.10 Scope and limitations 27
1.11 Research roadmap 28
2 **Literature Review and Theoretical Framework** 29
2.1 Blockchain and DeFi foundations 29
2.2 Ethereum, Layer-2 rollups, and Arbitrum One 30
2.3 ERC-20 tokens and the allowance lifecycle 31
2.4 Graph-based on-chain reputation 32
2.5 PageRank formalism 34
2.6 Crawl ordering 36
2.7 Reputation systems lineage 37
2.8 ERC-20 allowances as endorsement 38
2.9 What this research does not claim 39
2.10 Rank correlation theory 39
2.11 Theoretical framework 40
2.12 Terminology 42
2.13 Domain alignment metrics 44

2.14 Worked example: PageRank on a four-node endorsement graph 44
2.15 DeFi risk taxonomy and where reputation fits 45
2.16 Related measurement traditions 45
2.17 Accounts, gas, and why logs are the measurement surface 46
2.18 Approvals as security-relevant events 47
2.19 AWP in depth: what the transfer-graph baseline measures 47
2.20 Comparison of graph reputation methods 48
2.21 Summary of the reviewed literature 48
2.22 Research gap 50
2.23 Open problems motivating this research 52
3 **Research Methodology** 54
3.1 Notation 55
3.2 Data sources and sampling 57
3.3 Two-phase collection strategy 60
3.4 Pipeline architecture 61
3.5 Event decoding 61
3.6 Graph construction 65
3.7 Discovery-order test 72
3.8 Temporal holdout protocol 75
3.9 Construct-check design 79
3.10 Operational definitions of proxies 79
3.11 Statistical tests 81
3.12 Computational benchmark (Claim C6) 83
3.13 Scaling and robustness extensions 84
3.14 Reproducibility 86
3.15 Sampling considerations and missingness 87
3.16 Implementation notes 88
3.17 Ethics 88
• Privacy and anonymity 88
• Data minimization 89
• Data usage and compliance 89

• Fairness and transparency 89
• Responsible disclosure and impact 90
• Summary of methodological commitments 90
4 **Implementation and Empirical Results** 91
4.1 Case study setting: Arbitrum One 91
4.2 Computational benchmark (Claim C6) 95
4.3 Same-window alignment (Claim C5) 97
4.4 Rank divergence (H1 / RQ1) 100
4.5 Robustness and sensitivity checks 103
4.6 Discovery order (RQ2) 108
4.7 Temporal holdout of the single-layer walks (RQ3) 113
4.8 The coupled walk C-PR (RQ4, RQ5) 117
4.9 Claims (C1–C6) 126
5 **Discussion** 129
5.1 Findings by research question 129
5.2 Achievement of the research objectives 135
5.3 Theoretical implications 135
5.4 Practical implications and deployment scenarios 139
5.5 Societal and practical impact 142
5.6 Sybil attack cost model on allowance graphs 142
5.7 Regulatory and ethical considerations 148
5.8 Threats to validity 149
5.9 Limitations 154
6 **Conclusion and Future Work** 156
6.1 Answers to the research questions 156
6.2 Original contribution to knowledge 158
6.3 Future work 160
6.4 Closing reflection 161
7 **References** 162
8 **Appendix** 172
8.1 **Evaluation Pipeline and Reproducibility** 172

8.1.1 Pipeline stages 172
8.1.2 BigQuery extraction design 175
8.1.3 Account-type classification 175
8.1.4 Benchmark run logs 176
8.1.5 Parameter summary 176
8.2 **SQL Templates** 177
8.2.1 Event selection pattern 178
8.2.2 Latest-allowance and pair-table pattern 179
8.3 **Elementary Properties of PageRank and Rank Correlation** 180
8.3.1 Existence and uniqueness of PageRank 180
8.3.2 Sparsity and per-iteration cost 181
8.3.3 Properties of Spearman's ρ and Kendall's τ 181
8.4 **Data Description** 182
8.4.1 Extracted events and graphs 182
8.4.2 Tokens, prices and allowance amounts 182
8.4.3 Contract graph: account types, smart accounts and balances 183
8.4.4 Discovery-order test: universes, coverage, sensitivity and diagnostics 185
8.5 **EndorseRank and AWP Side by Side** 188
8.5.1 Same window 189
8.5.2 Proxy detail 192
8.5.3 Out of window 194
8.6 **The Registered Analysis** 195
8.7 **Ethics Clearance Certificate** 196
8.7.1 Certificate 196

List of Figures

2.1 ERC-20 allowance lifecycle: approve (or permit), optional `transferFrom`, and revoke. 32
2.2 Random-surfer intuition for PageRank on the four-node graph of Section 2.14. . . . 35
2.3 Conceptual model: the contract graph, the scores and the four evaluations. 43

3.1 Blockchain data used in this thesis: sources, months covered, held-out labels and uses. 58
3.2 Pipeline overview: from public logs to scores, checks and tables. 62
3.3 Logistic time-decay $\sigma(\Delta t)$ of AWP and EndorseRank ($k = 0.01$, $t_0 = 180$ days). . 67
3.4 One step of the coupled walk C-PR. 71

4.1 Runtime of EndorseRank and AWP on subsamples of the cohort and on the whole contract graph. 98
4.2 Marginal distributions of EndorseRank and AWP scores on the matched cohort ($n = 6{,}443$ contracts), on a $\log_{10}$ scale for contracts with a positive score. 101
4.3 Rank–rank scatter plot of EndorseRank vs. AWP for the matched cohort (random sample of $3{,}000$ contracts). 102
4.4 Sensitivity of family-mean Kendall τ over $d \in \{0.75, 0.85, 0.95\}$ (Table 4.7 values). 104
4.5 Discovery-order test: share of the hot contracts visited against the visits in the deciding run. 111

8.1 Discovery-order test: share of the hot contracts visited when a visit reveals out-edges only. 191

List of Tables

I Abbreviations used in this research. 11
II Mathematical notation used in this research. 12

1.1 The scores evaluated in this thesis, each computed on the whole contract graph of a freeze. 17
1.2 Summary of hypotheses, research questions, and empirical claims. 20

2.2 Summary of the reviewed literature: methodology, strengths, limitations, and relation to this study. 48
2.1 Comparison of selected graph reputation methods (conceptual). 51

3.1 Notation of this chapter's equations beyond the list of Table II. 56

4.1 The two layers of the contract graph at both freezes. 93
4.2 Where the members of each cohort lie in the contract graph. 94
4.3 Computational benchmark on the contract graph. 96
4.4 Runtime on subsamples of the matched cohort and on the whole contract graph. . . 97
4.5 Same-window family-mean Kendall τ for every score of this study. 99
4.6 Same-window family means with missing contracts scored zero or kept as isolated nodes. 101
4.7 Robustness: family-mean Kendall τ under PageRank damping d. 103
4.8 Robustness: family-mean Kendall τ under ten times smaller and larger value slopes. 105
4.9 Sensitivity of the family means to the evaluation set. 105
4.10 EndorseRank with uniform and with activity-weighted restarts, next to AWP. . . . 106
4.11 Sensitivity to the balance cap. 107
4.12 Discovery-order test: the decision contrasts and their reading. 109
4.13 Discovery-order test: $\mathrm{AUC}_{10\%}$ of every ordering in the four runs. 110
4.14 Discovery-order test: the contrasts reported outside the decision. 112
4.15 Temporal holdout W0 on the spender cohort. 114
4.16 Paired bootstrap differences on the spender holdout W0. 115
4.17 Spender holdout W0 restricted to spenders that had received a transfer before the freeze. 116
4.18 EndorseRank and its variant with activity restarts in window W1. 118

4.19 Same-window paired contrasts of C-PR and S-PR against their two layers and of C-PR against EndorseRank and AWP. . . . 119
4.20 Window W1: Kendall τ on the two spender labels. . . . 122
4.21 Window W1: F1 and F2, two of rule B's registered criteria, with their verdicts and the sensitivity analyses. . . . 123
4.22 C-PR against every other score and the raw counts out of window. . . . 124
4.23 The main out-of-window contrasts inside the union graph. . . . 125

5.1 Research objectives, the criterion stated in Chapter 1, and the outcome. . . . 136
5.2 The Sybil cost model on the contract graph and its sensitivity to a balance threshold. 148

8.1 Every timed solver call of the benchmark. . . . 177
8.2 Default parameters used in the evaluation. . . . 178
8.3 The five tokens, their allowances between contracts and their prices. . . . 183
8.4 Account types of the contract graph. . . . 184
8.5 Latest allowances between contracts and the balance cap. . . . 186
8.6 The labels by the class of the account they count. . . . 186
8.7 Sizes of the contract graph and of the universes of the discovery-order test. . . . 187
8.8 Coverage of the two importances by the contract graph and the universes. . . . 188
8.9 Discovery-order test: sensitivity of the decision contrasts. . . . 189
8.10 Discovery-order test: ties, wasted visits and hub visits. . . . 190
8.11 Same-window paired contrasts between EndorseRank and AWP. . . . 190
8.12 Domain alignment of EndorseRank and AWP with the transfer proxies. . . . 192
8.13 Domain alignment of EndorseRank and AWP with the allowance proxies. . . . 193
8.14 Domain alignment of EndorseRank and AWP with the Sybil-adjusted stability proxies. . . . 193
8.15 Paired bootstrap differences between EndorseRank and AWP out of window. . . . 194
8.16 Decision contrasts of the registered analysis and of the contract-graph analysis. . . . 196

Abbreviations and Notation

Table I: Abbreviations used in this research.

Abbreviation	Meaning
AUC	Area under the curve; $AUC_{10\%}$ is the normalized area under a crawl's discovery curve over the first 10% of its visits
AWP	Adaptive Weighted PageRank
BFS	Breadth-first search (the crawl ordering that visits contracts in the order of their discovery)
C-PR	Coupled PageRank (per-node mixture of the allowance and transfer layers, weight λ)
DeFi	Decentralized Finance
EIP	Ethereum Improvement Proposal (EIP-2612 defines `permit`; an EIP-7702 account is an externally owned account that delegates to contract code)
EOA	Externally owned account (an address controlled by a private key)
ER	EndorseRank (in table headings)
ERC-20	Ethereum Request for Comments 20 (fungible token standard)
ERC-4337	Ethereum Request for Comments 4337 (account abstraction; an ERC-4337 account is a smart account that names a canonical EntryPoint contract)
EVM	Ethereum Virtual Machine
FC	Financial Cryptography and Data Security
L1, L2	Layer 1 (the Ethereum mainnet) and Layer 2 (a network, such as Arbitrum One, that posts its transaction data to Layer 1)
PR	PageRank
RPC	Remote procedure call (the interface of a node of Arbitrum One through which account code and contract state were read)
RQ	Research Question
SCC	Strongly connected component (here the giant one of the union graph of a freeze)
SoK	Systematization of Knowledge
S-PR	Endorsement-Seeded PageRank (transfer layer walked with restarts drawn from the allowance walk)
USD	United States dollar, the unit of every amount in the main analysis
USENIX	USENIX Security Symposium
UTC	Coordinated Universal Time
W0	Holdout window: scores frozen on 31 March 2026, labels counted from April to June 2026
W1	Registered window: scores frozen on 30 June 2026, labels counted from July to September 2026
WCC	Weakly connected component (here the one that holds the giant SCC)
WETH, USDC, USDT0, ARB, WBTC	The five tokens of the main analysis: wrapped ether, USD Coin, Tether USD on Arbitrum One, the Arbitrum governance token, wrapped bitcoin

Table II: Mathematical notation used in this research. Each symbol is defined where it first appears, and Table 3.1 adds the symbols that only the equations of Chapter 3 use.

Symbol	Meaning
G_{approve}	Directed endorsement graph from latest ERC-20 allowances between contracts (owner $\to$ spender), each counted up to its owner's balance
G_{transfer}	Directed transfer graph between contracts (sender $\to$ receiver; AWP input)
$\lvert V\rvert$	Number of nodes in a reputation graph
$\lvert E\rvert$	Number of directed edges in a reputation graph
w_{ij}	Weight of the directed edge from node i to node j
o_i	Out-strength of node i, $o_i = \sum_j w_{ij}$ (written $o(u)$, or $o_A(u)$ and $o_T(u)$ per layer, in Chapter 3)
P	Transition matrix, $P_{ij} = w_{ij}/o_i$ for $o_i > 0$; a dangling row is uniform in Equation (2.1) and zero in Equation (3.2), where its mass is redistributed separately
G_{PR}	Google matrix $dP^\top + (1-d)\,\mathbf{1}\mathbf{1}^\top/\lvert V\rvert$ (Equation (2.2))
$\mathbf{r}$, r_i	PageRank vector and its entry for node i; $\mathbf{r}^{(t)}$ is the iterate after t power-iteration steps
$\boldsymbol{\pi}$	Teleportation (restart) distribution: uniform for EndorseRank and C-PR, proportional to activeness X_u for AWP, and the normalized allowance-walk score for S-PR
d	PageRank damping factor (default 0.85); the teleport probability is $1-d$
ε	Convergence tolerance of power iteration (10^{-8}, on the $\lVert\cdot\rVert_1$ change between iterates)
$I_{\max}$	Iteration cap of power iteration (300)
λ	Allowance-layer weight of the coupled operator C-PR ($\lambda = 0.5$ primary; 0.25 and 0.75 sensitivity; at $\lambda = 1$ and $\lambda = 0$ one layer is walked alone)
$P^{(\lambda)}$	Coupled transition matrix of C-PR with allowance-layer weight λ (Equation (3.9))
$\alpha_A(u)$, $\alpha_T(u)$	Weights of the allowance row and the transfer row in the coupled transition row of node u
k	Logistic time-decay steepness (0.01)
t_0	Logistic time-decay midpoint (180 days)
Δt	Age in days of a transfer or approval before the decay anchor (T_{obs}; t_1 in W0)
$\sigma(\Delta t)$	Logistic time weight of a transfer (AWP) or latest approval (EndorseRank)
$V(z)$	Bounded value weight of AWP and EndorseRank on a US-dollar amount z, $2/(1+e^{-bz})-1$ with $b = \ln 3/m$, so that the median amount m ($m_T = 160$ dollars for transfers, $m_A = 0.071$ dollars for capped allowances) gets $V = 1/2$
U	Cap on one allowance in the allowance layer of C-PR and in the in-approve value (16,000 dollars)
X_u	Activeness of address u, AWP's restart weight
n	Number of contracts in a correlation (matched cohort $n = 6{,}443$; spender cohorts $n = 7{,}864$ in W0 and $7{,}915$ in W1)
ρ	Spearman's rank correlation coefficient; in the discovery-order test, the share by which a crawl's visits grow before it recomputes its ordering ($1/50$)
τ	Kendall's rank correlation coefficient; every reported value is τ_b
T_{obs}	End of the score window of the same-window evaluation and score freeze of W1 (30 June 2026, 23:59:59 UTC)
t_1, t_2	Score freeze of W0 (31 March 2026, 23:59:59 UTC) and end of its label window (30 June 2026)
$\mathcal{W}$	Set of contracts in the evaluation cohort
$s(w)$	Reputation score of contract w under a given method
$p(w)$	Proxy value for contract w
$\mathcal{U}$	Universe of a run of the discovery-order test, the contracts that every crawl of the run visits
I_T, I_A	Importance of a contract in that test: the US dollars that user accounts transferred to it, and the balance-capped US dollars that they authorized it to spend (at most $U_{\text{user}} = 31{,}000$ dollars per account)
$H_q(I)$, AUC_x	Hot set, the share q of the universe with the largest importance I, and the normalized area under a crawl's discovery curve over its first $x\lvert\mathcal{U}\rvert$ visits
C-PR$'$, IB'_U; D_1, D_2	A prime marks a walk or count computed on the contracts a crawl has read so far; decision contrasts of $\text{AUC}_{10\%}$, C-PR$'$ minus the breadth-first crawl and C-PR$'$ minus the backlink count IB'_U

1 Introduction and Research Problem

A public blockchain that runs smart contracts holds very many of them. On Arbitrum One, the `Approval` and `Transfer` events of five tokens between April 2025 and September 2026 alone involve 1,829,719 addresses that this study types as contracts (Table 8.4). Wallet applications, block explorers, auditors and integrators cannot examine them all and must choose which contracts to look at first.

This thesis asks whether a PageRank walk over the relations between contracts finds the contracts that users rely on, as PageRank ranked the pages of the web by the links they receive (Page et al., 1999). The relations are ERC-20 transfers from one contract to another and allowances, by which an owner contract authorizes a spender contract to move its tokens. The method of the thesis, a coupled PageRank (C-PR), walks both edge types in one chain and mixes them at each contract with a fixed weight λ. Two single-layer walks serve as its references: EndorseRank, PageRank on the allowance edges alone with the edge weights of the transfer-based Adaptive Weighted PageRank (AWP) of T. Do et al. (2023), and AWP itself on the transfers. Every node of every graph is a contract. The walks are evaluated beside the raw counts that they smooth: by how soon a crawl ordered by each score reaches the contracts that user accounts pay and approve most, by the approvals and inflow that arrive after the scores are frozen, and under the decision rules of an analysis plan and a registration.

1.1 Background: contracts on a rollup and how trust in them is formed

Public blockchains make programmable settlement possible without a central authority (Nakamoto, 2008), and protocols stack smart contracts (Szabo, 1997) into decentralized markets for lending, automated market making and perpetual futures (Werner et al., 2022), composing new financial services from existing ones (Kitzler et al., 2023). On Arbitrum One and other Layer-2 (L2) net-

works fees are low, and every transfer, approval and protocol event can be inspected address by address.

Whoever deals with these contracts has to decide which of them deserve attention, and today that decision rests on evidence about one contract at a time. A wallet application asks its user to confirm an approval for a spender that the user may never have seen, and phishing sites lead users to sign transactions that let scammers take their assets (He et al., 2023). Code analysis looks in a contract's code for malicious functions (Z. Lin et al., 2024), an audit examines one protocol, and lists of known contracts are curated by hand. None of these reads which contracts rely on which, although the logs make that structure public.

The web met the same problem at a larger scale. PageRank ranks a page by the pages that link to it and weights each link by the rank of its source (Page et al., 1999). Cho et al. (1998) put such rankings to a test of discovery: a crawler knows only the links of the pages it has read and must choose which known page to read next. They compared orderings of that choice, breadth-first order, the number of backlinks and PageRank, the last two computed on the pages seen so far, and on a stored copy of the Stanford University web the crawl ordered by PageRank reached the hot pages, those with many backlinks, sooner than the crawl ordered by the backlink count (Section 2.6).

On public ledgers, PageRank has been applied to transfers: H.-D. Do and Do (2023) run PageRank and HodgeRank on Ethereum transactions, and AWP (T. Do et al., 2023) weights each transfer by its value and age, restarts the walk at addresses that sent recently, and is validated by its correlation with inbound transfer count and value. These walks run over the transfers among all addresses, user accounts included, and user accounts supply much of what a contract receives: more than four in five of the new approval pairs and new transfer senders that the spender contracts of this study gain in its label windows come from user accounts (Table 8.6). Anyone can create such accounts at almost no cost, the condition of a Sybil attack (Douceur, 2002), and the study does not measure who controls them. A contract costs a deployment, and an edge between two contracts records a relation that one program holds toward another. An ERC-20 allowance between two contracts is such a relation, a standing authorization under which the spender can move the owner's tokens

(Vogelsteller & Buterin, 2015), and the transfer walks do not read it.

1.2 Motivation for on-chain reputation

Three considerations motivate a ranking of contracts that is computed from public logs.

- Attention: wallet applications, explorers, auditors and integrators need an order in which to examine contracts. A ranking that puts first the contracts that users rely on tells them where to start; it replaces no examination (Section 5.4).
- Auditability: anyone can recompute a score that a fixed rule derives from public event data, unlike automated scores that outsiders cannot inspect (Packin & Lev-Aretz, 2024).
- Cost: a score that is refreshed often should be cheap to recompute, and one that user accounts can move is cheap to manipulate. In a graph of contracts every node costs a deployment, and an allowance counted only up to its owner's balance has to be backed by tokens (Section 5.6).

A ranking that proved reliable in these respects could later serve as one input to credit decisions about contracts; the thesis does not test that use.

Transfer-based AWP is public and auditable, but its graph grows with every payment (T. Do et al., 2023). An allowance graph is built differently. For an owner, a spender and a token, each new approval supersedes the older ones (Vogelsteller & Buterin, 2015), so the latest approval stands for the history of that pair. An allowance can exceed the owner's balance or be unlimited, yet the spender can move at most the balance, so each latest allowance is counted only up to its owner's balance of the token at the freeze (the balance cap, Section 3.6). Between contracts the resulting graph G_{approve} has far fewer edges than the transfer graph G_{transfer}, so its PageRank solve takes less time, a property of the smaller graph and not of the algorithm (Table 4.3; Claim C6). The cap also leaves the layer thin: most latest allowances between contracts belong to owners that hold no balance of the token and form no edge (Table 8.5).

Allowances also mean something different. A transfer might reflect routing, market making, or a passive receipt, whereas an allowance is an active grant of permission to spend funds, so treating

allowances as endorsement edges carries PageRank's citation reading (Page et al., 1999) over to a trust relationship that protocols already depend on. Whether one walk over both kinds of edge finds the contracts that users rely on is examined on the contract graph of Arbitrum One and on a matched cohort of $n = 6{,}443$ of its contracts.[1]

1.3 Research problem

The work reviewed offers those who must choose which contracts to examine no ranking computed from the relations among contracts alone, and it is not known whether such a ranking would find the contracts that users rely on. AWP infers trust from the transfer history of all addresses and assigns no role to spending authorization (T. Do et al., 2023). EndorseRank, the allowance reference of this thesis, keeps AWP's edge weights and solver but builds the directed graph from the latest allowance of each owner–spender pair and token. An allowance points from the owner to the spender, so EndorseRank scores the spenders that receive approvals and gives every owner that receives none the same value. Each of the two walks therefore follows one kind of act: AWP the payments that reach a contract, EndorseRank the spending rights it is granted.

Because both edge types live on the same contracts, a single walk can follow both, mixing them node by node with a fixed weight λ. This walk, the Coupled PageRank (C-PR), is the method of the thesis; at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. The main research question is: does one PageRank walk over allowance and transfer edges between Arbitrum One contracts find the contracts that users rely on sooner, and predict their later use better, than single-layer walks and raw counts? Research questions RQ1–RQ5 (Section 1.6) break it down. The analysis plan fixed one decision rule, and its object is C-PR (Section 3.8).

The first half of the question has a direct test. In the discovery-order test (Section 3.7) a crawl starts at a seed contract, reads the edges of each contract it visits and chooses the next contract by an ordering computed on what it has read so far, as the crawler of Cho et al. (1998) chose

[1] Every method scores the same contracts: those that received a non-zero approval of WETH, USDC, USDT0, ARB or WBTC from at least three distinct owners, of any account type, between 1 July 2025 and 30 June 2026 (Section 3.2).

its next page; a prime, as in C-PR′, marks such an ordering. Importance is the demand of user accounts, which no ordering reads: the US dollars that user accounts transferred to a contract (I_T, user-account inflow) and the balance-capped US dollars that their allowances let it spend (I_A, user-authorized value), each capped per account. The decision compares how soon the crawl ordered by C-PR′, a breadth-first crawl and a crawl ordered by the number of contract backlinks reach the contracts with the most of each importance. The test was defined after the registered results were known and decides no rule of the plan or the registration.

The scores appear in Table 1.1. Table 1.2 lists hypotheses (H1–H3), research questions (RQ1–RQ5), and empirical claims (C1–C6) from Chapter 4.

Table 1.1: The scores evaluated in this thesis, each computed on the whole contract graph of a freeze.

Score	Input graph	Role
C-PR	Both layers of the contract graph, in US dollars with each allowance counted up to its owner's balance and capped, mixed per node with weight λ	The method of this thesis, object of the decision of the discovery-order test (as the ordering C-PR′), of rule A and of the two applied criteria of rule B. Its layers walked alone, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), are weighted by amounts, so they are not EndorseRank and AWP. Its ablation S-PR restarts a transfer walk from the allowance walk
EndorseRank	Latest ERC-20 allowances between contracts on five tokens, in US dollars up to the owner's balance (G_{approve}), weighted as in AWP	Allowance delegation; the single-layer reference on allowances. Its variant with restarts at approving owners is reported beside it
AWP	Transfers of five tokens between contracts in US dollars (G_{transfer})	Transfer flow; the single-layer reference on transfers, the published transfer-graph method of T. Do et al. (2023) (Section 3.6)
Raw counts	G_{approve}, G_{transfer} and their union, counted per node	In-approve degree and value, transfer in-degree and union in-degree (the number of contract backlinks), the local counts that the walks smooth: proxies in the same window (the first three), baselines out of window and comparators in the discovery-order test

The main terms are defined as follows:

- *Smart contract:* Self-executing program code that runs on a blockchain and carries out an agreement without a trusted intermediary (Szabo, 1997).

- *Wallet / Address:* A blockchain account that holds funds and interacts with smart contracts, identified by its address (Buterin, 2014): an externally owned account (EOA), controlled by a private key, or a contract account governed by its code (Antonopoulos & Wood, 2018). No off-chain identity is assumed or recovered.
- *User account and contract:* A user account is an address that sent a transaction, an address without code, an EIP-7702 account (Buterin et al., 2024) or a smart account, that is, a Safe (Safe Ecosystem Foundation, n.d.) or an ERC-4337 account (Buterin et al., 2021); every other address with code is a contract (Section 3.5.1). Every node of every graph is a contract, and user accounts count only in the cohort rule, the labels and the importance measures.
- *On-chain reputation score:* A number computed from on-chain data, without off-chain identity information, that expresses a wallet's relative position for financial activity under pseudonymity (H.-D. Do & Do, 2023). The scores here are ordinal, and their only off-chain inputs are the five-token list and daily US-dollar prices.
- *PageRank:* A graph ranking algorithm that spreads influence through a directed graph with a damping factor (Page et al., 1999). Adaptive Weighted PageRank (AWP) is PageRank on the transfer graph, with edges weighted by the value and age of each transfer and restarts at recently active senders (T. Do et al., 2023). EndorseRank applies AWP's edge weights to the latest-allowance graph and restarts uniformly (Section 3.6).
- *Coupled PageRank (C-PR):* One random walk over both the allowance and the transfer layer, which at each contract mixes the two rows with weights λ and $1-\lambda$ over the layers that contract occupies. Endorsement-seeded PageRank (S-PR) keeps the transfer walk but restarts it from the allowance walk, as TrustRank (Gyöngyi et al., 2004) restarts from a seed set.
- *ERC-20 allowance / Approve / Permit:* Contract methods that record how many of the owner's tokens a spender may spend on the owner's behalf (Vogelsteller & Buterin, 2015); EIP-2612 `permit` makes the same state change as `approve` through a signature (Lundfall, 2020). The graphs use, for each owner–spender pair and token, the latest non-zero allowance up to the

owner's balance of the token at the freeze (the balance cap); an allowance whose owner holds no balance is no edge.

- *Domain alignment:* The degree to which a reputation ranking agrees with an observable proxy ranking, measured with Spearman's ρ (Spearman, 1904) and Kendall's τ (Kendall, 1938) and read as evidence on construct validity (Cronbach & Meehl, 1955); with proxies built from a score's own edges it is only an intended-construct check.

1.4 Temporal holdout

The contemporaneous allowance Kendall τ (Table 4.5) is an intended-construct check: PageRank on G_{approve} should agree with spender-side approval degree, because it smooths that degree globally. To test whether the score conveys information beyond the same-window graph, the scores are frozen at t_1 (31 March 2026, 23:59:59 UTC) and compared with two labels that only become available in April–June 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). In this first window, W0, the holdout cohort ($n = 7{,}864$) consists of every contract spender that holds, at t_1, at least one positive latest allowance of one of the five tokens set in the twelve months before, among the spenders listed for that freeze (Section 3.8). Each walk is also compared with the raw count it smooths at t_1. The labels count every account, most of them user accounts, which no score reads (Table 8.6).

A second window, W1, was registered after the W0 results were known and before the study extracted any log dated after 30 June 2026. Its scores are frozen on 30 June 2026 and its labels counted from July to September 2026, on the $n = 7{,}915$ contract spenders of that freeze (Section 3.8.1).

The analysis plan and the registration of W1 define the registered analysis, with amounts in raw base units of every ERC-20 token, scores from every event since 1 October 2023 and graphs that hold every account of the cohort's events. After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole

chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. This revised analysis, the contract-graph analysis, is the main one (Section 3.5.3). Its verdicts under rule A and under the two criteria of rule B that the thesis applies are a re-analysis, not registered tests, and the registered analysis gave the same readings (Appendix 8.6).

Table 1.2: Summary of hypotheses, research questions, and empirical claims.

Item	Addresses	Reported in
H1–H3 / RQ1	The two single-layer walks: different orderings, agreement with the checks of each edge type, and the cost of each walk	§4.2, §4.3 and §4.4; Tables 4.5, 4.6, 4.3 and 4.4; Claims C5 and C6
RQ2	The discovery-order test: the crawl ordered by C-PR′ against a breadth-first crawl and a crawl ordered by contract backlinks	§4.6 and Appendix 8.4.4; Tables 4.12, 4.13 and 4.14; Figure 4.5; Claim C1
RQ3	Out-of-window prediction of new approval pairs and new transfer senders; each walk against the raw count of the edges it smooths	§4.7 and §4.8.4; Tables 4.15, 4.16, 4.17, 4.20, 4.18, 4.22 and 4.23; Claims C2 and C4
RQ4	The coupled walk against each of its layers walked alone, in window and out of window (rule A)	§4.8 and §4.8.2; Tables 4.5, 4.19 and 4.16; Claim C3
RQ5	The allowance layer added to a transfer walk on new data (two of rule B's registered criteria, F1 and F2)	§4.8.3; Tables 4.20 and 4.21; Claim C3
C1	In the deciding run the crawl ordered by C-PR′ reaches the contracts with the most user-account demand sooner than a breadth-first crawl but not sooner than a crawl ordered by contract backlinks, which is sooner: the answer to RQ2 is no. Outside the decision, in the other reveal mode, C-PR′ leads both comparators	Tables 4.12, 4.13 and 4.14
C2	Out of window, EndorseRank ranks new approval pairs below the in-approve degree; AWP ranks new senders slightly above the transfer in-degree on the whole cohorts but below it among the W0 spenders that another contract had paid; C-PR lies above the union in-degree on approval pairs, below it on senders, and below the degree of each label's own edge type	Tables 4.15, 4.16, 4.17, 4.18, 4.22 and 4.23
C3	Rule A, applied to the scores of the contract graph, is not met in either part; of rule B's two applied criteria F1 holds and F2 fails, so rule B is not met	Tables 4.16, 4.19, 4.21 and 8.16
C4	Beside the rules, C-PR ranks new approval pairs below EndorseRank and above AWP and S-PR, and new senders above EndorseRank and below AWP	Table 4.22; §4.8.4
C5	Same-window agreement of each single-layer walk with the checks of its own edge type, partly built in, and different orderings of the cohort by the two walks	Tables 4.5 and 4.6; Figure 4.3
C6	Runtime and graph size of each score: speed follows graph size and the iterations each walk needs, not the algorithm; C-PR takes slightly longer than AWP	Table 4.3

1.5 Research objectives

The research has five objectives, one for each of RQ1–RQ5. Each is stated with its measurement and with the criterion that Chapter 5 applies when judging whether it was met.

O1. Establish whether EndorseRank and AWP order the contracts of the same cohort differently, how closely each score agrees with each construct in the same window, and what each walk costs, C-PR included (RQ1). Measurement: Kendall τ between the two score vectors ($n = 6{,}443$) and family-level Kendall τ for every score and construct, with 95% bootstrap intervals; solve time on the whole contract graph at T_{obs} over five timed runs, memory, iterations, $|V|$ and $|E|$ under one protocol. Criteria: the interval of the agreement between the two walks contains neither 0 nor 1; a score agrees with a construct when its interval on that family excludes 0, and agreement with the construct of its own edges is read as an intended-construct check; a time ratio is reported only with the ratio of $|E|$.

O2. Test whether a crawl ordered by C-PR on the contracts read so far reaches the contracts with the most user-account demand sooner than a breadth-first crawl and than a crawl ordered by the number of contract backlinks (RQ2). Measurement: crawls of the contract graph from 20 seed contracts under each ordering; for I_T and for I_A, $\text{AUC}_{10\%}$, the normalized area under the curve of hot contracts reached over the first 10% of visits, the hot contracts being roughly the top 1% of the crawl's universe by that importance; two paired contrasts, D_1 (C-PR′ minus breadth-first) and D_2 (C-PR′ minus the backlink count IB'_U), with 95% bootstrap intervals. Criterion: the decision rule stated in the protocol of the test (Section 3.7), read at the freeze of 30 June 2026 in the primary reveal mode: the verdict on an importance is V1 when both intervals lie above 0, V2 when only that of D_1 does and V3 otherwise, and the answer is the weaker of the two verdicts.

O3. Test whether scores frozen at a freeze predict the new approval pairs and new transfer senders that spender contracts gain afterwards, and whether any walk beats the raw count of the edges it smooths (RQ3). Measurement: Kendall τ with 95% intervals between every score at the

freeze, the raw counts included, and both labels, in W0 ($n = 7{,}864$ contract spenders) and W1 ($n = 7{,}915$), with paired contrasts of EndorseRank against the in-approve degree, of AWP against the transfer in-degree and of C-PR against the union in-degree and both single-layer degrees. Criterion: a score predicts a label when its interval excludes 0, and a walk beats a count only when the interval of the paired difference lies above 0.

O4. Test whether one walk across both edge types beats each of its layers walked alone (RQ4). Measurement: paired intervals on $\Delta\tau$ between C-PR at $\lambda = 0.5$ (with 0.25 and 0.75 as sensitivity values) and its two layers walked alone, and the same for its ablation S-PR. Criterion: rule A, fixed in the analysis plan before any log of the study was extracted: C-PR does no worse than its allowance layer on new approval pairs and better than its transfer layer on new senders in W0, and in the same window lies within the interval of the leading layer on the transfer and allowance families and above both layers on Sybil stability. The rule is applied to the scores of the contract graph as a re-analysis.

O5. Test whether adding the allowance layer to a walk over the transfers improves it on new data (RQ5). Measurement: paired contrasts of C-PR with its transfer layer walked alone in W1. Criterion: two of the criteria of rule B, registered after the W0 results were known and before the study extracted the W1 logs: under F1 C-PR beats the transfer layer on new approval pairs, and under F2 it is not worse by more than 0.01 on new senders. They are applied to the scores of the contract graph as a re-analysis, with the two registered sensitivity analyses (AWP in place of the transfer layer; contracts outside a graph kept as isolated nodes). The other contrasts of C-PR are reported beside both rules and decide neither.

1.6 Research questions

Three hypotheses go with the first research question. H1: on the same set of contracts, EndorseRank and AWP produce different orderings. H2: each score agrees with the checks built from the edges it walks on. H3: on the same contract graph, a score's computation time grows with the size

of the layers it walks, so EndorseRank, on the sparser allowance layer, is the fastest to compute. H1–H3 hold largely by construction and act as checks of the instruments. The falsifiable tests are RQ2–RQ5. RQ4 and RQ5 follow the rules of the analysis plan and of the registration, applied to the scores of the contract graph as a re-analysis; RQ2 and RQ3 follow criteria defined after the registered results were known.

RQ1. Do EndorseRank and AWP, the walks over allowance edges alone and over transfer edges alone, order the same contracts differently, does each agree most with current checks of its own edge type, and what does each walk, the coupled one included, cost?

RQ2. Does a crawl ordered by C-PR on the contracts read so far reach the contracts with the most user-account demand sooner than a breadth-first crawl and than a crawl ordered by the number of contract backlinks? (the discovery-order test)

RQ3. Do scores frozen at a freeze predict later new approval pairs and new transfer senders, and does any walk, the coupled one included, beat the raw count of the edges it smooths?

RQ4. Does the coupled walk rank contracts better than either of its layers walked alone, in window and out of window? (rule A)

RQ5. Does adding the allowance layer improve a transfer walk on new data, gaining on later approvals without losing on later inflow? (two of rule B's registered criteria, F1 and F2)

RQ2 and RQ3 measure on the chain, by the demand of user accounts and by later labels, because the study had no label of a contract's importance from outside it. Rule B, the rule for RQ5, came after the W0 labels had been seen, so it is judged only on W1 (Section 3.8.1).

1.7 Summary of the literature review

Chapter 2 reviews four areas. In graph ranking, PageRank (Page et al., 1999) ranks nodes by recursively propagated endorsement, EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) adapt it to peer-to-peer trust and web spam, and crawl-ordering experiments ask how soon an ordering leads a crawler to the important pages (Cho et al., 1998). In on-chain reputation,

H.-D. Do and Do (2023) apply PageRank and HodgeRank to Ethereum transactions, and AWP (T. Do et al., 2023) weights transaction edges by value and age and checks its ranking against inbound transfer count and value, statistics drawn from the same data as its graph. D. Lin et al. (2025) propagate account risk through the transaction network, and surveys of blockchain trust and reputation systems (Bellini et al., 2020) and of reputation systems built on smart contracts (Almasoud et al., 2020) describe designs that rest on transfers or explicit ratings. Deployed DeFi tooling such as Human Passport instead aggregates on-chain and off-chain credentials to decide whether an address belongs to a distinct person (Human Passport, n.d.). None of this work ranks contracts on the relations among contracts alone or takes the ERC-20 allowance (Vogelsteller & Buterin, 2015) as a ranking edge; address clustering uses approvals only to link addresses of one entity (Victor, 2020). In DeFi risk and measurement, studies of wash trading (Victor & Weintraud, 2021), extractable value (Qin et al., 2022), scam tokens (Xia et al., 2021) and airdrop farming (Luo et al., 2025) show that much transfer activity is artificial. In measurement theory, construct validity (Cronbach & Meehl, 1955) separates what a score measures from external criteria, and the bootstrap (Efron, 1979) gives distribution-free intervals for rank statistics. The gap (Section 2.22) is a ranking that reads the relations among contracts with a walk over allowance and transfer edges, and a test of whether it finds the contracts that users rely on, beside the raw counts, with confidence intervals, paired differences, a fixed timing protocol and temporal holdouts.

1.8 Summary of the research methodology

Chapter 3 applies a quantitative, comparative design to public ledger data. ERC-20 `Approval` and `Transfer` logs on Arbitrum One from 1 October 2023 to 30 September 2026 are extracted from Google BigQuery in two scans, the second after W1 was registered. The main analysis takes from them every event of five tokens from 1 April 2025, types each address by its transactions and by its code at one pinned block, and keeps the events between two contracts in the twelve months before each freeze, in US dollars, with each allowance counted up to its owner's balance; these definitions

were made after the registered results were known (Section 3.5.3). One power-iteration PageRank implementation computes every walk. The discovery-order test crawls the contract graph of each freeze under each ordering and compares how soon the crawls reach the contracts with the most user-account demand. Each score is also compared with Spearman's ρ and Kendall's τ against same-window transfer, allowance and Sybil-stability proxies and, out of window, against new approval pairs and new transfer senders on the spender contracts of W0 and W1, with the raw counts at each freeze as baselines. Except in the value-slope and sample-definition checks, every coefficient has a 95% interval from 400 paired resamples (seed 42), and method comparisons are paired. The code, configuration and machine-readable summaries behind every table are in a public repository (Section 3.14), and the study uses only public data (Section 3.17).

1.9 Contributions

The thesis makes five contributions to the study of on-chain reputation. EndorseRank and AWP both apply weighted PageRank (Page et al., 1999) to a single edge type, each with its own restart distribution; the coupled walk C-PR changes the transition rule itself. Most results are negative for the walk and are stated with each contribution.

- Coupled walk over authorization and payment (C-PR): one random walk over a US-dollar allowance layer, with each allowance counted up to its owner's balance and capped, and a US-dollar transfer layer, which at each contract mixes the two transition rows with a fixed weight λ over the layers that contract occupies; S-PR is its seeded ablation. No reputation method reviewed couples the two relations (Section 2.22).
- A discovery-order test for a ledger ranking: the crawl-ordering experiment of Cho et al. (1998) repeated on a graph of contracts, with importance measured on the events of user accounts, which no ordering reads, and a decision rule stated in its protocol. The answer to RQ2 is no. The verdict is V2 on both importances: a crawl ordered by C-PR$'$ reaches the contracts with the most user-account demand sooner than a breadth-first crawl but not sooner than a crawl

ordered by the number of contract backlinks, and the backlink ordering was sooner (Claim C1). Outside the decision, the allowance layer adds a small, resolved amount to C-PR′, and in the reveal mode that shows out-edges only, which carries no verdict, C-PR′ is sooner than both comparators.

- Its test under the rules of the plan and the registration: rule A, fixed with λ in the analysis plan, and the two applied criteria of rule B set C-PR against its own layers walked alone; only these contrasts change the coupling and nothing else. Applied to the scores of the contract graph, rule A is not met in either part. Rule B is not met. The thesis applies two of its registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds. F1, the gain over the transfer layer on new approval pairs in W1, holds, and F2 fails because the loss on new senders exceeds the margin (Claim C3). Beside the rules, C-PR ranks later approval pairs below EndorseRank and above AWP and its ablation S-PR, and later senders above EndorseRank and below AWP (Claim C4).
- The balance-capped allowance between contracts as a ranking edge (EndorseRank), the allowance reference: an edge $u \to v$ stands for the latest in-window ERC-20 allowance that the spender contract v holds from the owner contract u, valued in US dollars up to the owner's balance and weighted as AWP weights a transfer. The edge records a delegation of spending rights that the owner can back, not a history of payments, and no reputation method reviewed ranks on it (Section 2.22). Between contracts the layer is thin: most latest allowances are capped to zero, and at T_{obs} one owner contract holds most of the pairs that remain (Tables 8.5 and 4.1). At depth one the walk has a closed form, $c\,(1 + d\,\delta(v))$ with $\delta(v)$ the sum of the transition probabilities into the spender v, c the score common to nodes without in-edges and d the damping factor; it holds to within 1% for every depth-one spender of the cohort but for only a minority of the deeper ones (Section 5.3.2).
- Evaluation of every walk against the count it smooths: every score is checked against the same proxies on one cohort under the protocol used with AWP (T. Do et al., 2023), against the approval pairs and senders that arrive after a freeze, and in discovery order, each time

beside the raw counts, with 95% bootstrap intervals and paired contrasts (Efron, 1979). On these data a count wins more often than a walk: the count of contract backlinks reaches the important contracts sooner than C-PR′, EndorseRank ranks later approval pairs below the in-approve degree with either restart rule, and C-PR ranks later approval pairs above the union in-degree but later senders below it. AWP is the exception: its crawl is sooner than the crawl ordered by transfer backlinks, and it ranks later senders slightly above the transfer in-degree on the whole cohorts, though below it among the W0 spenders that another contract had paid (Claims C1 and C2).

1.10 Scope and limitations

The study extracts Arbitrum One from 2023-10-01 to 2026-06-30, W1 adds July to September 2026, and the main analysis scores the twelve months before each freeze, on the matched cohort and on the contract spenders of each freeze. The graphs hold the five-token events between contracts of the whole chain, and no user account moves a score. Account type is read from an address's transactions and from its code at one pinned block, not at the time of each event, and the rule for smart accounts recognizes Safes and ERC-4337 accounts, so kinds that it misses stay in the graphs as contracts (Appendix 8.4.3). The allowance layer between contracts is thin, and at T_{obs} one owner contract holds most of its pairs (Table 4.1). About a third of the matched cohort and more than a quarter of each spender cohort end no pair between contracts and score zero under every score and baseline (Table 4.2). The verdict of the discovery-order test is read in the reveal mode that the rule of its protocol made primary; in the other mode the contrast with the backlink count has the opposite sign (Table 4.12). No claim is made that the results generalize beyond Arbitrum One, the five tokens or the months studied. The amounts in US dollars on five tokens, the twelve-month score windows, the contract graph with its balance cap and the discovery-order test were defined after the registered results were known (Section 3.5.3). Adversarial manipulation remains a threat although every node costs a deployment: Section 5.6 shows that a closed farm of contracts collects

several times its population share of EndorseRank, and that a balance threshold raises the capital a farm must hold while it removes most of the honest graph (Table 5.2).

1.11 Research roadmap

Chapter 2 reviews graph-based on-chain reputation, crawl ordering, ERC-20 allowance semantics and the reputation tooling deployed in DeFi, and develops the theoretical framework. Chapter 3 defines the account types, the tokens and prices, the contract graph and its balance cap, the scores, the discovery-order test (Section 3.7), the temporal holdouts, the construct checks, the timing protocol, and the standards for reproducibility and ethics. Chapter 4 reports the data of the contract graph, the benchmark (Claim C6), the same-window alignment of every score and the divergence of the two single-layer walks (C5), robustness checks, the discovery order (Section 4.6; C1), the temporal holdouts of the single-layer walks (C2), and the coupled walk under its rules and beside every other score (C3 and C4); Appendix 8.4.3 describes the account types and balances, Appendix 8.4.4 gives the further tables of the discovery-order test, Appendix 8.5 sets EndorseRank and AWP side by side, and Appendix 8.6 reports the registered analysis. Chapter 5 interprets the findings by research question and objective, works out their implications, models the cost of Sybil attacks on allowance graphs, and weighs threats to validity and limitations. Chapter 6 answers RQ1–RQ5, states the contribution to knowledge and sets out future work.

2 Literature Review and Theoretical Framework

The research problem set out in Chapter 1 is that a rollup holds far more contracts than anyone can inspect, and that the usual ways of forming trust in a contract do not read which contracts other contracts authorize and pay. The thesis asks whether a coupled PageRank over the allowance and transfer edges between contracts, C-PR, finds the contracts that users rely on, as a PageRank ordering was found to reach important web pages sooner than other orderings (Cho et al., 1998). Its reference walks are EndorseRank, over the allowance edges alone, and Adaptive Weighted PageRank (AWP), over the transfer edges alone; the objectives O1–O5 and research questions RQ1–RQ5 follow from that question. The gaps the literature leaves are taken up in Chapters 3 and 4.

2.1 Blockchain and DeFi foundations

A public blockchain records transactions in hashed blocks, and each block is secured by the ones added after it (Nakamoto, 2008). Once a majority of the nodes has accepted a block, rewriting it is costly, so parties can settle on the ledger without a known legal identity (*trust-minimized* settlement).

Bitcoin built an electronic cash network on such a ledger (Nakamoto, 2008); Ethereum lets an account hold a program alongside its balance. A transaction can then call one of the program's functions and change the contract's state (Buterin, 2014), and these functions run on the Ethereum Virtual Machine (EVM) (Wood, 2025). Such programs are called smart contracts (Szabo, 1997); any honest node that replays a function call gets the same output (Wood, 2025). Financial terms can therefore be enforced by the contract rather than by a trusted third party (Schär, 2021).

Together these contracts make up Decentralized Finance (DeFi), doing the work of exchanges,

lending platforms, derivatives markets and asset managers; Werner et al. (2022) define the protocol classes, and Schär (2021) describes how protocols are layered on one another. Because protocols interact, new financial instruments can be composed from existing ones (Kitzler et al., 2023), and contracts come to authorize and pay one another, the relations that the graphs of this thesis record. The same property lets a failure spread across protocols, as in flash-loan exploits (Qin et al., 2021) and price-oracle manipulation (Zhou et al., 2023). Transaction histories on these chains are public, yet the addresses in them are pseudonymous, which hampers the identity checks of conventional finance (D. Lin et al., 2025).

2.1.1 Pseudonymity and the choice of contracts

On a public blockchain the basic unit of analysis is an address (a wallet), which may stand for a legal entity, a trust, an algorithmic agent or a smart contract. One entity can own several addresses (Douceur, 2002), and statistical clustering can recover ownership only in part (Victor, 2020). One consequence concerns credit: with no legal entity to tie a credit history to, on-chain lending asks for collateral worth more than the loan (Bartoletti et al., 2021), and proposed on-chain credit scores estimate repayment from on-chain behavior, with prospect theory (Hassija et al., 2020) or machine learning on lending data (Palaiokrassas et al., 2024).

A second, more basic consequence is the one this thesis takes up. Anyone can deploy a contract, an address says nothing about the code behind it, and the ledger does not say which of its very many contracts matter. A user asked to approve a spender, an explorer that labels addresses, an auditor and an integrator must each decide which contracts to look at first (Section 2.4).

2.2 Ethereum, Layer-2 rollups, and Arbitrum One

Most DeFi applications run on the Ethereum mainnet, which under proof of work could not raise its block size or shorten its block interval far without weakening its security (Gervais et al., 2016), so growing use raised gas costs and constrained throughput. Layer-2 (L2) protocols try to keep

Ethereum's security guarantees while making transactions cheaper: some execute transactions off-chain, others batch them and post a succinct summary to the mainnet (Gudgeon et al., 2020), and Thibault et al. (2022) survey and compare rollups. Optimistic L2s such as Arbitrum treat each submitted batch as valid unless it is challenged within a dispute period (Kalodner et al., 2018).

Arbitrum One is such an optimistic L2 and executes EVM smart contracts. We use it because the public BigQuery dataset includes both its ERC-20 `Approval` and `Transfer` events, and because its fees are low enough that the windows studied contain dense activity. It also holds very many contracts: Table 8.4 counts those in the five-token events of this study alone. We make no claim that our results generalize to other L2s or to Ethereum mainnet without further validation. The framework can be applied to any chain that emits the same logs.

2.3 ERC-20 tokens and the allowance lifecycle

The ERC-20 standard defines a common interface for fungible tokens, made up of `transfer`, `approve`, `transferFrom`, `allowance` and the associated events (Vogelsteller & Buterin, 2015). After the owner calls `approve(spender,amount)`, the token contract records that the spender may move up to `amount` out of the owner's balance. The spender can then move the tokens with `transferFrom`. With EIP-2612 `permit`, an allowance can be updated through an off-chain signature that settles in a single transaction, with the same semantics of authorization (Lundfall, 2020).

Figure 2.1 traces an ERC-20 allowance through its lifecycle. An allowance is an authorization and moves no tokens. An owner, whether an externally owned account or a contract, can grant allowances to routers, vaults or trading contracts that it trusts to act within set limits. Revoking (approving zero) and updating (setting a new non-zero allowance) both change the allowance currently in force, and `transferFrom` reduces it; some tokens record the remainder in a new `Approval` event. EndorseRank keeps only the most recent approval of each owner, spender and token, and that approval loses weight as it ages even while it stays in force (Section 2.8).

Between contracts, allowances are much sparser than transfers (Table 4.1): a single approval can

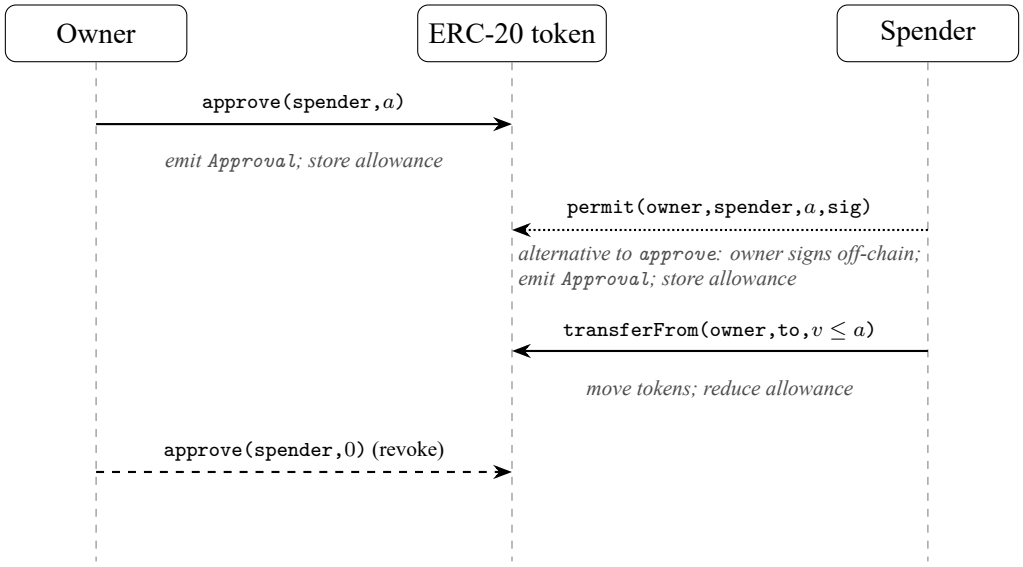


Figure 2.1: ERC-20 allowance lifecycle: `approve`, or its signature-based alternative `permit` (dotted), an optional `transferFrom`, and revoke. EndorseRank keeps the most recent `Approval` event of each owner, spender and token in the score window and weights it by its age and by its US-dollar amount up to the owner's balance.

cover many transfers, most contracts approve no other contract on the five tokens, and most allowances that contracts do grant come from owners that hold none of the token at the freeze (Table 8.5).

2.4 Graph-based on-chain reputation

PageRank and its variants rank nodes by the influence that flows to them along directed edges (Page et al., 1999), and recent studies apply them to blockchain transaction graphs. H.-D. Do and Do (2023) run PageRank and HodgeRank on a sample of Ethereum transactions as a measure of social credit; their nodes are wallet addresses, whatever kind of account lies behind them, and the highest-ranked addresses they identify are exchanges, routers, token contracts and other protocol contracts. T. Do et al. (2023) propose Adaptive Weighted PageRank (AWP), which scores on-chain entities from their transaction histories and is evaluated here beside EndorseRank (Section 2.19). D. Lin et al. (2025) propagate risk through the transaction network to rate accounts without labels, and Wu et al. (2022) detect phishing addresses from a network embedding of the transaction graph.

Bellini et al. (2020) survey blockchain-based trust and reputation management, and Almasoud et al. (2020) review reputation systems built on smart contracts, in which users rate providers. Data mining on cryptocurrency transactions likewise works from transaction histories (Liu et al., 2021). In graphs of this kind every address in the data is a node. Mukhia et al. (2026) sort more than 81 million ERC-20 transfers on Ethereum by whether sender and recipient are externally owned accounts (EOAs) or smart contracts, and find that transfers between smart contracts follow other statistical patterns than those of EOAs. In the events of this study, user accounts (EOAs and smart accounts; Section 2.17) are most of the addresses and most of the accounts that newly approve or pay a spender contract (Tables 8.4 and 8.6). A user account costs no deployment, and the study does not measure who controls one, so the graphs of this thesis hold contracts only.

Outside the research literature, trust in a contract is formed in ways that each look at one contract or one account at a time. A wallet application shows a prompt before a user approves a spender; of the interfaces of 22 decentralized applications and 9 wallet applications that Wang et al. (2022) examined, 3 explained the approval mechanism. Code analysis reads the contract itself (Z. Lin et al., 2024); audit reports, which the incident study of Zhou et al. (2023) draws on, record a review of its code; and labels name known addresses but are sparse (D. Lin et al., 2024).

Attestation tooling summarizes evidence about the holder of an address. Human Passport (the former Gitcoin Passport) collects weighted verifiable credentials, called stamps, for checks such as government identity, biometrics, vouching and past web2 and web3 activity, and an address passes when the weighted sum reaches a threshold (Human Passport, n.d.). According to the same documentation, Passport also classifies EVM addresses on several networks, Arbitrum among them, as human or Sybil from their on-chain activity. Its purpose is Sybil resistance, the aim of the proof-of-personhood protocols that Siddarth et al. (2020) and Bordeianu and Popescu (2026) review, so it works as a gate and not as a graded ordering. None of these ways of forming trust, as reviewed here, uses the record of which contracts other contracts authorize and pay; the scores of this thesis are built from that record alone (Table 2.1).

A propagated score must also beat a simpler one. Where degree correlations are weak, as on the

Web graph, PageRank is on average proportional to in-degree and can be approximated from it (Fortunato et al., 2008). New links concentrate on nodes that already have many, as preferential attachment assumes (Barabási & Albert, 1999) and measured network growth shows (Newman, 2001), so link prediction, on Ethereum transaction networks too (D. Lin et al., 2022), uses degree-based predictors as standard baselines (Liben-Nowell & Kleinberg, 2007).

2.5 PageRank formalism

Consider a directed graph $G = (V, E)$ with $|V|$ nodes, and write $w_{ij} > 0$ for the weight of the edge from i to j. That weight stands for influence from i to j, in the endorsement or the transfer sense. Let W be the corresponding weighted adjacency matrix, and let $o_i = \sum_j w_{ij}$ be the out-strength of node i. The row-stochastic transition matrix P is then

$$P_{ij} = \begin{cases} w_{ij}/o_i & \text{if } o_i > 0, \\ 1/|V| & \text{if } o_i = 0 \end{cases} \tag{2.1}$$

(dangling nodes spread their mass evenly). Given a damping parameter $d \in (0, 1)$, the Google matrix is

$$G_{\text{PR}} = d\, P^{\top} + \frac{1-d}{|V|}\, \mathbf{1}\mathbf{1}^{\top}, \tag{2.2}$$

and the PageRank vector $\mathbf{r}$ satisfies $\mathbf{r} = G_{\text{PR}}\mathbf{r}$ with $\mathbf{r} \geq 0$ and $\sum_i r_i = 1$ (Langville & Meyer, 2006). Element by element, with the dangling rows of P written as a separate middle term, this becomes

$$r_j = d \sum_{i:(i\to j)\in E} r_i \frac{w_{ij}}{o_i} + \frac{d}{|V|} \sum_{i:\, o_i = 0} r_i + \frac{1-d}{|V|}. \tag{2.3}$$

2.5.1 Random-surfer interpretation

Equation (2.3) can be read probabilistically. At each step a surfer at a node with out-edges follows one of them with probability d, choosing it in proportion to its weight, or teleports to a uniformly

chosen node with probability $1 - d$ (Figure 2.2). At a node without out-edges the surfer always jumps to a uniformly chosen node; this is the dangling rule of Equation (2.1). The stationary distribution then ranks nodes by how often they are visited in the long run. Damping keeps rank sinks (groups of nodes with no edge leaving them) from absorbing all the mass. With $d < 1$ and uniform teleportation, every entry of G_{PR} is positive, so G_{PR} is primitive and its Perron vector is unique (Langville & Meyer, 2006). Appendix 8.3.1 gives the argument.

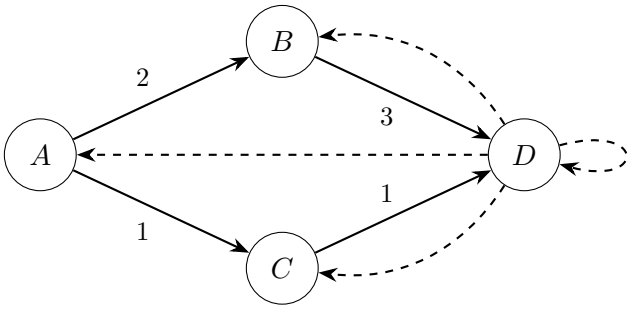


Figure 2.2: Random-surfer intuition for PageRank on the four-node graph of Section 2.14; edge labels are the allowance weights. Solid arrows are graph transitions, followed with total probability d in proportion to their weights. Dashed arrows show uniform teleportation, which reaches every node with probability $(1 - d)/|V|$; the dashed fan is drawn only from D, although every node has one. D has no outgoing edge, so under the dangling rule of Equation (2.1) a surfer at D always jumps to a uniformly chosen node.

2.5.2 Power iteration and complexity

In practice, $\mathbf{r}$ is usually obtained by power iteration from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/|V|$,

$$\mathbf{r}^{(t+1)} = G_{\text{PR}}\mathbf{r}^{(t)}, \tag{2.4}$$

until $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or until the iteration cap $I_{\max}$ is reached. On a sparse graph, each iteration costs $O(|E|)$ arithmetic operations (Leskovec et al., 2020). A method with fewer edges, such as latest-allowance snapshots in place of full transfer histories, therefore lowers the cost per iteration and sometimes the memory.

By default this work sets damping to the conventional $d = 0.85$ (Page et al., 1999) and uses two study-specific solver settings, a convergence tolerance $\varepsilon = 10^{-8}$ and a maximum of $I_{\max} = 300$

iterations.[1] Damping is varied over $d \in \{0.75, 0.85, 0.95\}$ in the robustness section.

2.6 Crawl ordering

A crawler has read only some pages and must choose which of the pages they link to it reads next; if it stops early, that order decides which pages it holds. Cho et al. (1998) study this problem of URL ordering: to visit the important pages first with an ordering that uses only what the crawl has seen. Importance is, among other measures, a page's backlink count over the whole collection or its PageRank; the orderings are breadth-first order and the same two quantities computed on the pages seen so far; and an ordering is judged by the share of the *hot* pages, those above an importance threshold, visited after a given number of pages. On a stored copy of the Stanford University web, about 179,000 pages reachable from the home page where every crawl began, the crawl ordered by PageRank (damping factor 0.9) reached the pages with at least 100 backlinks sooner than the crawl ordered by the backlink count. The authors call the result counter-intuitive and trace it to the start of a crawl, where the backlink count keeps the crawler inside one cluster of pages. With hot pages defined by PageRank, the PageRank ordering led by more. The authors name the single collection as a limitation.

Najork and Wiener (2001) crawled 328 million pages in breadth-first order, found that pages with a high PageRank were downloaded early, and argue that this order is the better trade-off because recomputing PageRank during such a crawl is too costly; they ran no other ordering. Baeza-Yates et al. (2005) simulated crawls of the Chilean and the Greek web, between two and four million pages each, with PageRank as the importance: orderings by the backlink count and by a partially updated PageRank did worse than a random crawl, breadth-first order was close to the best orderings only over the first 20 to 30 percent of the pages, and the authors place PageRank recomputed in batches on the pages seen so far among the best, together with two cheaper strategies that they recommend. In both studies that compare orderings, the backlink count thus trails a PageRank that is recomputed

[1] They keep run-times reproducible and are not part of the classical PageRank formulation.

in full on the pages seen so far.

The discovery-order test of this thesis repeats the experiment of Cho et al. (1998) on the contract graph (Section 3.7). A page becomes a contract and a link an allowance or transfer edge between two contracts; a visit reads a contract's edges, and the next contract is chosen by an ordering computed on what has been read so far: breadth-first order, the number of contract backlinks, or a walk such as C-PR. Importance is taken from outside the graph: it is the demand of user accounts, the US dollars they transferred to a contract and the balance-capped US dollars they authorized it to spend, which no ordering reads. Importance by backlink count and by PageRank, as in the original experiment, is an intended-construct check and is reported second.

2.7 Reputation systems lineage

PageRank belongs to a wider family of spectral and propagation-based reputation mechanisms. Kleinberg's HITS algorithm gives each page a hub score and an authority score (Kleinberg, 1999). EigenTrust extends the idea to peer-to-peer networks and normalizes local trust, so a malicious node cannot raise its own score at will (Kamvar et al., 2003). To suppress web spam, TrustRank starts from a seed set of trusted nodes and propagates their trust outward (Gyöngyi et al., 2004). Where trust flows only from a vetted seed set, a newcomer cannot raise its score without endorsements from nodes that already hold reputation. With uniform teleportation, which EndorseRank uses (Section 3.6.2), that protection is weaker. Gyöngyi and Garcia-Molina (2005) derive the link farms and farm alliances that most raise a target's PageRank, and Cheng and Friedman (2005) show that symmetric reputation functions such as PageRank can be manipulated with Sybils, whereas some asymmetric functions based on flows or paths from a trusted node resist them. The Sybil cost model of Section 5.6 prices such a farm on the allowance layer of the contract graph.

On-chain, the Sybil attack remains a central concern (Bordeianu & Popescu, 2026). Every node of the graphs of this thesis is a contract, which costs a deployment, and its allowance counts only up to the balance it holds. The Sybil-adjusted stability proxies used in this work (inbound counter-

party ratio, tenure, active months) partly capture diversity of interaction, but do not prevent their manipulation.

2.8 ERC-20 allowances as endorsement

Through the ERC-20 `approve` function (Vogelsteller & Buterin, 2015) and EIP-2612 `permit` (Lundfall, 2020), an owner states explicitly how much a spender may move on its behalf. EndorseRank reads the latest allowance from an owner contract to a spender contract as one endorsement. On each of the five tokens of the study (Section 3.5.3), the latest in-window approval of the pair contributes its time weight times the bounded value weight of its US-dollar amount, as AWP weights a transfer; a pair whose latest value is zero is no edge, and the terms are summed over tokens (Equation (3.7) in Section 3.6.2). PageRank on the resulting graph G_{approve} yields the EndorseRank scores $\mathbf{s}_{\text{ER}}$.

An allowance may exceed the owner's balance, and an unlimited one always does, yet a spender can never move more than the owner holds. Each latest allowance therefore counts up to its owner's balance of the token at the freeze, in US dollars at the price of that day, and an allowance whose owner holds none of the token is no edge (the balance cap; Section 3.6).

Treating allowances as endorsements is a deliberate choice, and an imprecise one. An allowance between two contracts is a working relation between programs, not a judgment that the spender is trustworthy, and an allowance manager such as Uniswap's Permit2, which owners approve once so that applications can spend through it, collects the endorsements meant for those applications (Section 3.6.2). Even so, an allowance signals approval more explicitly than a received transfer, which could come from market making, wash trading (Cong et al., 2023), airdrop farming (Luo et al., 2025) or routing without any intent to endorse.

Victor's self-authorization heuristic reads ERC-20 approvals as a relation between addresses and clusters those that it attributes to one entity (Victor, 2020). That reading links addresses and ranks nothing. It is also a caution, since an approval between two addresses of one entity is a self-

endorsement, the edge that the Sybil cost model of Section 5.6 prices.

2.9 What this research does not claim

The evidence in this thesis is of three kinds, and none of them an outcome for the users of a contract. Importance in the discovery-order test is the demand of user accounts; it shows that users rely on a contract, not that they are right to. The out-of-window labels count later events of the two kinds from which the graphs are built. Construct validity expects a score to agree with checks of the construct it was built to measure (Cronbach & Meehl, 1955), so each walk's lead on the same-window checks of its own edges is partly built in. No score in this thesis is offered as a measure of the safety of a contract, of the gains or losses of its users, or of whether a party will repay what it owes. Nor is any score a security audit: a contract can contain malicious functions whatever its standing among owners (Z. Lin et al., 2024), and the scores read neither code nor upgrade permissions. Apart from the token list and daily prices, the scores use no off-chain information.

2.10 Rank correlation theory

Reputation scores and the proxies of the construct checks are ordered quantities without a natural boundary for classification. Following the AWP validation design (T. Do et al., 2023) and classical psychometrics (Cronbach & Meehl, 1955), this research measures two kinds of rank agreement, monotonic and pairwise. Bootstrap confidence intervals quantify the sampling uncertainty of every coefficient of the main analysis (Efron, 1979).

Spearman's rank correlation (Spearman, 1904) is the Pearson correlation of the ranks. For paired ranks (R_i, S_i) of n items without ties, it reduces to

$$\rho = 1 - \frac{6 \sum_{i=1}^{n} (R_i - S_i)^2}{n(n^2 - 1)}. \tag{2.5}$$

With ties, Spearman's ρ is the Pearson correlation of average ranks (mid-ranks), in which tied items

share the mean of their positions, and every reported ρ is computed this way.

Kendall's τ (Kendall, 1938) rests on counts of concordant and discordant pairs. Of the $n_0 = n(n-1)/2$ pairs (i, j) with $i < j$, n_c are concordant (both rankings order them the same way) and n_d are discordant (the rankings order them oppositely); a pair tied in either ranking is neither. Let $n_1 = \sum_k t_k(t_k - 1)/2$, where t_k is the size of the k-th group of tied values in the first ranking, and define n_2 in the same way for the second ranking. The τ-b variant is then

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_0 - n_1)(n_0 - n_2)}}, \tag{2.6}$$

and without the tie corrections it becomes $\tau_a = (n_c - n_d)/n_0$. Every τ reported in this thesis is τ_b, computed on the raw scores and proxies. The difference matters for heavily tied scores: the raw degrees are integer counts, and a walk gives one value to every contract that has no edge in its graph, which for EndorseRank is most of the matched cohort (Section 4.4). Kendall's τ is usually the easier coefficient to interpret for pairwise orderings, and Chapter 4 uses it as the main family-level summary. Its values are compared with those of other methods on the same cohort; no uncited absolute benchmark is used.

2.11 Theoretical framework

The study first compares the two single-layer walks, EndorseRank and AWP, on the same contracts: whether they order them differently (H1), whether each agrees with the checks of its own edge type (H2) and what each costs (H3); together these make up RQ1. It then tests the coupled walk C-PR, the method of the thesis, by the order in which a crawl that follows it reaches the contracts with the most user-account demand (RQ2), by the prediction of later approvals and senders against the raw counts (RQ3), and under its rules (RQ4 and RQ5). RQ4 and RQ5 follow the rules of the analysis plan and of the registration, applied to the scores of the contract graph as a re-analysis; RQ2 and RQ3 follow criteria defined after the registered results were known (Section 3.5.3). Six arguments frame these steps.

- Hyperlink–citation semantics: In the original PageRank algorithm (Page et al., 1999), an inbound hyperlink counts as a citation. H.-D. Do and Do (2023) apply this reasoning to Ethereum transactions, and AWP (T. Do et al., 2023) weights each transfer edge by the age and the value of its transfers (Section 3.6.1). EndorseRank extends citation-based propagation to ERC-20 approve and permit edges: an allowance is an explicit, on-chain delegation of spending authority (Vogelsteller & Buterin, 2015).
- Contracts as nodes: Link farms raise a target's PageRank (Gyöngyi & Garcia-Molina, 2005), so a ranking by links depends on what a node and a link cost to make. A contract costs a deployment, and its allowance counts only up to the balance it holds, so the graphs hold contracts only and no score or baseline reads an edge of a user account. User accounts enter only through the labels and as the importance that the discovery-order test asks an ordering to find (Section 2.6).
- Domain alignment framework: Measurement theory uses construct alignment to ask how closely an indicator follows the construct it is meant to track (Cronbach & Meehl, 1955), and Messick (1995) widens the idea into one unified judgment about what a score means and what using it implies. Graph coherence is checked with rank correlations (Section 2.10) against same-window transfer and allowance statistics; the out-of-window checks are temporal holdouts on the spender cohorts, new approval pairs and new transfer senders in W0 and in the registered window W1.
- Computational complexity: Each PageRank iteration costs $O(|E|)$ (Section 2.5.2), so on the sparser allowance graph an iteration needs less time and memory; this argument underlies H3. The number of iterations also depends on the damping factor (Langville & Meyer, 2006), the graph and the restart vector. Every walk uses the same solver, so runtime differences come from the graphs and restart vectors and are not an algorithmic contribution (Appendix 8.3.2).
- Local counts as a baseline: Where contracts that already have many counterparties gain the most new ones (Section 2.4), the raw count is a strong predictor of future arrivals, and in the

crawl-ordering experiments PageRank is measured against the backlink count. A PageRank differs from the count in three ways: the endorsers' own scores propagate along the edges, the edges are weighted by age and amount, and the walk restarts by a rule of its own. RQ2 and RQ3 therefore set each walk against the count of the edges it smooths, and neither attributes a difference to propagation alone.

- Layer coupling: Multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) treat several relations as layers over one node set. Mixing the layers node by node, as C-PR does, leaves a question open for layers of very different density: a node with no out-edge in the sparse layer follows the dense one, so the signal of the sparse layer may or may not survive. RQ4 therefore asks whether the mixture beats both layers (rule A), and RQ5 whether it improves on its transfer layer walked alone (two of rule B's registered criteria, F1 and F2; Section 3.8.1). Both layers weight edges by US-dollar amounts without AWP's bounded value weight and are walked with uniform restarts, so neither walk is AWP or EndorseRank, and C-PR is compared with AWP separately (Section 4.8.4). Coverage matters there: a walk over one layer leaves every contract without an edge of that layer in one tie, and the other layer may order some of them.

Figure 2.3 sets out these parts. Only EndorseRank and AWP enter H1; H2 and H3 concern every score.

2.12 Terminology

The formal constructs are defined in Chapter 1; the terms below are used in interpreting results:

- Contract and user account: every node of every graph is a contract; a user account (an EOA, an EIP-7702 account or a smart account) is a node of no graph (Section 2.17).
- Single-layer walks: EndorseRank (allowance layer) and AWP (transfer layer), the two PageRank scores that walk on one edge type; C-PR walks both (Section 3.6.3).
- Balance cap: an allowance counts up to its owner's balance of the token at the freeze.

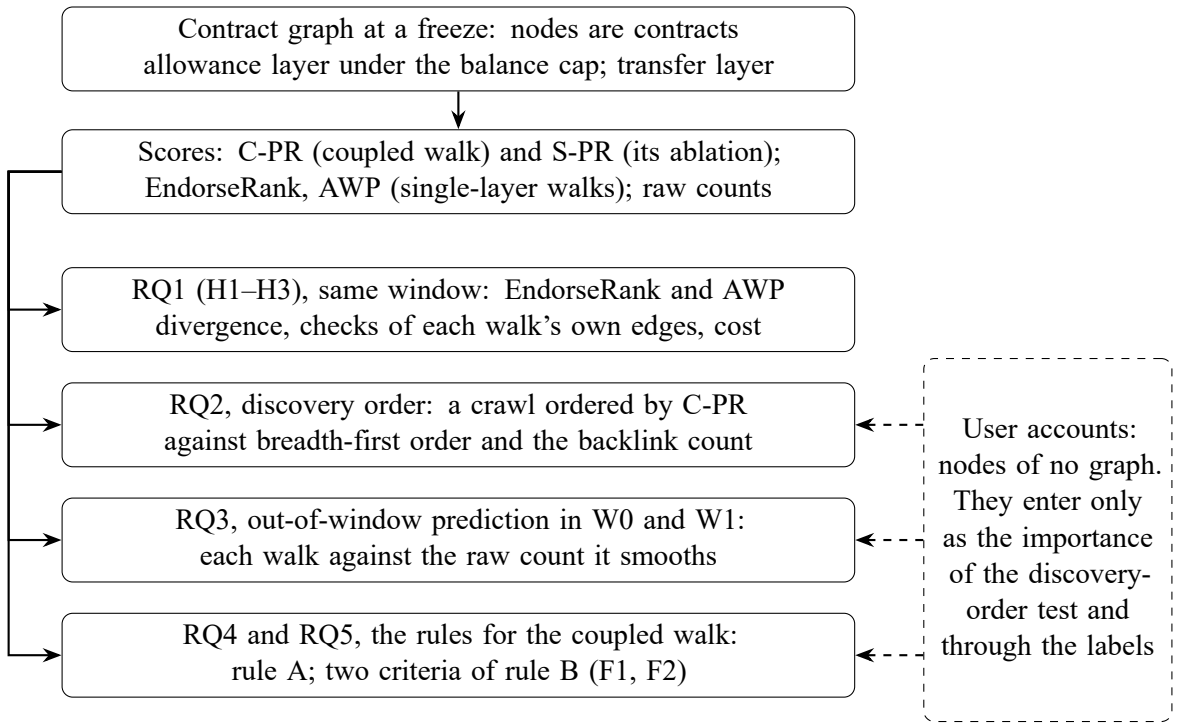


Figure 2.3: Conceptual model. The contract graph of a freeze, whose nodes are contracts only, is the only input of the scores, and the scores enter four evaluations, from top to bottom in the order of the research questions. User accounts (dashed) are nodes of no graph: they enter only as the importance of the discovery-order test and through the out-of-window labels, which count every account.

- Intended-construct check: agreement of a score with a same-window statistic of its own edges (Cronbach & Meehl, 1955); it is not offered as validation.
- Discovery-order test: crawls of the contract graph, each ordered by a score computed on the part read so far, compared by how soon they reach the contracts with the most user-account demand (Section 3.7).
- Out-of-window prediction (temporal holdout): in the first window, W0, scores are frozen at t_1 (31 March 2026), and new owner–spender approval pairs and new transfer senders are counted up to t_2 (30 June 2026); a registered second window, W1, freezes scores on 30 June 2026 and counts labels from July to September 2026 (Section 3.8.1). The matched cohort and the spender cohorts are the contract sets of Section 4.1.1.

2.13 Domain alignment metrics

Construct alignment is measured by Spearman's ρ and Kendall's τ between scores and check rankings (Section 2.10), on same-window checks and on the out-of-window labels; the discovery-order test has a statistic of its own (Section 3.7). Chapter 3 gives the operational definitions.

2.14 Worked example: PageRank on a four-node endorsement graph

Take four contracts $\{A, B, C, D\}$ joined by the latest-allowance edges

$$A \to B\ (2), \quad A \to C\ (1), \quad B \to D\ (3), \quad C \to D\ (1),$$

with damping $d = 0.85$. The out-strengths are $o_A = 3$, $o_B = 3$, $o_C = 1$ and $o_D = 0$ (the last is dangling). The edges $B \to D$ and $C \to D$ are the only out-edges of their nodes, so each is followed with probability one and its weight has no effect on the scores. Teleportation adds $(1 - d)/4 = 0.0375$ to every node at every step, the first one included. One power-iteration step from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/4$ gives $\mathbf{r}^{(1)} = (0.0906,\ 0.2323,\ 0.1615,\ 0.5156)$ for (A, B, C, D). The iteration converges to the stationary vector $\mathbf{r} = (0.1375,\ 0.2154,\ 0.1765,\ 0.4706)$, so D comes first, B and C follow, and A is last. No contract endorses A. Apart from teleportation, its whole score comes from the dangling rule, which spreads D's mass evenly over the four nodes ($d\, r_D/4 = 0.1000$). The example shows that a node gains score by *receiving* endorsements from nodes that are endorsed in turn, which is the citation logic of PageRank (Page et al., 1999).

Transfers among the same contracts would in general join other pairs, or the same pairs with other totals, and so give another transition matrix P and possibly a different ranking. Hypothesis H1 is this situation in miniature: EndorseRank and AWP need not agree. They also restart differently, so H1 compares edges and restarts together; Section 4.5.4 walks the allowance graph with AWP's restarts.

2.15 DeFi risk taxonomy and where reputation fits

Werner et al. (2022) separate the technical security of DeFi protocols from their economic security, and Zhou et al. (2023) place the attacks and accidents seen in practice in one reference frame. Bugs in contract code are smart-contract risks (Li et al., 2022); a stale price, or a feed under someone's control, is an oracle risk (Angeris & Chitra, 2020); volatility and liquidation cascades are market risks (Perez et al., 2021); and voting power held by a few addresses is a governance risk (Fritsch et al., 2024). Each layer starts from a given protocol and asks what can go wrong with it. The thesis asks the question that comes first: among very many contracts, which ones users rely on. A ranking of contracts by the relations between them is no substitute for auditing or oracle design; it is meant to say where such methods should be applied first, not to predict exploits or governance attacks.

Artificial activity makes any ranking from ledger data harder to trust. Wash trading inflates volume on decentralized exchanges (Victor & Weintraud, 2021) and, on a larger scale, on unregulated centralized exchanges (Cong et al., 2023), extractable value creates transfers that follow ordering strategies rather than relationships (Daian et al., 2020), and scam tokens add transactions without sustained relationships (Xia et al., 2021). The transfer layer is sensitive to such volume, and the study does not measure it. The allowance layer is sensitive to approvals granted only to raise a score; the contract-only graph and the balance cap raise their cost, and Section 5.6 prices what remains.

2.16 Related measurement traditions

Outside blockchains, reputation systems have been studied in e-commerce (Resnick & Zeckhauser, 2002), peer-to-peer networks (Kamvar et al., 2003) and web spam (Gyöngyi et al., 2004). Four insights from this work apply here: (i) reputation is a property of relations; (ii) when identities are easy to create, Sybil attacks become possible (Douceur, 2002); (iii) seeding and normalization limit the effect of inflation; and (iv) evaluation needs external benchmarks, because internal coherence

alone is not enough (Cronbach & Meehl, 1955). The thesis takes up (i) directly and answers (ii) in part, with graphs whose nodes are contracts. Of (iii) it takes up normalization but not seeding (Section 2.7). On (iv) it goes part of the way: the same-window checks test coherence, the out-of-window prediction tests later events of the same two kinds, and the discovery-order test uses a benchmark that no score reads.

Topic-sensitive PageRank (Haveliwala, 2003) and personalized PageRank (Jeh & Widom, 2003) bias the teleportation vector toward a chosen group of nodes, as TrustRank does with its seed set (Gyöngyi et al., 2004) and AWP with sending activity (T. Do et al., 2023). Multiplex PageRank lets a node's centrality on one layer shape the walk on another (Halu et al., 2013), and the multilayer framework of De Domenico et al. (2015) lets one walker move within and between the layers of an interconnected network. C-PR takes the simplest form of this idea, a per-node weighted sum of two transition matrices with one weight, and S-PR uses the allowance score as the teleportation vector (Section 3.6.3).

2.17 Accounts, gas, and why logs are the measurement surface

Ethereum has two kinds of account: externally owned accounts (EOAs) and contract accounts. A private key controls an EOA, whereas a contract account is governed by the code it stores (Antonopoulos & Wood, 2018). The line between a user's account and a contract does not follow this division exactly: an EOA can delegate to code under EIP-7702 (Buterin et al., 2024), and a smart account, such as a Safe (Safe Ecosystem Foundation, n.d.) or an ERC-4337 account (Buterin et al., 2021), is a contract account that acts for a user. This thesis therefore separates *user accounts* (EOAs, EIP-7702 accounts and smart accounts) from *contracts* (every other contract account; Section 3.5.1) and scores contracts only (Section 3.2). Every state-changing transaction consumes gas. Logs come from a contract's `LOG` instructions, which run when its code emits an event (the Solidity `emit` statement); writing to storage produces no log. A reputation system that means to be auditable must prefer logs to off-chain databases, because logs are a permanent part of

the blockchain history that any full node or index provider can recompute (Wood, 2025).

2.18 Approvals as security-relevant events

Unlimited approvals (`type(uint256).max`) save the owner from approving again, at a cost in safety. If the spender is compromised, the owner's whole balance of the token can be stolen until the approval is reset. An allowance is therefore a capability and not a record: whoever holds it can move the owner's tokens until it is spent or revoked (Vogelsteller & Buterin, 2015). Transaction-based phishing exploits this: fake websites lead users to sign transactions that let scammers take their assets (He et al., 2023).

Wang et al. (2022) find that unlimited approvals make up the majority of ERC-20 approvals on Ethereum and quantify the risk they carry. They are the majority of the positive latest allowances between the contracts of this study as well (Table 8.5). Under the balance cap such an approval counts for what its owner holds of the token at the freeze, like any allowance above the balance; in the allowance layer of C-PR a second cap, in US dollars, bounds what one large balance can add (Section 3.6.3).

2.19 AWP in depth: what the transfer-graph baseline measures

AWP, proposed by T. Do et al. (2023), ranks the addresses of a transaction graph. Each transaction adds to its edge a logistic function of its time multiplied by a bounded function of its value, so that no transaction adds more than one to the edge weight. The walk restarts at each address in proportion to its activeness, the sum over the addresses it has sent to of the time weight of its latest transaction to each. Ranking accuracy is then validated against inbound degree and inbound value, taken as proxies for the economic attention a wallet attracts. The form of the decay (Equation (3.3)) and the in-degree/in-value proxies in Chapters 3 and 4 are taken from the AWP paper. The paper gives no parameter values; Section 3.6.1 gives the values used here.

EndorseRank keeps AWP's weight functions under PageRank and its rank-correlation check, and

changes what the edges mean and where the walk restarts: EndorseRank restarts uniformly, AWP at active senders (Section 3.6.2), which on the contract graph are contracts.

2.20 Comparison of graph reputation methods

Table 2.1 compares the scores of this thesis with the other approaches of this chapter by edge semantics, temporal model and validation design.

As Table 2.1 indicates, the contribution of the thesis lies in the graph, the operator and the evaluation. The graph holds contracts only and reads two relations between them, one of which, the allowance, the reviewed methods do not use as an edge. The operator, C-PR, couples the two relations into one walk, with a seeded ablation to measure how much the coupling contributes. The evaluation sets the walk against the demand of user accounts, against the raw counts and against its rules (Section 2.11). EndorseRank's contribution is the use of the latest allowance as the edge of a ranking; it is neither the ranking algorithm nor the reading of approvals as a link between addresses (Section 2.8).

2.21 Summary of the reviewed literature

Table 2.2 collects the sources underlying this study, with the strengths this thesis draws on and the limitations it tries to address.

Table 2.2: Summary of the reviewed literature: methodology, strengths, limitations, and relation to this study.

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Page et al. (1999); Langville and Meyer (2006)	Eigenvector ranking of a hyperlink graph, solved by power iteration	Interpretable; $O(\|E\|)$ per iteration; a unique Perron vector under damping	Edges mean hyperlinks; no notion of time	Same solver and complexity argument for every walk

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Cho et al. (1998); Najork and Wiener (2001); Baeza-Yates et al. (2005)	Web crawls ordered by breadth-first order, backlink count or PageRank on the pages seen	Test an ordering under partial knowledge	Importance taken from the links themselves; results vary with the collection	Model of the discovery-order test, with user-account demand as importance
Kamvar et al. (2003); Gyöngyi et al. (2004)	Trust aggregation in P2P networks; propagation from seeds to demote spam	Normalization and seed sets curb inflation; explicit adversary	Need ratings or curated seeds; not applied on-chain	Inform the Sybil cost model and the choice of uniform restarts
H.-D. Do and Do (2023)	PageRank and HodgeRank on Ethereum transactions, as social credit	Public data; the top-ranked addresses are exchanges and protocol contracts	Addresses of every kind in one graph; checked against transaction counts	Ledger PageRank over all addresses; here the graph holds contracts only
T. Do et al. (2023), AWP	Transaction PageRank with value weights, logistic recency and activity restarts	Validates on ranks; models recency	Proxies come from the same transfers as the graph; no parameter values reported	Transfer-layer reference; parameters set here (Section 3.6.1)
D. Lin et al. (2025); Wu et al. (2022)	Risk propagation and network embedding on Ethereum transaction graphs	Rate or detect accounts that carry no label	Need labels of illicit accounts; aimed at fraud	Show that graph position is informative; not reproduced here
Kitzler et al. (2023); Mukhia et al. (2026)	DeFi protocol networks; ERC-20 transfers by EOA and contract ends	Protocols interact mainly in one strongly connected component; contract-to-contract transfers differ	No ranking of contracts	Support a graph of contracts only
Hassija et al. (2020); Kotb (2023)	Credit scoring from blockchain data	Aimed at lending decisions	Need labels or off-chain features	Background; no such score is built here

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Bellini et al. (2020); Almasoud et al. (2020)	Surveys of blockchain-based trust and reputation systems	Taxonomies of the design space	Reputation from users' ratings; no authorization edges; nothing measured on L2	None of the surveyed designs uses allowance edges
Victor (2020); Victor and Weintraud (2021); Cong et al. (2023)	Address clustering, partly through ERC-20 approvals; wash-trading detection	Measure multi-address and artificial activity	No ranking proposed	Closest prior use of approvals; transfer signals can be inflated
Human Passport (n.d.); Siddarth et al. (2020); Bordeianu and Popescu (2026)	Weighted credentials about an address holder; reviews of proof-of-personhood	In deployed use; adds off-chain evidence against Sybil attacks	A gate, not an ordering; relations between addresses play no part	How trust is formed in practice; the scores here give no personhood guarantee
Cronbach and Meehl (1955); Messick (1995)	Psychometric construct validity	Separate what a score is designed to measure from external criteria	Not specific to blockchains	Basis for reading same-window agreement as an intended-construct check
Efron (1979); DiCiccio and Efron (1996)	Bootstrap resampling and confidence intervals	Uncertainty for rank statistics without distributional assumptions	Units must be exchangeable	Source of the 95% intervals and paired contrasts of Chapter 4

2.22 Research gap

Five gaps emerge from Table 2.2 and Sections 2.6 and 2.16. First, the ledger rankings reviewed here rank addresses in general from their transactions (PageRank and HodgeRank (H.-D. Do & Do, 2023), AWP (T. Do et al., 2023)), and the tooling through which trust is formed in practice

Table 2.1: Comparison of selected graph reputation methods (conceptual).

Method	Edge semantics	Temporal model	Typical validation
PageRank (web) (Page et al., 1999)	Hyperlinks	None / static snapshot	Retrieval quality; crawl order (Cho et al., 1998)
EigenTrust (Kamvar et al., 2003)	Local trust ratings	Iterative aggregation	P2P file authenticity
TrustRank (Gyöngyi et al., 2004)	Links from seed goods	Static / seeded	Spam demotion
PageRank/ HodgeRank on Ethereum (H.-D. Do & Do, 2023)	Transactions among wallet addresses	Static sample	Rank correlation with transaction counts
AWP (T. Do et al., 2023)	Transactions, bounded value weights	Logistic decay of edges and restarts	In-degree / in-value
RiskProp (D. Lin et al., 2025)	Transactions, with a de-anonymization score	Propagation of risk	Labeled illicit accounts
Multiplex PageRank (Halu et al., 2013); multilayer ranking (De Domenico et al., 2015)	Several relations over one node set	Static layers	Versatility of top-ranked nodes
Human Passport (n.d.)	No edges; weighted credentials about the address holder, on-chain and off-chain	Current credential set	Admission threshold; known human and Sybil addresses
EndorseRank (this study)	Latest allowances between contracts, up to the owner's balance; AWP's weight functions	Logistic decay of each latest approval; uniform restarts	Allowance checks; later approval pairs
C-PR / S-PR (this study)	Allowance edges mixed with, or seeding, transfer edges between contracts; allowances capped	Allowance snapshot and logistic decay of transfers	Discovery order; out-of-window labels; rule A and two criteria of rule B

judges one contract or one account at a time (Human Passport, n.d.). Relations between contracts have been measured (Kitzler et al., 2023), but in the work reviewed here no ranking is computed from them on a graph whose nodes are contracts only; this gap sets the graph of every research question. Second, none of the reviewed methods uses the ERC-20 allowance, the on-chain record of a delegation of spending power, as a ranking edge. Approvals have served address clustering and security analysis (Sections 2.8 and 2.18), and it is unknown whether, as edges between contracts counted against what the owner holds, they order contracts differently from transfers and at what cost (O1 and RQ1). Third, the experiment in which a PageRank ordering reached important web pages sooner than other orderings (Cho et al., 1998) has, in the work reviewed here, no counterpart on a ledger: the ledger rankings are checked by rank correlation with statistics of their own edges, which shows coherence and not that a ranking leads to the contracts that users rely on (O2 and RQ2). Fourth, a walk is seldom compared out of window with the raw count that it smooths, although such counts are the standard baselines of link prediction (Liben-Nowell & Kleinberg, 2007) (O3 and RQ3). Finally, multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) have not, in the work reviewed here, been applied to two on-chain relations over one set of contracts, and no study reports whether an authorization layer improves a payment layer under rules fixed before the outcome is seen, as preregistration asks (Nosek et al., 2018) (O4 and RQ4, O5 and RQ5); the thesis fixed such rules for its registered analysis (Appendix 8.6) and applies them to the main analysis as a re-analysis. C-PR, the method of the thesis, is built for the first, second and fifth gaps; the third and fourth shape its evaluation.

2.23 Open problems motivating this research

The literature leaves these problems in on-chain reputation unresolved:

1. Selection: Among the very many contracts of a rollup, which should be looked at first when labels are sparse (D. Lin et al., 2024)?
2. Edge semantics: Which on-chain relations between contracts show that one relies on another

(Kitzler et al., 2023)?

3. Evaluation: What shows that a ranking leads to the contracts that matter, when statistics of its own edges agree with it by construction (Cronbach & Meehl, 1955)?
4. Computational cost: As graphs grow, can scores still be refreshed at interactive latencies (Langville & Meyer, 2006)?
5. Adversarial robustness: How much does it cost to forge standing when addresses are cheap (Douceur, 2002)?

The thesis takes on (1)–(4) with a graph of contracts, a walk that couples its two edge types, the discovery-order test, out-of-window prediction against the raw counts, and runtime benchmarks. Problem (5) is analyzed at the model level in Chapter 5.

Chapter 3 turns this gap statement into an operational design.

3 Research Methodology

Chapter 2 ended with five gaps. This chapter gives each a measurable procedure and sets out how the coupled walk C-PR, the method of the thesis, and its two single-layer references, EndorseRank and Adaptive Weighted PageRank (AWP), are evaluated as scores that should find the contracts of Arbitrum One that users rely on. EndorseRank runs on the ERC-20 allowance graph and AWP on the transfer graph (T. Do et al., 2023); C-PR walks both in one chain (Section 3.6.3). Every walk is solved with weighted PageRank (Page et al., 1999).

The evaluation has three parts: the discovery-order test, which asks whether a crawl ordered by a score reaches the contracts with the most user-account demand sooner than simpler orderings (Section 3.7); the out-of-window prediction of later approvals and inflow (Section 3.8); and the same-window evaluation, which follows construct (domain) alignment: each ranking method is one indicator of a reputation construct that cannot be observed directly, in the sense of construct validity (Cronbach & Meehl, 1955). The claim is comparative throughout: every score is set against the raw counts it smooths.

Four principles guide each operational choice.

- Same-seed comparability: every score is computed on an identical set of contracts, so no difference between scores can come from sampling.
- Fixing choices in advance: the analysis plan and the versioned configuration were committed to the authors' Git repository before the study extracted any log (Section 3.14). They fix the windows, the cohort and account-type rules, every score with its parameters, the values of λ, the proxies, the W0 labels, the bootstrap, the benchmark protocol, the robustness checks and rule A, the plan's one decision rule, whose object is C-PR (RQ4; Section 3.8). Scan A, the cohort, the ego extraction and the same-window and W0 results followed. The registration of W1 was committed next, with rule B and its margin (Section 3.8.1); scan B and the rest

of the registered analysis came after it (Appendix 8.6). Chapter 4 reports the main analysis, whose definitions Section 3.5.3 lists with their status. RQ4 and RQ5 follow the rules of the analysis plan and of the registration, applied to the scores of the contract graph as a re-analysis; RQ2 and RQ3 follow criteria defined after the registered results were known. Every contrast outside the rules was computed after its data were known and is reported as such, with its interval.

- Construction and timing: each cohort's edge lists are built once, before any timer starts, and the benchmarks time the solver call.
- Scope: RQ1, with H1 to H3 and Claims C5 and C6, concerns the matched cohort in the same window and the cost of each walk on the graphs of T_{obs}. RQ2 and Claim C1 concern crawls of the contract graph at each freeze. RQ3 to RQ5 and Claims C2 to C4 draw on the spender cohorts of W0 and W1, with the same-window part of rule A on the matched cohort.

3.1 Notation

The notation used throughout the thesis is listed in Table II; Table 3.1 adds the symbols that only the equations of this chapter use. Vectors are set in bold and sets in calligraphic type.
Except where stated otherwise, PageRank means the weighted variant of Page et al. (1999), whose update at iteration t is

$$\mathbf{r}^{(t+1)} = (1-d)\,\boldsymbol{\pi} + d\,P^{\top}\mathbf{r}^{(t)} + d\left(\mathbf{r}^{(t)\top}\mathbf{1}_{\text{dang}}\right)\boldsymbol{\pi}, \tag{3.1}$$

where P is the transition matrix of Equation (3.2) below, $\boldsymbol{\pi}$ is the restart distribution over the nodes, uniform ($\pi_u = 1/|V|$) unless a method specifies another (AWP in Section 3.6.1, EndorseRank with activity restarts in Section 3.6.2, S-PR in Section 3.6.3), and $\mathbf{1}_{\text{dang}}$ marks the dangling nodes (Table 3.1). With $w(u,v)$ denoting the weight of the directed edge from u to v and $o(u) = \sum_v w(u,v)$

Table 3.1: Notation of this chapter's equations beyond the list of Table II.

Symbol	Meaning
$G = (V, E)$	Graph passed to the PageRank solver: every node is a contract, and V holds the end points of the retained edges E
V_A, E_A; V_T, E_T	Nodes and edges of the EndorseRank allowance graph $G_{\text{approve}} = (V_A, E_A, w_A)$ and of the AWP transfer graph $G_{\text{transfer}} = (V_T, E_T, w_T)$
$a^\star(u, v, \text{token})$	Latest in-window allowance of owner u to spender v on one token (US dollars)
$\text{bal}(u, \text{token})$	Balance of owner u on the token at the freeze (US dollars)
$\hat{a}(u, v, \text{token})$	Capped allowance, $\min(a^\star, \text{bal})$
$\Delta t^\star(u, v, \text{token})$	Age of the approval that set $a^\star$
$w_A(u, v)$	EndorseRank edge weight, owner $u \to$ spender v: $\sigma(\Delta t^\star)V(\hat{a})$ summed over tokens
$w_T(u, v)$	AWP edge weight, sender $u \to$ receiver v: $\sigma(\Delta t_e)V(x_e)$ summed over transfers
x_e	Amount of transfer e (US dollars)
b_T, b_A	Slopes of the value transform V for transfers and for allowances, $b = \ln 3/m$ with $m = m_T$ and $m = m_A$
X_u^A	Activeness of contract u in the allowance graph, the restart weight of EndorseRank with activity restarts
$\tilde{w}_A(u, v)$, $\tilde{w}_T(u, v)$	Weights of the two layers of C-PR, in US dollars
$D_\lambda(u)$	Common denominator of the layer weights $\alpha_A(u)$ and $\alpha_T(u)$ in node u's C-PR row
$\mathbf{1}[\cdot]$	Indicator: 1 if the condition holds, 0 otherwise
$\mathbf{1}_{\text{dang}}$	Indicator vector of the dangling nodes, entries $\mathbf{1}[o(u) = 0]$
$\mathbf{s}_M$, $\mathbf{p}_c$	Score vector of method M and value vector of check c over the cohort
R_i, S_i	Mid-ranks of contract i's score and proxy value (tied values share their average position)
n_c, n_d; n_0, n_1, n_2	Concordant and discordant contract pairs; all pairs, $n(n-1)/2$, and the pairs tied in the first and in the second ranking

the out-strength, the transition matrix is

$$P_{u,v} = \begin{cases} w(u, v)/o(u) & \text{if } o(u) > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.2}$$

The rows of dangling nodes are zero, so P is substochastic; the mass lost at those nodes returns through the third term of Equation (3.1) and is spread according to $\boldsymbol{\pi}$. Filling each zero row with

$1/|V|$ gives the row-stochastic matrix of Equation (2.1), and with uniform $\boldsymbol{\pi}$ Equation (3.1) is then the Google-matrix form of Equation (2.2). The iteration starts from $\mathbf{r}^{(0)} = \mathbf{1}/|V|$ and stops once $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or after $I_{\max}$ iterations, and the final vector is normalized to sum to one over V.

The default damping of 0.85 is the classical web setting (Page et al., 1999), and the robustness sweep varies it with edge weights held fixed (Section 3.13). The strict $\varepsilon = 10^{-8}$ and $I_{\max} = 300$ keep the measured runtimes tied to the structure of the graph and not to early termination.

3.2 Data sources and sampling

Figure 3.1 places every kind of on-chain record used in the thesis on one calendar.

- Primary chain: All events come from Arbitrum One (chain ID 42161), an optimistic rollup on Ethereum (Thibault et al., 2022) whose logs are in the public BigQuery export. One chain avoids cross-chain aliases and gives every address the same fees and latencies.
- Extraction and score windows: Scan A extracts the events from 2023-10-01 00:00:00 UTC to 2026-06-30 23:59:59 UTC, and scan B those of the label window W1, July to September 2026, so the data cover three years. Every score of the main analysis is computed from the twelve months before its freeze: from 2025-07-01 00:00:00 UTC to T_{obs} = 2026-06-30 23:59:59 UTC in the same-window evaluation and in W1, and from 2025-04-01 00:00:00 UTC to t_1 = 2026-03-31 23:59:59 UTC in W0 (Section 3.8). A window of one year bounds the age of every event that enters a score, a proxy, a baseline or a label history and gives the two freezes windows of the same length. The time decay is anchored at the freeze, so an event on the first day of a window is 365 days old. The registered analysis (Appendix 8.6) scores every event of scan A from 2023-10-01.
- BigQuery source: Logs are read from the public BigQuery table

  `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`.

  Scans A and B copy the events the study needs into its own Google Cloud project, and later

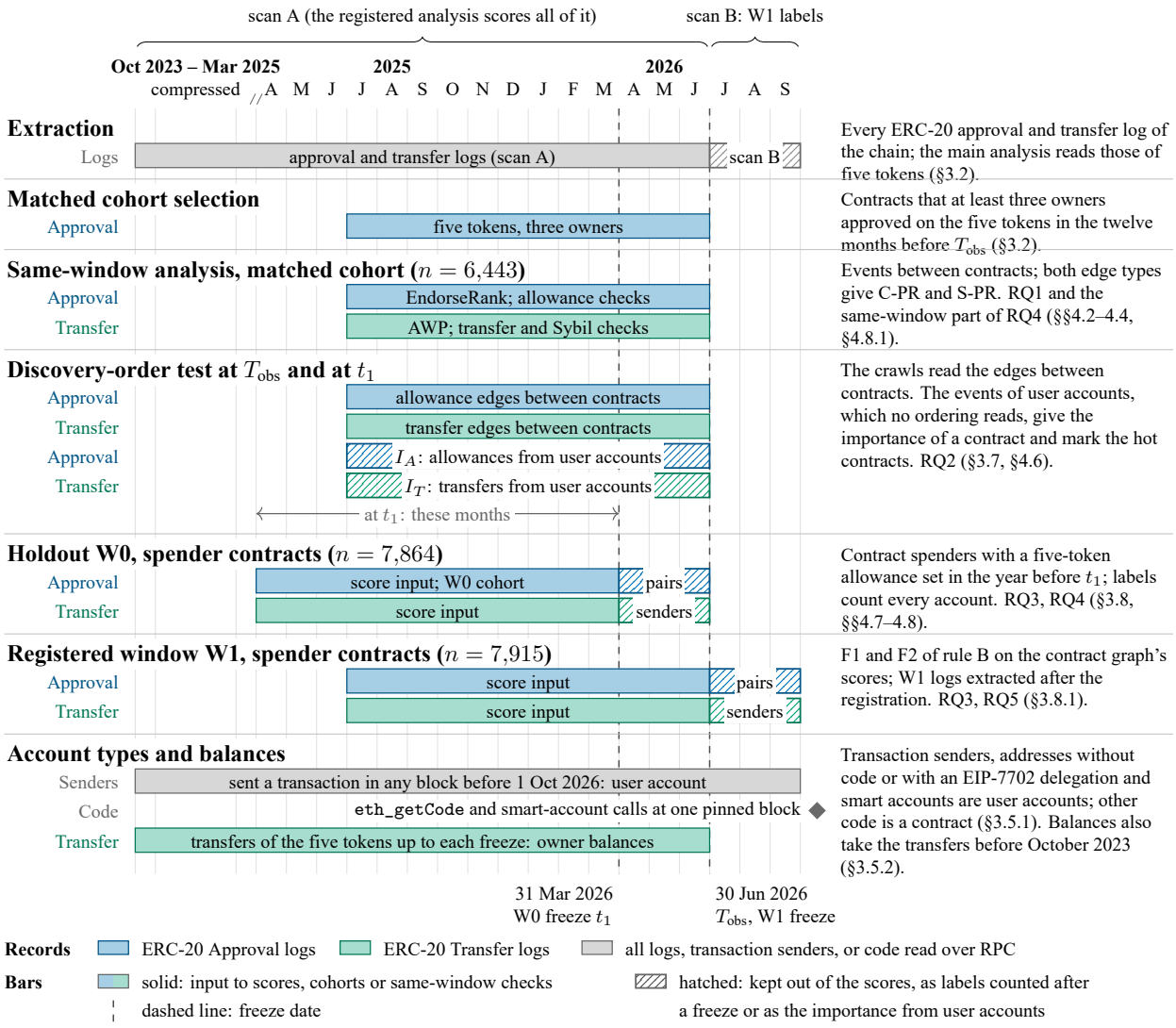


Figure 3.1: Blockchain data used in this thesis, all from Arbitrum One. Each lane is one analysis, and each bar is one kind of on-chain record placed on the months it covers; the 18 months before April 2025 are drawn at a compressed scale. Solid bars are inputs to scores, to cohorts or to same-window checks. Hatched bars are kept out of the scores: the labels counted after a freeze date and, in the discovery-order test, the events of user accounts that give the importance of a contract. The July–September logs were extracted only after the registration of W1 was committed. The approval logs of five tokens in the twelve months before each freeze select the cohort of that freeze; the graphs of the main analysis hold the events of those tokens between two contracts in those months, read from the whole chain; the registered analysis scores every event of scan A.

queries read those copies (Section 3.3); the same public dataset gives the addresses that sent a transaction and the transfers before scan A that enter the balances. Appendix 8.2 shows the query patterns, which anyone with a Google Cloud account can rerun.

- Reputation events: EndorseRank uses ERC-20 `Approval` logs (Vogelsteller & Buterin, 2015), which also cover approvals made through EIP-2612 `permit` (Lundfall, 2020), and AWP uses `Transfer` logs. Both are kept only for logs with exactly three topics and for the five tokens of Section 3.5.3. The main analysis reads every such `Approval` and non-zero `Transfer` of the chain, not only those around a cohort, from 1 April 2025 to 30 September 2026 (the typing window).
- Nodes and scored addresses: Every node of every graph is a contract, and only contracts are scored. A user account is an address that sent a transaction, an address without code, an EIP-7702 account, which an externally owned account (EOA) becomes by setting a delegation while keeping its key (Buterin et al., 2024), or a smart account (Section 3.5.1), that is a Safe (Safe Ecosystem Foundation, n.d.) or an ERC-4337 account (Buterin et al., 2021). No user account is a node, and no score or baseline reads an event of one: an EOA costs no deployment, and the study does not measure who controls a user account, whereas a contract costs a deployment and acts only as its code states. What user accounts do is kept apart, as the demand that the discovery-order test measures and that the labels count. The graphs mix contracts of every role and no analysis is stratified by role, so differences between roles enter every coefficient.
- Matched cohort: Alignment is applied to the $n = 6{,}443$ contracts that received a non-zero `Approval` of one of the five tokens from at least three distinct owners inside the score window of T_{obs}: the rule of the plan, whose cohort is the $27{,}844$ contracts approved on any token from 2023-10-01, applied on the five tokens within that window. The rule counts every approving account, and of the addresses it lists those typed as user accounts are not in the cohort. W0 draws a cohort of its own by the same rule inside its own score window, so that no event after t_1 chooses its contracts. The owners of a cohort contract may all be user accounts, whose

approvals no graph reads, so a cohort contract need not appear in either graph and then scores 0 (Section 4.1.1 counts these contracts).

- Cohort events: The cohort rule, the spender lists and the labels count the events of every account around the addresses that the rule lists for a freeze, the events in which one of these addresses is owner or spender, sender or recipient. A spender that the rule does not list thus enters through an allowance that a listed address granted it, and its labels count only its events with listed addresses. The graphs hold every event between two contracts on the chain (Section 3.6).
- Exclusions: Mempool transactions, off-chain order books, centralized exchange accounts and bridges lie outside the data, and no attempt is made to merge addresses controlled by one operator into one entity. Tokens other than the five are left out, and native ether enters only as WETH.

3.3 Two-phase collection strategy

Collection and evaluation run as two phases, so that the tables, benchmarks and robustness runs can be rebuilt without cloud costs by anyone who has run the extraction.

- Phase 1 (reputation streams): Scan A copies the approval and transfer events from 1 October 2023 to 30 June 2026 into the study's project, and scan B those of July to September 2026 once the registration of W1 is committed. One query lists the spenders approved by at least three owners, the code check fixes the plan's cohort, and a further query extracts its ego events, from which the cohort, the spender list and the labels of each freeze are drawn on the five tokens. For the graphs, the end points of the five-token events are typed (Section 3.5.1), the owners' balances are computed (Section 3.5.2), and the events between two contracts are kept. At each anchor date the approvals of the score window are reduced to capped latest allowances and the transfers are aggregated per sender–recipient pair (Section 3.6), with amounts in US dollars; the proxies and the importance measures are computed alongside. These tables, of

which a last query checks 64 invariants, are the only inputs to graph construction.

- Phase 2 (local evaluation): From the downloaded Phase 1 tables a local machine computes the scores, the alignment statistics, the crawls of the discovery-order test, the holdouts, the benchmarks and the robustness runs, without querying BigQuery.
- Cost guardrails: Every query is first run as a dry run, and a query whose estimate would take the spending past the study's budget of USD 60 (USD 50 in the plan's configuration) is refused, and the jobs of the main analysis share a cap of their own within that budget.[1] Every job is written to a cost ledger with its billed bytes and cost (Appendix 8.1).

3.4 Pipeline architecture

Figure 3.2 traces the whole pipeline, from public logs to the research tables. The remote steps write tables in the study's BigQuery project, into which the account types read over RPC are loaded, and the downloaded files are never changed.

3.5 Event decoding

Decoding turns each log, whose topics and data carry no field names, into a typed record; an error here would corrupt both graphs silently.

- Log anatomy: Each log records the emitting contract, a sequence of topics and an ABI-encoded data blob; the first topic is the keccak-256 hash of the event signature, and the indexed parameters follow (Vogelsteller & Buterin, 2015). ERC-721 tokens emit `Transfer` and `Approval` under the same signatures but put the token ID in a fourth topic, so the condition of exactly three topics keeps them out. SQL filters by topic and decodes the ERC-20 fields.
- ERC-20 `Approval`: Its topic hash is

  `0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925`,

[1] This budget is enforced by the study's own pipeline; BigQuery does not impose it.

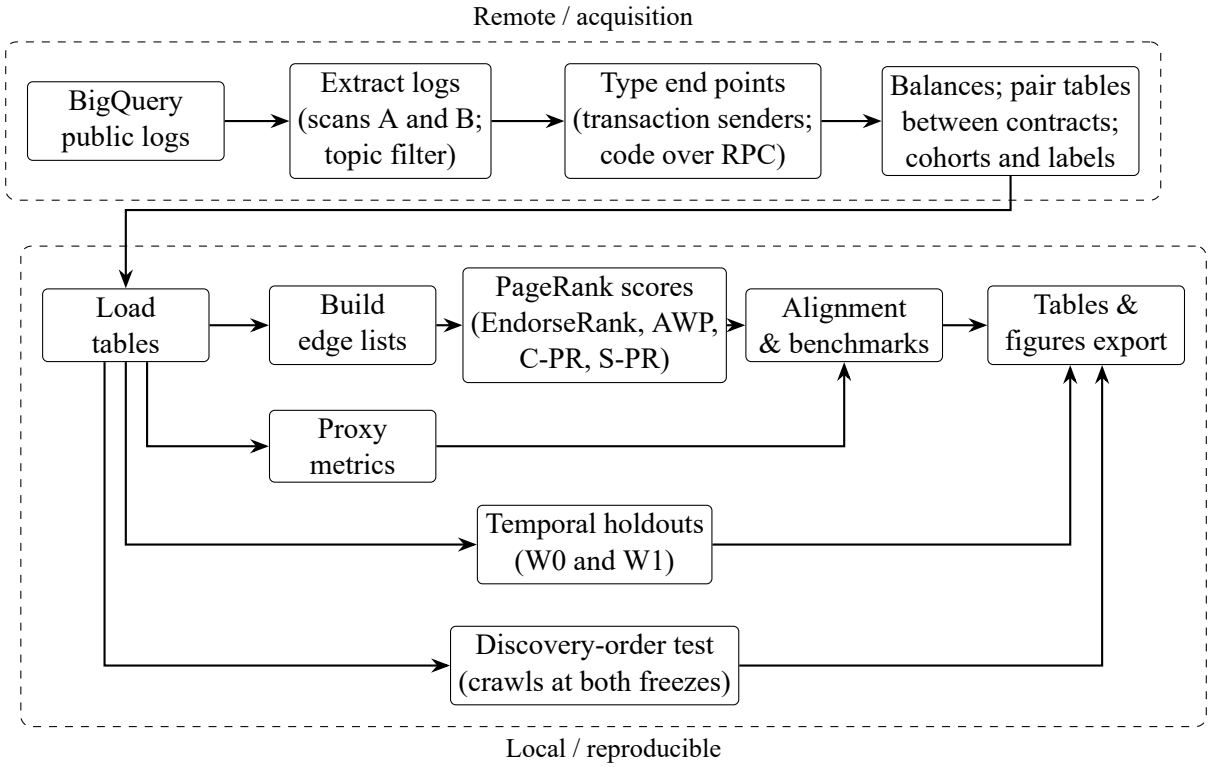


Figure 3.2: Pipeline overview. Approval and transfer logs are copied remotely in two scans; the end points of the five-token events are typed, the owners' balances are computed, and the events between contracts are aggregated into pair tables. Locally the pair tables become edge lists, from which the scores are computed; the alignment and benchmark calculations, the temporal holdouts and the crawls of the discovery-order test follow, and the tables and figures are exported last.

the keccak-256 hash of `Approval(address,address,uint256)`. Owner and spender occupy Topics 1 and 2, and the allowance is decoded from the data field in the token's base units and valued in US dollars (Section 3.5.3). An allowance of zero is kept, since it resets the allowance in force.

- ERC-20 `Transfer`: Its topic hash is

  `0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef`,

  the keccak-256 hash of `Transfer(address,address,uint256)`. Sender and recipient are indexed, and the amount is valued in US dollars at the price of its day. Zero-value transfers are not kept. A mint or a burn names the zero address, which is neither a contract nor a user

account: it forms no edge and counts only in the balances (Section 3.5.2). Self-transfers stay in the events table but are left out of the inbound-counterparty computations (Section 3.10), since they reveal no counterparty.

- Address normalization and units: Addresses are lower-cased before the join. Amounts are the `uint256` value as a double-precision float, divided by ten to the power of the token's decimals and multiplied by its US-dollar price, so an unlimited allowance, $2^{256} - 1 \approx 1.16 \times 10^{77}$ base units, is a very large dollar amount until the balance cap bounds it (Section 3.6).

3.5.1 Account types

An address other than the zero address that is an end point of a five-token event of the typing window is typed by four rules, applied in this order (Appendix 8.1.3; Table 8.4). (1) An address that sent a transaction in any block before 1 October 2026 is an EOA, and so a user account, since an address with deployed code cannot send one (Feist et al., 2021). (2) Every other address that could end an edge between contracts, that is one with a counterparty that is neither a transaction sender nor the zero address, and every member of a cohort or spender list, is read with `eth_getCode` at block $513{,}216{,}506$; no code, or an EIP-7702 delegation designator, means an EOA. (3) An address with other code is a smart account, and so a user account, if at the same block it answers as a Safe, with a threshold and owners, or as an ERC-4337 account, with a canonical EntryPoint contract and, in its code, the selector of `validateUserOp` and no paymaster selector. Two lists of code families extend this test to accounts whose proxy hides that code or whose EntryPoint getter has another name; they were drawn up from the code of the families alone, without reading a label, a cohort or spender membership or a score of the addresses they move. (4) Any other address with code is a contract, unless it is a system address (a precompile), which is never a node. Section 3.15 states the limits of these rules.

3.5.2 Balances

The balance that an owner held of a token at a freeze is reconstructed from the `Transfer` logs as the net flow of the token to the owner up to the freeze, mints and burns included: the public logs before scan A, scan A and the five-token events of the typing window. A negative value and a value below 10^{-6} token are set to zero. Balances are computed at t_1 and at T_{obs} for every (token, owner) pair of a five-token approval from 1 April 2025 to T_{obs}. As a check, $1{,}500$ sampled pairs are compared with the token's `balanceOf` at the last block of 30 June 2026; the pipeline stops unless 98% of them match. The check passed (Section 4.1).

3.5.3 Tokens, prices and the definitions of the main analysis

The analysis plan fixed amounts in raw base units, every ERC-20 token and $b = 1$ in Equation (3.4), on graphs that hold every account of the cohort's events. There the amount plays almost no part in EndorseRank or AWP, since $V(z)$ is within 10^{-4} of 1 for almost every event, the layers of C-PR add base units across tokens with different decimals, and any token contract can emit logs that name an owner or a sender that took no part.

After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. It is the main analysis of the thesis, called the contract-graph analysis where it stands beside the registered one.

Of the 133 tokens that Chainlink's directory of data feeds maps on Arbitrum One, 64 have a market-price feed, for the token or the asset it represents, with a low or medium market-risk category (Chainlink, n.d.), and 46 of these have a daily price at DefiLlama (DefiLlama, n.d.); the analysis keeps the five with the most rows among the events of the plan's cohort: WETH, USDC, USDT0, WBTC and ARB. A transfer is valued at the DefiLlama daily close of its UTC day, and a latest

allowance and a balance at the price of the anchor's day. The constants are set on the W0 score window of the contract graph, so that no event after t_1 enters them, and rounded to two significant digits. The slope of V is $b = \ln 3/m$, so that an amount of m dollars weighs one half: $m_T = 160$ dollars is the median transfer between contracts, and $m_A = 0.071$ dollars the median positive capped allowance between contracts at t_1, unlimited allowances included. $U = 16{,}000$ dollars, the 99th percentile of the same transfers, is the most that one allowance counts in the allowance layer of C-PR, and $U_{\text{user}} = 31{,}000$ dollars, the 99th percentile of the transfers from user accounts to contracts, the most that one user account adds to the importance of one contract.

Rule A and two of rule B's registered criteria are applied to the scores of the contract graph with the thresholds of the plan and the registration (Sections 3.8 and 3.8.1); the verdicts are a re-analysis and not registered tests. Appendix 8.6 sets the registered analysis beside the main one, and Appendices 8.4.2 and 8.4.3 give the tokens, the price checks, the account types and the balance cap.

3.6 Graph construction

EndorseRank and AWP weight their edges the same way and run the same PageRank solver, Equation (3.1); they differ in the events that make the edges and in where the walk restarts. An edge needs a contract at both ends, and the graph of a freeze holds every such edge of its score window, whatever cohort its ends belong to; a contract that ends no edge of positive weight in a method's graph scores 0 under that method.

Edges in EndorseRank stand for endorsements and edges in AWP for transfers (Section 2.8). Both methods weight an event by its age and pass its US-dollar amount through a bounded function, so that one event counts at most once.

3.6.1 AWP transfer graph

AWP is the method of T. Do et al. (2023). A transfer e between two contracts worth x_e US dollars, made Δt_e days before the score window closes at its anchor (T_{obs}, or t_1 in W0), enters the edge

from its sender to its receiver with the weight $\sigma(\Delta t_e)\, V(x_e)$. The time weight is the logistic decay

$$\sigma(\Delta t) = \frac{1}{1 + \exp\big(k(\Delta t - t_0)\big)}, \qquad k = 0.01, \quad t_0 = 180 \text{ days}, \tag{3.3}$$

and the value weight is

$$V(z) = \frac{2}{1 + e^{-bz}} - 1, \qquad b = \frac{\ln 3}{m}, \tag{3.4}$$

which is 0 at $z = 0$, $1/2$ at the median amount $z = m$ and rises towards 1 as the amount grows. For transfers $b = b_T = \ln 3/m_T$ (Section 3.5.3). The edge weights are sums over transfers:

$$w_T(u, v) = \sum_{e:\, u \to v} \sigma(\Delta t_e)\, V(x_e). \tag{3.5}$$

The sum runs over every transfer from u to v in the window, whatever the token. A self-transfer ($u = v$) is kept and becomes a self-loop on u. The walk restarts at a contract in proportion to its activeness,

$$X_u = \sum_{v \neq u} \max_{e:\, u \to v} \sigma(\Delta t_e), \tag{3.6}$$

and the restart distribution of Equation (3.1) is $\pi_u = X_u / \sum_{v \in V_T} X_v$, and it also takes the mass of the dangling nodes. A contract that never sends to another contract has $X_u = 0$ and receives no restart mass, so its score comes from its in-edges alone.

The paper scores Ethereum transactions and gives no parameter values; its time weight is the logistic function of Equation (3.3) written in calendar time and not in age, and k, t_0 and b are this study's choices, as is the restriction of the nodes to contracts. A transfer of twice the median amount weighs 0.8. Within a score window the decay discounts older events strongly (Figure 3.3): the weight falls from $\sigma(0) = 0.858$ through $\sigma(182) = 0.495$ to $\sigma(365) = 0.136$ for an event from the first day of the window.

Algorithm 1 states the construction and scoring steps formally.

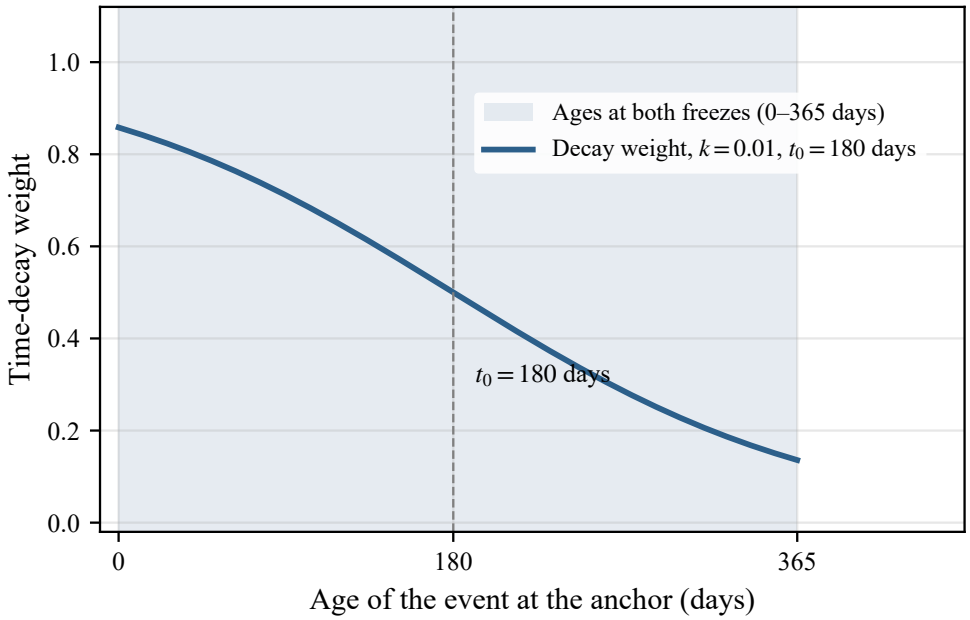


Figure 3.3: Logistic time-decay $\sigma(\Delta t)$ of AWP and EndorseRank ($k = 0.01$, $t_0 = 180$ days). The horizontal axis shows the age of an event in days before the anchor, $T_{\text{obs}} = 30$ June 2026 or, in W0, $t_1 = 31$ March 2026, and the vertical axis shows the time weight of a transfer or an approval. The shaded band marks the ages that occur at either freeze (0 to 365 days), over which the weight falls from 0.86 to 0.14.

Algorithm 1 AWP graph construction and scoring

Require: Transfers $\mathcal{T}$ between contracts in US dollars, evaluated contracts $\mathcal{W}$, observation end T_{obs}, parameters $k, t_0, b_T, d, \varepsilon, I_{\max}$

Ensure: Scores $\mathbf{s}_{\text{AWP}}$ on $\mathcal{W}$

$E \leftarrow \emptyset$; $X_u \leftarrow 0$ for every contract u

for each transfer $e = (u, v, x_e, t_e)$ in $\mathcal{T}$ **do**

  $\Delta t \leftarrow (T_{\text{obs}} - t_e)$ in days

  accumulate $\sigma(\Delta t)\, V(x_e)$ on edge $(u \to v)$ ▷ Equations (3.3)–(3.5); $u = v$ gives a self-loop

end for

for each sender u **do**

  $X_u \leftarrow \sum_{v \neq u} \max_{e:\, u \to v} \sigma(\Delta t_e)$ ▷ Equation (3.6)

end for

remove non-positive edges

$\pi_u \leftarrow X_u / \sum_v X_v$ for the nodes u of the retained edges

$\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equations (3.1)–(3.2)

return $\mathbf{s}_{\text{AWP}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

3.6.2 EndorseRank allowance graph

EndorseRank puts AWP's edge weights on the allowance graph. Let $\mathcal{A}$ denote the set of latest allowances between contracts after preprocessing. For each (token, owner–spender) triple only the most recent `Approval` event inside the score window is kept, in (block number, log index) order, and a zero allowance deletes the edge. The surviving allowance $a^\star(u, v, \text{token})$ counts up to the balance $\text{bal}(u, \text{token})$ of its owner at the freeze (Section 3.5.2), both in US dollars at the price of the freeze day, so the capped allowance is $\hat{a} = \min(a^\star, \text{bal})$. It was set $\Delta t^\star(u, v, \text{token})$ days before T_{obs} and enters the edge as one event of that age and of the capped amount:

$$w_A(u, v) = \sum_{\text{token}} \sigma\big(\Delta t^\star(u, v, \text{token})\big)\, V\big(\hat{a}(u, v, \text{token})\big), \tag{3.7}$$

with σ, V, k and t_0 as for AWP and $b = b_A = \ln 3/m_A$. A directed edge $u \to v$ means that owner contract u endorses spender contract v. The spender can move at most the owner's balance, whatever the allowance says, so the capped amount is what the endorsement puts at stake; an owner that holds none of the token at the freeze gives no edge, however much passed through it in the window. The balance and not the allowance sets most capped amounts (Table 8.5), so m_A is small: a capped allowance of a dollar or more enters with its full time weight to four decimals, and V separates only the smaller ones.

The walk restarts uniformly over the contracts of the graph. AWP's restart rule, carried over to allowances, restarts the walk at owner u in proportion to $X_u^A = \sum_{v \neq u} \max_{\text{token}} \sigma\big(\Delta t^\star(u, v, \text{token})\big)$, summed over the spenders v of u's positive capped allowances. That rewards granting endorsements, while EndorseRank is meant to rank the contracts that receive them. The analysis plan fixed uniform restarts and named this variant, EndorseRank with activity restarts, as one to report next to EndorseRank (Table 4.10).

The weight keeps the capped amount and the time of the latest approval; it is not the allowance in force at the end of the window. An allowance last set before the score window began is not

an edge, even if it is still in force, so the allowance graph holds the authorizations that owner contracts granted or renewed in the year before the freeze. An owner who approves an allowance-manager contract such as Uniswap's Permit2 appears only with an edge to that contract, not to the application that later spends through it. Some tokens also emit `Approval` from inside `transferFrom` with the reduced allowance. For them the latest record can be the remainder after the spender's last use, dated at that spend, so the time weight partly records use, an allowance spent in full ends at zero and gives no edge, and an authorization granted before the score window and used in a label window can appear as a new approval pair. How often the latest record is such a remainder was not measured.

Algorithm 2 lists the steps.

Algorithm 2 EndorseRank graph construction and scoring

Require: Latest allowances $\mathcal{A}$ between contracts in US dollars, owner balances bal, evaluated contracts $\mathcal{W}$, observation end T_{obs}, parameters $k, t_0, b_A, d, \varepsilon, I_{\max}$
Ensure: Scores $\mathbf{s}_{\text{ER}}$ on $\mathcal{W}$

$E \leftarrow \emptyset$
for each latest allowance $(u, v, \text{token}, a^\star, t^\star)$ in $\mathcal{A}$ **do**
    $\Delta t^\star \leftarrow (T_{\text{obs}} - t^\star)$ in days; $\hat{a} \leftarrow \min\big(a^\star, \text{bal}(u, \text{token})\big)$
    accumulate $\sigma(\Delta t^\star)\, V(\hat{a})$ on edge $(u \to v)$ ▷ Equation (3.7)
end for
remove non-positive edges
$\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \text{uniform } \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equations (3.1)–(3.2)
return $\mathbf{s}_{\text{ER}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

Allowance edges between contracts are far fewer than transfer edges (Table 4.1); since one PageRank iteration costs $O(|E|)$, this sparsity is the structural basis of H3.

Edge lists are stored as (from, to, weight) triples, with duplicate edges merged by summing their weights.

3.6.3 Coupled operator over both layers (C-PR) and the seeded ablation (S-PR)

EndorseRank and AWP each walk on a single edge type, so each ranks a contract by one kind of act: the spending rights it is granted or the payments it receives. C-PR walks both edge types of

the same contracts in one chain. The analysis plan fixed the coupled operator on two layers that weight each event by its amount; the main analysis values the amounts in US dollars and caps each allowance at its owner's balance:

$$\tilde{w}_A(u,v) = \sum_{\text{token}} \min\big(\hat{a}(u,v,\text{token}), U\big), \qquad \tilde{w}_T(u,v) = \sum_{e:\, u\to v} x_e\, \sigma(\Delta t_e). \tag{3.8}$$

The allowance layer has the edges of EndorseRank and the transfer layer those of AWP, apart from the few pairs whose value weight is zero in floating point (Section 4.1.1), which only the layers of C-PR keep; otherwise only the weights differ. Neither layer passes an amount through V, and only the transfers decay with age. Both rules for C-PR were fixed on these layers in raw base units. The two layers form one graph on the union node set $V = V_A \cup V_T$, with transition matrices P_A and P_T from Equation (3.2). For node u the coupled row is a convex mixture of the two layer rows, renormalized over the layers in which u has out-edges:

$$\begin{aligned} P^{(\lambda)}_{u,\cdot} &= \alpha_A(u)\, P_{A,u,\cdot} + \alpha_T(u)\, P_{T,u,\cdot}, \\ \alpha_A(u) &= \frac{\lambda\, \mathbf{1}[o_A(u) > 0]}{D_\lambda(u)}, \qquad \alpha_T(u) = \frac{(1-\lambda)\, \mathbf{1}[o_T(u) > 0]}{D_\lambda(u)}, \end{aligned} \tag{3.9}$$

where $D_\lambda(u) = \lambda\, \mathbf{1}[o_A(u) > 0] + (1-\lambda)\, \mathbf{1}[o_T(u) > 0]$, and $o_A(u)$ and $o_T(u)$ are the out-strengths of u in the two layers. A layer of weight 0 is left out with its edges and nodes, and a node with $D_\lambda(u) = 0$ is dangling, its mass handled by the third term of Equation (3.1). Teleportation is uniform over the contracts of V, $d = 0.85$, and $\mathbf{s}_{\text{CPR}}$ is the PageRank vector of Equation (3.1) with $P = P^{(\lambda)}$, under the solver settings of the single-layer methods. Figure 3.4 shows one step of the walk, and Algorithm 3 lists the construction and scoring steps.

At $\lambda = 1$ the operator walks the allowance layer alone and at $\lambda = 0$ the transfer layer alone; we write these walks C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). They are the comparators of both rules for C-PR (RQ4 and RQ5; Sections 3.8 and 3.8.1). The family is not continuous at the end points: for $0 < \lambda < 1$ a node with out-edges in only one layer follows it whatever λ is, whereas at $\lambda = 1$ a node

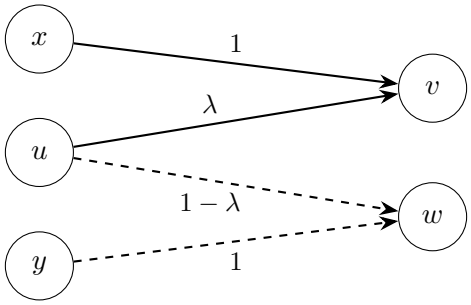


Figure 3.4: One step of the coupled walk C-PR (Equation (3.9)); solid arrows are allowance edges and dashed arrows transfer edges, labeled with their transition probabilities before damping. The contract u has out-edges in both layers and mixes them with weights λ and $1-\lambda$; x and y have out-edges in one layer only and follow it whatever λ is. With probability $1-d$ the walk restarts at a uniformly chosen contract.

Algorithm 3 C-PR graph construction and scoring

Require: Capped latest allowances $\mathcal{A}$ and transfers $\mathcal{T}$ between contracts in US dollars, evaluated contracts $\mathcal{W}$, observation end T_{obs}, cap U, weight λ, parameters $k, t_0, d, \varepsilon, I_{\max}$
Ensure: Scores $\mathbf{s}_{\text{CPR}}$ on $\mathcal{W}$

allowance layer: accumulate $\min(\hat{a}, U)$ on $(u \to v)$ for each latest allowance in $\mathcal{A}$ ▷ Equation (3.8)
transfer layer: accumulate $x_e\,\sigma(T_{\text{obs}} - t_e)$ on $(u \to v)$ for each transfer in $\mathcal{T}$
remove non-positive edges; leave out a layer of weight 0
$V \leftarrow V_A \cup V_T$; $P_A, P_T \leftarrow$ row-normalized layers ▷ Equation (3.2)
for each node $u \in V$ with $D_\lambda(u) > 0$ **do** ▷ the others are dangling
    $P^{(\lambda)}_{u,\cdot} \leftarrow \alpha_A(u)\,P_{A,u,\cdot} + \alpha_T(u)\,P_{T,u,\cdot}$ ▷ Equation (3.9)
end for
$\mathbf{s} \leftarrow \text{WeightedPageRank}(P^{(\lambda)}, \text{uniform } \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equation (3.1)
return $\mathbf{s}_{\text{CPR}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)

whose out-edges are all transfers is dangling or absent, and the same holds at the other end. The values $\lambda = 0.5$ (primary) and $\lambda \in \{0.25, 0.75\}$ (sensitivity) were fixed in the plan's configuration. Without its two caps the allowance layer would be very uneven: about two in three of the positive latest allowances between contracts are unlimited (Table 8.5), and each would take almost the whole row of its owner. The balance cap makes every allowance finite, and U keeps one large balance from taking the row. EndorseRank and AWP bound every amount through V.

In the S-PR ablation the transfer layer is walked, but the restart vector $\boldsymbol{\pi}$ of Equation (3.1) is not uniform: it is the score of C-PR ($\lambda = 1$), restricted to V_T and renormalized over it. Let $\mathbf{r}_A$ be the

PageRank vector of the allowance layer walked alone on V_A, extended by $r_{A,u} = 0$ for $u \notin V_A$. Then

$$\pi_u = \frac{r_{A,u}}{\sum_{v \in V_T} r_{A,v}} \qquad (u \in V_T).$$

If no node of V_T lies in V_A, the denominator is zero and $\boldsymbol{\pi}$ falls back to the uniform distribution on V_T. The result is the TrustRank construction (Gyöngyi et al., 2004), but with allowance-derived seeds in place of hand-labeled ones. A comparison of S-PR with C-PR separates what the allowance edge contributes as a propagation medium from what it contributes as a seed. Like AWP and C-PR ($\lambda = 0$), however, S-PR scores zero every contract without a transfer edge, about three spenders in ten at each freeze (Table 4.2). Most of them end no edge at all, but about 3% of each spender cohort has an allowance edge only, so a contrast of a transfer-only walk with C-PR, F1 of rule B included, mixes ranking with coverage.

Because C-PR walks on $E_A \cup E_T$, one of its iterations costs as much as an iteration on each layer, and its runtime is expected to be at or above AWP's; the speed result of Claim C6 is therefore EndorseRank's alone. S-PR needs one solve on each layer.

3.7 Discovery-order test

The discovery-order test repeats on the contract graph the crawl-ordering experiment of Cho et al. (1998) (Section 2.6). Its question is that of RQ2: ordered by C-PR computed on the contracts read so far, does a crawl from a random contract of the giant strongly connected component reach the contracts with the most user-account demand sooner than a breadth-first crawl and sooner than a crawl ordered by the number of contract backlinks? The test was defined after the registered results were known and decides no rule of the plan or the registration; its protocol is `experiments/docs/discovery_protocol.md` in the repository.

A page is a contract and a link an edge of the graph G of one freeze, in either layer; T_{obs} is the primary freeze and t_1 the secondary one. A crawl starts at a seed and visits one contract at a time, knows only the edges that its visits have revealed, and chooses the next visit from its frontier,

the contracts discovered and not yet visited. The union graph holds the distinct ordered pairs of different contracts with an edge in either layer. In the `out` mode a visit reveals the out-edges of the contract, the approvals it granted and the transfers it sent, and the universe $\mathcal{U}$ is the giant strongly connected component of the union graph with every contract reachable from it, SCC $\cup$ OUT in the bow-tie decomposition of Broder et al. (2000). In the `both` mode a visit reveals the in-edges as well, and $\mathcal{U}$ is the weakly connected component that holds the giant SCC (WCC). The protocol makes `both` the primary mode if SCC $\cup$ OUT holds fewer than half the contracts of the WCC in the structure pilot below, as it does, and `out` is then a sensitivity (Table 8.7).

The seeds are the 20 contracts of the giant SCC with the lowest SHA256 hash of a fixed string and the address. An ordering other than the random one scores every frontier contract: after K visits the scores are recomputed, and the $\max(1, \lceil \rho K \rceil)$ highest-scored frontier contracts are visited before the next recompute, ties going to the earlier discovery and then to the lower address. A prime marks what is computed on the contracts read so far. The orderings are breadth-first (BFS), which follows the order of discovery; random; three counts of the visited contracts with an edge to the frontier contract, the raw degrees that the walks smooth (IB'_A on allowance edges, IB'_T on transfer edges and IB'_U on either, the contract backlinks); and the walks of Section 3.6 on the graph seen so far (EndorseRank′, EndorseRank-activity′, AWP′, C-PR′ at $\lambda = 0.5$, 0.25 and 0.75, its two layers and S-PR′), each recompute starting from the previous vector, a warm start that is part of the method. The ideal order and the orders by C-PR and by union in-degree of the full graph are references that are not crawls, and the normalized in-strength δ', the depth-one closed form of the walk, is a diagnostic outside every comparison.

Importance is user-account demand, which no ordering reads. For a contract c of $\mathcal{U}$ and the score window of the freeze, $I_T(c) = \sum_j \min(x_{j\to c}, U_{\text{user}})$ and $I_A(c) = \sum_j \min\big(\hat{a}(j,c), U_{\text{user}}\big)$ with $\hat{a}(j,c) = \sum_{\text{token}} \hat{a}(j,c,\text{token})$, where j runs over the user accounts, $x_{j\to c}$ is the US dollars that j transferred to c, and $\hat{a}(j,c,\text{token})$ is the latest allowance of j to c on a token capped at the balance of j, as in Equation (3.7). The numbers of accounts with at least one dollar, N_T and N_A, are secondary. Importance in the manner of Cho et al., here the union in-degree IB_U and C-PR

(IR) of the full graph, is measured on the edges that the orderings read, so agreement with it is an intended-construct check, reported second.

The hot set $H_q(I)$ holds the contracts of $\mathcal{U}$ whose importance is positive and among the $\lceil q|\mathcal{U}| \rceil$ largest, ties included. With t_h the visit at which a crawl reaches hot contract h, AUC_x is the mean over the hot contracts and the seeds of $\max(0, 1 - t_h/\lceil x|\mathcal{U}| \rceil)$, the normalized area under the curve of the share of the hot set visited during the first $x|\mathcal{U}|$ visits; the random reference curve, $K/|\mathcal{U}|$ after K visits, gives $x/2$. The primary values are $x = 10\%$ and $q = 1\%$. Intervals come from a paired bootstrap over the hot contracts ($B = 400$, seed 42, one index matrix for every ordering).

The decision is read at T_{obs} in the primary mode, on $H_q(I_T)$ and on $H_q(I_A)$, from two contrasts of $\mathrm{AUC}_{10\%}$: D_1, C-PR′ minus BFS, and D_2, C-PR′ minus IB'_U. Section 6 of the protocol words the verdict on each importance:

- V1, both intervals above 0: "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most [user-account inflow | user-authorised value] sooner than a breadth-first crawl and than a crawl ordered by the number of contract backlinks."
- V2, the D_1 interval above 0 and the D_2 interval not above 0: "… sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks."
- V3, the D_1 interval not above 0: "… did not reach the contracts of the universe with the most [user-account inflow | user-authorised value] sooner than a breadth-first crawl."

An interval below 0 adds a sentence naming the ordering that was faster. "The answer to the question of section 1 is the weaker of the two verdicts (V2 is weaker than V1, and V3 than V2), so it is yes only when both are V1"; the question of the protocol's first section is that of RQ2. At t_1 and in the other mode the same contrasts are reported without a verdict.

Outside the decision the protocol reports, in a fixed order: each walk′ against the count it smooths; contrasts among the walks, first C-PR′ against C-PR′ ($\lambda = 0$) and against AWP′, which show what the allowance layer adds; the other values of λ; the hot sets of IB_U and IR, where C-PR′ minus IB'_U

on $H_q(\mathrm{IB}_U)$ repeats the comparison of Cho et al.; sensitivities, the other reveal mode among them; and the freeze t_1, which shares nine months of its score window with T_{obs}.

A structure pilot on the t_1 graph, which reads the pairs of each layer without weights and computes no importance, score or crawl, sets three values by rules of the protocol, for both freezes: the primary mode, $\rho = 0.02$ (the smallest of 0.01, 0.02 and 0.05 with a projected run time of at most twelve hours) and $q = 1\%$. The pilot also raised the stop flag that the protocol sets on a thin allowance layer, since 0.54% of the contracts of SCC $\cup$ OUT have a capped allowance in-edge against a threshold of 5%, and the test went on by the author's decision, which the protocol records with its reasons, because it is the only direct test of the question.

3.8 Temporal holdout protocol

In the holdout W0, scores use the events between contracts of the W0 score window, from 1 April 2025 up to t_1 = 31 March 2026, 23:59:59 UTC. Labels are counted on events from 1 April 2026, 00:00:00 UTC, up to $t_2 =$ 30 June 2026, 23:59:59 UTC. The cohort consists of the $n = 7{,}864$ contracts among the spenders that hold at least one positive latest allowance of one of the five tokens, set inside the window, at t_1, in the events around the addresses listed for W0 (Section 3.2); these are listed from the approvals of the same window, so no event after t_1 chooses the spenders. A spender that ends no edge of the contract graph stays in the cohort and scores zero under every score and baseline. Every score in the holdout table (EndorseRank, AWP, C-PR at the three values of λ and at its two end points, S-PR and four raw counts) is computed from the contract graph of this window alone, with the time decay anchored at t_1, and so is EndorseRank with activity restarts (Table 4.10).

Two labels are counted per spender, one for each construct, so that each single-layer method is tested on its own construct and the hybrids on both:

1. New approval pairs: the count of owner–spender pairs, attributed to the spender, that receive a positive approval in the label window but were absent from the latest-positive snapshot at

t_1. This label measures the endorsement construct.

2. New transfer senders: the count of distinct addresses that send the spender a transfer in the label window without having sent it one between the start of the score window (1 April 2025) and t_1, self-transfers excluded. It measures the flow construct.

Both labels count every account in the events around the listed addresses, and most of what they count comes from user accounts (Table 8.6), whose events no score reads: the labels ask whether the standing of a contract among contracts predicts later demand from all accounts. The baselines are raw counts on the contract graph at t_1: each spender's in-approve degree (the distinct owner contracts with a positive capped allowance to it), its in-approve value (Equation (3.13)), its transfer in-degree (the distinct sender contracts other than itself) and its union in-degree (the distinct other contracts with an edge to it in either layer). Each single-layer PageRank smooths one of the first three and C-PR the fourth. The increment of a PageRank over its own raw count answers RQ3; the PageRank coefficient on its own does not.

Every difference between two coefficients on the same label is a paired difference on the resamples of Section 3.11. The contrasts fixed in the plan are the two of rule A below; C-PR against the union in-degree was named when the main analysis was defined and belongs to no rule. Every other contrast in this window was computed after the labels were known; the contrasts of C-PR with every other score are in Table 4.22, and the differences between EndorseRank and AWP in Appendix 8.5.

Rule A, fixed in the analysis plan, judges C-PR against its own layers walked alone and has two parts. The primary part uses W0: C-PR at $\lambda = 0.5$ should do no worse than C-PR ($\lambda = 1$) on new approval pairs (the interval of $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=1}$ contains zero or lies above it) and should be better than C-PR ($\lambda = 0$) on new transfer senders (the interval of $\tau_{\text{CPR}} - \tau_{\lambda=0}$ lies above zero). The secondary part uses the same window and the matched cohort. Its condition is that C-PR's point estimate lies within the 95% interval of the better of the two layers on the transfer family and on the allowance family, and that C-PR exceeds both layers on the Sybil-stability family. The paired

contrasts of Table 4.19 are reported beside this rule as a stricter check; they do not decide it. If its primary part fails, RQ4, whether C-PR beats a walk on either layer alone, receives a negative answer. RQ5, whether C-PR improves on a walk over the transfers alone, is not covered by rule A; two criteria of rule B test it in W1 (Section 3.8.1). Once labels have been observed, λ is not tuned. The pipeline also counts, over the label window, revocations, the pairs held at t_1 that receive a zero approval (for some tokens an allowance spent in full, Section 3.6.2), and an approval-then-outbound drain heuristic. Neither is offered as supporting evidence (Section 4.7).

Two descriptions beside the rules decide none of them: the main comparisons are repeated on the spenders inside the union graph, outside which more than a quarter of each spender cohort ties at zero under every score (Tables 4.2 and 4.23), and each label is split by the class of the account it counts (Table 8.6).

3.8.1 Registered window W1: July to September 2026

The W0 holdout also showed something that no rule had asked about: against its transfer layer walked alone, C-PR ranked the spenders that later gained new approval pairs higher, with an interval above zero, while it was worse on new transfer senders (in the registered analysis; Appendix 8.6). Noticed only after the labels were in, this counts as a hypothesis and not as a finding, and it is tested again on a window that the study had not extracted when the test was fixed. The registration document, written after the W0 results were known, and the configuration, with the status of the window set to registered, were committed to the authors' Git repository, and scan B, the first query to read logs dated after 30 June 2026, ran after that commit, an order that the extraction script enforces (Appendix 8.1.1).

The method is unchanged: C-PR at $\lambda = 0.5$, as fixed in the plan, with the two sensitivity values and S-PR, and the solver settings, label definitions and paired bootstrap of W0. Scores are frozen at T_{obs}, so they are the scores of the same-window evaluation: in the registration, which belongs to the registered analysis, they use every extracted event from 1 October 2023 onwards on graphs that hold every account of the cohort's events, and in the main analysis the events between contracts of

the score window of T_{obs}. Labels are counted from 1 July to 30 September 2026.

The spender cohort is defined as in W0: the $n = 7{,}915$ contracts among the spenders holding at least one positive latest allowance at the freeze, with the two labels of W0. It holds about five in six of the W0 spenders, so W1 is a later label window on largely the same contracts.

Rule B compares C-PR with its transfer layer walked alone, C-PR ($\lambda = 0$), through the paired difference $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=0}$. The thesis applies two of its registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds.

- F1, spenders, new approval pairs: the 95% interval of $\Delta\tau$ lies above zero;
- F2, spenders, new transfer senders: the lower end of the interval lies above $-\delta$.

If F1 or F2 fails, rule B is not met; if both hold, the reading is that both applied criteria hold, and no verdict is given on the rule as registered. The margin $\delta = 0.01$ is the half-width of the paired interval for the F2 contrast in W0 in the registered analysis, rounded up to two decimals, as the registration fixed it; in both analyses the spender cohorts resolve differences smaller than that (Table 8.16). Outside the decision the registration reports the primary part of rule A rerun on the new window (F6): C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs (F6a) and better than C-PR ($\lambda = 0$) on new transfer senders (F6b).

Two sensitivity analyses were registered in the same commit, before the study extracted any July–September log. Their spender parts are reported next to the two criteria, and neither enters the reading. The first repeats F1 and F2 with AWP (Section 3.6.1) in place of C-PR ($\lambda = 0$). The second repeats F1, F2 and F6 with the spenders that a graph leaves out kept as isolated nodes and not scored zero (Section 4.4), in the graph of every walk but S-PR. Other contrasts in W1, each PageRank against its own count and C-PR against every other score among them, were computed after the labels were known and are reported apart from the two criteria.

3.9 Construct-check design

Every score enters the same-window evaluation: EndorseRank, also with activity restarts, AWP, C-PR at its five values of λ and S-PR. For each score M and check c the pipeline reports Spearman's ρ and Kendall's τ between $\mathbf{s}_M$ and $\mathbf{p}_c$ with 95% bootstrap intervals (Section 3.11).

The same-window evaluation is organized around two contemporaneous families: transfer, the contract's inbound transfer activity (the proxies used to evaluate AWP (T. Do et al., 2023)), and allowance, its inbound approval activity as a spender. Both are graph-adjacent: each check is a local statistic, not a global score, of the events between contracts that make up G_{transfer} and G_{approve}. Alignment of AWP with the transfer checks, or of EndorseRank with the allowance checks, is an intended-construct check: it shows coherence within a construct and is not evidence of independent predictive power. A third family, the Sybil-adjusted activity summaries of Section 3.10.3, enters the secondary part of rule A, but no hypothesis rests on it.

The discovery-order test, whose importance comes from user accounts, and the temporal holdout, whose labels come after the freeze, do not rest on the edges that a score has read. Packin and Lev-Aretz (2024) warn that decentralized scoring can become opaque; publishing explicit formulas (Section 3.10) and keeping the same-window checks apart from the other two evaluations reduce that opacity.

3.10 Operational definitions of proxies

In what follows, $\mathcal{W}$ denotes the matched cohort, and every proxy is computed from the events between contracts in the score window of T_{obs}.

3.10.1 Transfer family

Write $\mathcal{T}_{\text{in}}(w)$ for the set of transfers that contract w receives from contracts in the window and u_e for the sender of transfer e. Self-transfers belong to $\mathcal{T}_{\text{in}}(w)$, so a contract that sends tokens to itself

counts itself as one sender, and amounts x_e are summed in US dollars across the five tokens. Define

$$\text{in_degree}(w) = \big|\{\, u_e : e \in \mathcal{T}_{\text{in}}(w) \,\}\big|, \tag{3.10}$$

$$\text{in_value}(w) = \sum_{e \in \mathcal{T}_{\text{in}}(w)} x_e. \tag{3.11}$$

Both are baselines defined in the AWP paper (T. Do et al., 2023). Unlike PageRank they are local and ignore the senders' reputations, so comparing AWP with them shows how far propagation, recency weighting and restarts at active senders move its ordering away from the raw inflow.

3.10.2 Allowance family

On the allowance side, w is the spender (owner $\rightarrow$ w). With $\hat{a}$ the capped latest allowances of Equation (3.7), define

$$\text{in_approve_degree}(w) = \big|\{\, u : \hat{a}(u, w, \cdot) > 0 \,\}\big|, \tag{3.12}$$

$$\text{in_approve_value}(w) = \sum_{u,\text{token}} \min\big(\hat{a}(u, w, \text{token}), U\big). \tag{3.13}$$

Like the transfer baselines, they are local and follow no multi-hop endorsement chain.

3.10.3 Sybil-adjusted stability family

The inbound counters leave out self-transfers ($u = w$). Let $N_{\text{tx}}(w)$ be the number of transfers that w receives from other contracts and $N_{\text{uniq}}(w)$ the number of distinct inbound counterparties. Tenure and active months use a second set of rows. Every transfer between contracts that involves a cohort contract is attributed to its sender if the sender is in the cohort, and otherwise to its recipient; a transfer between two cohort contracts thus counts only for the sender. We take $t_{\text{min}}(w)$ and $t_{\text{max}}(w)$ to be the earliest and latest timestamps among the rows attributed to w. $N_{\text{mon}}(w)$ is the larger of two counts of distinct calendar months: the months in which w received a transfer from another

contract, and the months of the rows attributed to w. Define

$$\text{inbound_counterparty_ratio}(w) = \frac{N_{\text{uniq}}(w)}{\max(N_{\text{tx}}(w), 1)}, \tag{3.14}$$

$$\text{transfer_tenure_days}(w) = \frac{t_{\max}(w) - t_{\min}(w)}{86400} \quad \text{(seconds to days)}, \tag{3.15}$$

$$\text{active_months}(w) = N_{\text{mon}}(w). \tag{3.16}$$

The counterparty ratio penalizes contracts that receive many transfers from few counterparties, a simple wash pattern. It does not penalize the usual Sybil pattern, in which many fresh contracts send one transfer each; a contract fed that way scores close to the maximum of one. Tenure and active months track persistence over the score window. These proxies are therefore weak Sybil checks, and colluding adversaries can satisfy all three.

3.11 Statistical tests

Concordance is measured with two rank correlations, as in the AWP evaluation protocol (T. Do et al., 2023), so that no conclusion hinges on one coefficient; both are invariant under monotonic re-scaling, and the few hubs of the heavy-tailed scores do not dominate them.

- Spearman's ρ: the Pearson correlation of the mid-ranks R_i and S_i of the raw score and the raw proxy value of contract i (Section 2.10).
- Kendall's τ: every reported value is the tie-corrected τ_b of Equation (2.6), computed on the raw scores and proxies. Without ties, $(1+\tau)/2$ is the probability that a randomly chosen pair is ordered the same way by both rankings. With ties that reading belongs to $\tau_a = (n_c - n_d)/n_0$ and not to τ_b. Ties are common: a contract outside a method's graph scores zero, so the scores on the allowance layer tie most pairs of the matched cohort and the other scores a smaller share (Section 4.4), and the raw counts and the labels tie as well. When the two sides of a paired contrast tie very different shares of pairs, τ_b gives the more tied side the smaller denominator, so the contrast can favor it under τ_b and not under τ_a; the contrasts, computed with τ_b alone,

are exposed to this.

- Family aggregation: A family is summarized by the plain average of the coefficients of its proxies. Conclusions are drawn from the confidence intervals, and no absolute τ cutoff is used as a decision criterion.
- Uncertainty: Sampling uncertainty is estimated with a paired bootstrap over contracts (Efron, 1979). The evaluated cohort is resampled with replacement $B = 400$ times (seed 42, fixed in the configuration), and one index matrix is reused for every method and proxy, so the resampled statistics are paired across methods. For each proxy τ, family mean and inter-method difference, the 2.5th and 97.5th percentiles of the bootstrap distribution form a 95% percentile interval (DiCiccio & Efron, 1996). The percentile interval is first-order accurate at the sizes of the cohorts if contracts are independent; contracts that share owners or belong to one protocol are not, so the intervals may be too narrow. With $B = 400$ the Monte Carlo error of an interval end is about 0.13 times the bootstrap standard error. The discovery-order test resamples the hot contracts in the same way, after averaging the crawls of the 20 seeds for each hot contract (Section 3.7); hot contracts are not independent either and the seeds are few, so a bootstrap over the seeds is reported as a sensitivity.
- Difference tests: Every comparison is a paired contrast on the same resamples: we compute $\Delta\tau = \tau_A - \tau_B$ resample by resample and report the 95% percentile interval of its bootstrap distribution, and a contrast counts as a difference when its interval excludes zero. The intervals of the many contrasts computed after the labels were known are not adjusted for multiplicity, and a few would exclude zero by chance alone; a post hoc contrast resolved in one window only is therefore read as a hypothesis, and one resolved in the same direction in both windows as less exposed to chance. The contrasts that decide a rule were fixed before the study extracted the data they are judged on: those of rule A (Section 3.8) before any log, and F1 and F2 of rule B before the W1 logs; on the scores of the contract graph they are applied as a re-analysis (Section 3.5.3). C-PR against its transfer layer walked alone on new approval pairs in W0 belongs to no rule and is marked as exploratory wherever it appears. In the same

window, eight contrasts set C-PR and S-PR against whichever layer, walked alone, leads each family, and six set C-PR against AWP and EndorseRank; none of these fourteen was fixed in the plan, and two of them were computed afterwards from the saved scores. Out of window, the contrasts of C-PR with every other score and with the four raw counts (Table 4.22) were computed after the labels were known, as were the five same-window contrasts of EndorseRank and AWP in Appendix 8.5.

- Directionality and imputation: Each proxy is oriented so that a higher value means a better reputation. A contract absent from a proxy source or from a graph takes the value zero, under every score alike, so each correlation uses the whole cohort.
- Non-inferiority: F2 of rule B (Section 3.8.1) asks only that C-PR not fall behind its transfer layer walked alone by more than the registered margin $\delta = 0.01$. The test is weaker than a difference test and says nothing about superiority on that label.
- What is excluded: The study attempts no causal inference, trains no supervised models on separate training and testing sets, and reports no p-values.

3.12 Computational benchmark (Claim C6)

The benchmark of Claim C6 times the solver call that turns an edge list into scores; a scoring service that refreshes its scores periodically would amortize the one-time cost of building the edge lists.

- Timed component: The edge lists and AWP's restart weights are built before the timer starts. The timed call indexes the nodes, assembles the sparse weighted adjacency matrix, row-normalizes it, forms the restart vector and runs power iteration with $d = 0.85$, $\varepsilon = 10^{-8}$ and $I_{\max} = 300$. Matrix assembly and each iteration take time that grows with $|E|$ and $|V|$, and a deployment would repeat both at every update.
- Measurement protocol: Every configuration is measured in one session. An untimed warm-up run, on which alone peak memory is tracked, is followed by five timed runs, from which we report the mean and standard deviation of the run times and the iteration count, with the

nodes and edges of the input graph. A graph and setting that appear in more than one table are measured once (Appendix 8.1.4). Every runtime table lists edge counts, since edges drive the cost of an algorithm with $O(|E|)$ work per iteration (Langville & Meyer, 2006).

- Graphs: The main benchmark runs on the whole contract graph of T_{obs}, which holds every pair between contracts and not only the pairs of the matched cohort: the allowance graph, the transfer graph and, for C-PR, their union. For the growth of runtime with graph size, three deterministic subsamples of the cohort keep the pairs incident to the sampled contracts (Section 3.13.1).
- Fairness: Every algorithm runs with the same solver, tolerance, iteration cap and damping (outside the damping sweep), so differences in cost come from the graphs and from AWP's restart vector, which can change the number of iterations, and not from the implementation. The cost of two scores is compared through time ratios, each reported beside the corresponding edge ratio.

3.13 Scaling and robustness extensions

None of the analyses of this section supplies an alternative primary sample: alignment claims in Chapter 4 refer to the matched cohort unless stated otherwise.

3.13.1 Cohort subsamples for runtime scaling

The scaling stages are drawn from the matched cohort by deterministic subsampling: contracts are sorted by the SHA256 hash of the string made of the seed `contract-benchmark-v1`, a colon and the lower-cased address, and a stage is the first n contracts in that order, so each stage contains the one before it and the subsamples cannot drift between library versions the way pseudorandom draws do. We report three stages, $n = 1{,}000$, $2{,}000$ and $5{,}000$, and after them the main measurement on the whole contract graph.

Three features of this design limit what the experiment can show. A stage keeps the pairs of the

contract graph at T_{obs} incident to its contracts, those with at least one end among them, so $|E|$ grows with the stage but not in proportion to n, since a contract with many counterparties brings many pairs. The last row is not a fourth stage of this construction: it is the whole contract graph, which also holds the pairs between contracts outside the cohort, and the graph of the pairs incident to the full cohort ($n = 6{,}443$) has no timing of its own. And no row goes beyond the contract graph of T_{obs}, so the experiment says nothing about larger graphs.

3.13.2 Damping sensitivity

On the matched cohort the damping is varied over $d \in \{0.75, 0.85, 0.95\}$, with the graphs and proxies held fixed, to see whether alignment on the allowance and transfer families depends on the teleportation probability.

3.13.3 Value-slope sensitivity

For this check EndorseRank and AWP are recomputed at T_{obs} with the slope b of V multiplied and divided by ten, which moves the amount of half weight to a tenth and to ten times the median; the plan's check on twenty-token subgraphs belongs to the registered analysis.

3.13.4 Sample-definition sensitivity

At each scaling stage of Section 3.13.1, and for the full cohort, the family means of EndorseRank and AWP are recomputed on the stage's contracts, scored on the pairs incident to them; the full cohort too is scored here without the pairs between contracts outside it, on a graph that has no timing of its own. A contract's score changes because those pairs are missing, so the per-stage coefficients are a sensitivity check on the sample definition and not evidence of stability, and no intervals were computed for them. This analysis was not in the plan.

3.13.5 Uncapped allowance layer

One sensitivity analysis shows what the balance cap changes: it repeats the comparisons of every score and baseline on the W0 and W1 spender labels with an uncapped allowance layer, in which every positive latest allowance between contracts counts at $a^{\star}$ in place of $\hat{a}$ in Equations (3.7) and (3.8). It is reported next to the main comparisons (Section 4.5.5), decides no rule, and the discovery-order test does not use it.

3.14 Reproducibility

No table in Chapter 4 was transcribed by hand. Once the Phase 1 tables are on disk, the evaluation drivers recompute every check, every alignment statistic with its interval, every crawl, every holdout result and the benchmark, and write them to machine-readable summaries, from which `scripts/export_latex.py` exports the tables and figures. One versioned configuration file holds the windows, cohort threshold, event-topic IDs, damping, decay parameters, values of λ, bootstrap count and seed, benchmark seed and cost budget; an overlay for the main analysis (variant `contract12m`) sets the start of each score window, its budget and the settings of the typing, of the balance check and of the discovery-order test. `scripts/run_contract12m_pipeline.py` runs the remote steps in order, and it and `scripts/run_discovery.py`, which runs the crawls, refuse to run unless the documents and the code they rest on are committed without local changes.

The plan and the registration of W1 were committed to the authors' Git repository, which is private. Its folder for this study is published with its commit history in the public repository https://github.com/abitocodes/ERC20-Allowance-PageRank-Wallet-Reputation, which also holds the documents, the code and the results of the main analysis; the file `HISTORY.md` maps the commit identifiers that the extraction manifest and the digest of the registered analysis record. There the plan and the registration are `experiments/docs/analysis_plan.md` and `experiments/docs/registration_w1.md`, and four files of `experiments/docs/` define the main analysis: `revision_price_weighting.md`, `revision_one_year_window.md`, `revision_contract_graph.md`,

with the lists of code families, and `discovery_protocol.md`. These files fix every element that could vary between runs: the subsampling of the cohort by SHA256 hashes, the solver settings, the bootstrap seed and the resample-index matrix of each cohort and hot set, the coupling weight λ and the holdout logic, the seeds, the schedule and the tie rule of the crawls, the pinned block of the account types and the extraction manifest of scan B. Appendix 8.1 names the commit of the summaries behind the tables of this thesis, the configuration file and the summary files with their digest.

3.15 Sampling considerations and missingness

The matched cohort is a constructed sample, not a random draw from all Arbitrum addresses. The same-window findings apply to the contracts that the cohort rule selects, whatever their transfer activity, and the holdout findings to the contract spenders in the events around them; user accounts and contracts with fewer approving owners are excluded by design (Section 3.2).

The cohorts are chosen by the approvals of every account and the graphs hold only the events between contracts, so about a third of each cohort and more than a quarter of each spender cohort end no edge and score zero under every score and baseline (Table 4.2); the comparisons inside the union graph set this effect apart (Table 4.23). The discovery-order test has a coverage of its own, the share of each importance that its universes hold (Table 8.8).

The typing rules have limits. A smart account that is neither a Safe nor an ERC-4337 account with a canonical EntryPoint stays a contract and a node: after the listed code families, 115,084 typed contracts still hold a `validateUserOp` selector and no paymaster selector, and no rule picks out the contracts that a service deploys for each user, payment or order, which, judged from their code, are several of the largest code families among the nodes (section 13 of `experiments/docs/revision_contract_graph.md`). The lists move whole code families, with the hidden code read for sampled addresses only, and every address is typed by its code at one block, nine days after the end of the data. An address that sent no transaction and has only transaction senders as counterparties

is not read: it cannot end an edge between contracts, and whatever it is, it lies outside every graph and every universe of the discovery-order test.

Data can be missing in a few ways. An amount that cannot be parsed is read as zero, which loses a transfer and resets an allowance. EndorseRank does not see approvals that emit no standard ERC-20 event, and AWP does not see transfers made off-chain or inside custodial accounts. A balance that changed without a `Transfer` log would escape the reconstruction of Section 3.5.2. The scores are thus biased toward on-chain, standard-token behavior, which is acceptable for an on-chain reputation study but matters once results are extrapolated to all contracts of a protocol.

3.16 Implementation notes

The PageRank code is written in Python, with a SciPy sparse transition operator and NumPy power iteration. Appendix 8.1.1 names the machine on which the benchmark ran. Randomness enters only through the bootstrap and the random ordering of the discovery-order test, whose seeds are fixed; given the downloaded tables and the configuration, evaluation is deterministic. Floating-point differences between platforms would be negligible against the three-decimal precision of τ.

3.17 Ethics

Only publicly available blockchain data are used in this research, and it involves no human subjects. The study was reviewed by the institutional ethics committee (Unisa SoC/CAES). Appendix 8.7 contains the ethics clearance certificate (reference [Unisa SoC/CAES ethics clearance ref: TBD], dated [TBD]).

- Privacy and anonymity: An address on Arbitrum One is a pseudonym and does not by itself reveal the real-world identity behind it (Packin & Lev-Aretz, 2024). Pseudonymity is not anonymity: a determined adversary could tie an address to a person through exchange deposits, online profiles or heuristics, and this work makes no such links. The scored addresses are contracts, and no result is reported for any user account. Our data collection is limited

to on-chain `Approval` and `Transfer` events, the addresses that sent a transaction and read-only calls for the code, the smart-account tests and the balance check, and we scraped no off-chain personally identifiable information (such as IP addresses, exchange KYC records, social media handles or deanonymization graphs). We did not cluster addresses into real-world entities, publish leaderboards of addresses likely to be personally identifiable, or try to join an address's transaction history with other databases to infer identity. The published results are aggregate measures only (correlation coefficients, runtimes, family means) and contain no individualized rankings of named participants.

- Data minimization: Scans A and B keep every ERC-20 `Approval` and `Transfer` event of their months in the study's cloud project. The tables downloaded for the main analysis hold only what graph construction and the checks require: pairs of contracts with their weights, sums over user accounts for each contract, the cohort lists with their labels, the class of each typed address and the balances of the check; none lists the approvals or transfers of a user account. The extracted tables are kept so that the research pipeline can be audited. They will not be released for the purpose of deanonymizing users.
- Data usage and compliance: Every query ran against public data hosted in the `bigquery-public-data.goog_blockchain_arbitrum_one_us` dataset and complied with the Google Cloud terms of service and rate limits (Section 3.3). Account types, smart-account tests, the balance check and token data were read with read-only calls to public RPC endpoints, and daily prices came from the public DefiLlama API. We sent no transactions and accessed no privileged or non-public indexing endpoints, and no call could interfere with the operation of the network.
- Fairness and transparency: Decentralized scoring raises concerns about fairness as well as opacity (Packin & Lev-Aretz, 2024). New, rarely used and unapproved contracts may be undervalued by graph-based reputation systems, EndorseRank and AWP among them: the allowance graph disadvantages an address that has received no spend authority, and the transfer graph favors addresses embedded in value-flow networks. On graphs of contracts, a contract

that only user accounts approve or pay ends no edge and scores zero. Chapter 5 discusses the limitations, among them cold-start effects and the Sybil susceptibility of approval graphs.

- Responsible disclosure and impact: The scores of this study may inform risk assessment in DeFi, but they are not financial advice or a consumer reporting product. Lending, trading and compliance decisions still need human oversight. Chapter 5 discusses adversarial manipulation, for example Sybil attacks on G_{approve} or wash transfers on G_{transfer}. Any future operational deployment would require additional governance, appeal mechanisms and fairness monitoring, all of which lie beyond the scope of this research.
- Summary of methodological commitments: The plan and the registration fixed their rules before the study extracted the data they are judged on, Section 3.5.3 gives the status of the definitions of the main analysis, and each PageRank is set against the raw count it smooths.

Chapter 4 reports what this protocol yields for Arbitrum One.

4 Implementation and Empirical Results

This chapter runs the design of Chapter 3 on Arbitrum One as a single case study: one chain and one score window per freeze, one matched cohort of contracts for every same-window comparison, and spender cohorts for the two label windows W0 and W1. It reports the contract-graph analysis, the main analysis of the thesis (Section 3.5.3): the data (Section 4.1); the cost, agreement, divergence and robustness of the two single-layer walks (Sections 4.2 to 4.5); the discovery-order test (Section 4.6); the out-of-window prediction of the single-layer walks against the counts they smooth (Section 4.7); and C-PR, the method of the thesis, under its rules and beside every other score (Section 4.8). Rules A and B were fixed in the analysis plan and in the registration of W1 for the raw-unit analysis; here they are applied to the scores of the contract graph as a re-analysis, and the registered analysis gave the same readings (Appendix 8.6, Table 8.16). Each Kendall τ comes with its confidence interval except in the value-slope and sample-definition checks, and each walk is read against the raw count it smooths.

4.1 Case study setting: Arbitrum One

Arbitrum One is a heavily used Ethereum rollup with mature DeFi protocols (Section 2.2), and its logs are in the public BigQuery export that the pipeline reads. The score windows run from 1 July 2025 to 30 June 2026 for the same-window evaluation and W1 and from 1 April 2025 to 31 March 2026 for W0, and the time decay is anchored at the freeze (Figure 3.3). After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the con-

tract graph; and the discovery-order test. That analysis is the one reported here, so no result of this chapter is a registered result.

From 1 April 2025 to 30 September 2026 the five tokens (WETH, USDC, USDT0, ARB and WBTC) have 1,240,275,042 `Approval` and non-zero `Transfer` rows, which name 25,537,280 addresses other than the zero address. The 17,477,779 that sent a transaction are user accounts. The addresses that could still end an edge between contracts, 6,230,270 with the cohort members and two audit samples, were read with `eth_getCode` at block $513{,}216{,}506$; those without code, EIP-7702 accounts, Safes and ERC-4337 accounts are user accounts too, which leaves 1,829,719 contracts (Section 3.5.1; Table 8.4 in Appendix 8.4.3). Only the 603,475,784 rows with a contract at both ends can make an edge: no user account is a node, and no score or baseline reads an edge of one.

The contract graph numbers 1,403,959 contracts over both freezes and has two layers (Table 4.1): transfer pairs join a sender contract to a recipient contract, and allowance pairs an owner contract to a spender contract whose latest allowance of the window is positive and stays positive under the balance cap (Section 3.6). The cap decides the size of that layer. At T_{obs}, 223,892 of the 339,282 positive latest allowances between contracts are unlimited (66.0%), 222,559 are capped to zero because the owner holds none of the token (65.6%), and 116,723 stay positive; at t_1, 289,299 of 308,922 are capped to zero (93.6%) and 19,623 stay positive (Table 8.5 in Appendix 8.4.3). For 1,493 of 1,500 sampled (token, owner) pairs (99.5%) the reconstructed balance (Section 3.5.2) equals `balanceOf` at the last block of 30 June 2026.

The allowance layer is far smaller than the transfer layer: 115,006 pairs against 2,580,709 at T_{obs} (4.5%) and 17,940 against 2,269,073 at t_1. At T_{obs} it is also concentrated: one owner contract, which this study did not identify, holds 98,073 of the 115,006 pairs, whereas the largest owner at t_1 holds 256. The allowance-side results at T_{obs}, those of the same window and of W1, are read with this in mind, and W0 gives the same comparisons on a layer without such an owner. The union graph, the distinct pairs of different contracts with an edge in either layer, has 2,575,632 pairs over 1,299,438 contracts at T_{obs} and 2,269,421 over 1,111,554 at t_1. The constants, set on the W0 score

Table 4.1: The two layers of the contract graph at both freezes: ordered pairs of contracts, the distinct contracts at their source (owners or senders) and destination (spenders or recipients), and the largest number of pairs that one source and one destination hold. Every end of every pair is a contract; the graph numbers 1,403,959 contracts over both freezes. At T_{obs} one owner contract holds 98,073 of the 115,006 capped allowance pairs; at t_1 the largest owner holds 256 of 17,940.

Layer	Pairs	Sources	Destinations	Largest source	Largest destination
At T_{obs}					
Allowance layer (capped)	115,006	9,459	100,112	98,073	2,097
Allowance pairs before the cap	302,559	141,974	108,825	98,073	28,082
Transfer layer	2,580,709	1,007,310	801,813	130,667	441,612
At t_1					
Allowance layer (capped)	17,940	10,153	2,050	256	2,008
Allowance pairs before the cap	275,917	140,327	89,727	78,382	25,810
Transfer layer	2,269,073	873,388	626,356	121,708	439,299

window of the contract graph, are those of Section 3.5.3: $m_T = 160$ and $m_A = 0.071$ dollars, $U = 16{,}000$ dollars for C-PR's allowance layer, and $U_{\text{user}} = 31{,}000$ dollars for the importance measures of Section 3.7.

4.1.1 Cohorts

The rules of the three contract sets below count every approving account, user accounts included, so membership does not depend on the contract graph; Table 4.2 shows where the members lie in it.

- Matched cohort ($n = 6{,}443$). The rule of Section 3.2, a non-zero `Approval` of one of the five tokens from at least three distinct owners in the score window of T_{obs}, lists 6,483 addresses with code; 40 of them are smart accounts and so user accounts, and the cohort is the other 6,443. Every same-window alignment, difference test and benchmark is computed on it. The same rule in the W0 score window lists 6,343 addresses, of which $6{,}292$ are contracts; the spenders and labels of each window are drawn from the events around the addresses listed for its freeze.
- Spender cohorts of W0 and W1 ($n = 7{,}864$ and $7{,}915$). In W0 the scores are frozen at t_1

(31 March 2026, 23:59:59 UTC), and the labels are counted for every spender that holds, in the events around the addresses listed for W0, a positive latest allowance of one of the five tokens at t_1, set inside the W0 score window. The cohort is the 7,864 contracts among these 7,962 spenders; 97 are user accounts and 1 is a system address. In W1 the scores are frozen at T_{obs}, and the same rule gives 7,915 contracts among 7,995 spenders. A spender needs only one owner, so these cohorts are not subsets of the matched cohort.

- Subsamples of the matched cohort ($n = 1{,}000$, $2{,}000$ and $5{,}000$). Deterministic SHA256 subsamples (seed `contract-benchmark-v1`), used only for runtime against size and for the sensitivity to sample definition (Section 3.13.1). Each stage keeps the pairs incident to its sampled contracts.

Table 4.2: Where the members of each cohort lie in the contract graph of their freeze: members that end a capped allowance pair, a transfer pair, a pair of the union graph, and members that end no pair between contracts. A member outside a graph stays in its cohort and scores zero under every score and baseline, so the scores tie on these members; Table 4.23 repeats the main comparisons on the members inside the union graph.

Cohort	Members	In the allowance layer	In the transfer layer	In the union graph	Outside the graph
Matched cohort at T_{obs}	6,443	1,250 (19.4%)	4,059 (63.0%)	4,270 (66.3%)	2,173 (33.7%)
W0 cohort at t_1	6,292	1,248 (19.8%)	3,893 (61.9%)	4,108 (65.3%)	2,184 (34.7%)
W0 spenders at t_1	7,864	1,901 (24.2%)	5,443 (69.2%)	5,700 (72.5%)	2,164 (27.5%)
W1 spenders at T_{obs}	7,915	1,975 (25.0%)	5,504 (69.5%)	5,759 (72.8%)	2,156 (27.2%)

The contract graph does not reach every member. Of the matched cohort, 1,250 contracts (19.4%) end a capped allowance pair, 4,059 (63.0%) a transfer pair and 4,270 (66.3%) a pair of the union graph; the other 2,173 have no transfer with another contract in the window and no allowance to or from one that stays positive under the cap. Of the W0 spenders 5,700 (72.5%) and of the W1 spenders 5,759 (72.8%) lie in the union graph. A member outside a graph stays in its cohort and scores zero under every score and baseline of that graph, so the scores tie on these members. Every coefficient of this chapter therefore reflects both how a score orders the members that the graph reaches and how it sets them apart from the others; Table 4.23 repeats the main out-of-window contrasts inside the union graph.

The solver graphs of EndorseRank and AWP in Table 4.3 do not hold the 9 allowance pairs and 328 transfer pairs whose value weight is zero in floating point, nor the 8 and 75 contracts that end no other pair; the layers of C-PR, whose weights are sums of amounts, hold them. Table 8.3 in Appendix 8.4.2 gives the five tokens with their allowances and price checks.

4.1.2 Implementation

Every algorithm runs on one Python implementation of PageRank ($d = 0.85$, $\varepsilon = 10^{-8}$, $I_{\max} = 300$); the methods differ in their edge lists and restart vectors. The tables of this chapter come from the summary files of the evaluation runs, the benchmark and the discovery-order test, as collected in the results digest. At $d = 0.85$ every solve of the same-window and out-of-window evaluations converged well inside the iteration cap. Appendix 8.1 lists the runs, the repository revision, the configuration file and the summary files.

4.2 Computational benchmark (Claim C6)

Claim C6 concerns what each score costs to compute on the same input. Table 4.3 reports the cost of each algorithm on the contract graph at T_{obs}, on which the matched cohort is scored, following the timing procedure of Section 3.12. Each graph and setting was timed once, so the same measurement reappears as the last row of Table 4.4 and as the $d = 0.85$ row of Table 4.7; Table 8.1 in Appendix 8.1.4 lists every timed run.

EndorseRank solves the allowance graph ($114{,}997$ edges) in 0.101 s and AWP the transfer graph ($2{,}580{,}381$ edges) in 2.82 s, so AWP takes 27.9 times as long, at an edge ratio of 22.4. The cost of an iteration grows with $|E|$ and with $|V|$ (Appendix 8.3.2). Per iteration, assembly included, AWP's solve takes 15.9 times as long as EndorseRank's (0.029 against 0.0018 s), between the node ratio of 11.9 and the edge ratio, and AWP needs 1.75 times as many iterations (98 against 56); together the two factors give the time ratio (for other d, Section 4.5.1). Both scores use the same solver, so the difference in speed comes from the size of each graph and from the iterations its walk needs

Table 4.3: Computational benchmark on the contract graph at T_{obs}, which holds every pair between contracts and not only the pairs of the matched cohort (AWP/EndorseRank runtime ratio 27.9× at an edge ratio of 22.4×). Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up, on which alone peak memory is traced; SD over the same 5 runs. A timed call turns an edge list into scores: node indexing, sparse-matrix assembly, row normalization, restart vector and power iteration. $|V|$ and $|E|$ are the nodes and distinct edges of each method's solver graph. C-PR walks on the union of the allowance layer and the transfer layer, so its $|E|$ counts the distinct pairs of both; S-PR runs on the transfer layer, and its runtime and iterations include the solve of the allowance layer that supplies its restart vector. Table 8.1 lists every run.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$\lvert V\rvert$	$\lvert E\rvert$
EndorseRank	0.101	0.009	13.7	56	109,381	114,997
AWP	2.821	0.189	228.2	98	1,298,649	2,580,381
C-PR ($\lambda = 0.5$)	3.080	0.184	204.8	92	1,299,479	2,591,253
S-PR	2.657	0.202	142.6	153	1,298,724	2,580,709

there, which the benchmark does not separate; it is no advantage of the EndorseRank algorithm.

C-PR walks on the union of both layers ($2{,}591{,}253$ distinct pairs, the pairs of a contract with itself included). Its solver keeps one matrix per layer and multiplies both in every iteration; it converges in 92 iterations and takes 3.08 s, 1.09 times as long as AWP. S-PR takes 2.66 s and 153 iterations, the solve of the allowance layer that supplies its restart vector included. Every algorithm that walks the transfer layer, alone or with allowance edges, pays for its size.

The time covers only the solver call on edge lists built in advance, not the reading of events, the typing of accounts, the balances or the edge lists, which all algorithms share wholly or in part.

4.2.1 Runtime on subsamples of the matched cohort

Table 4.4 times EndorseRank and AWP on three subsamples of the matched cohort and, in its last row, on the whole contract graph.

Few of the pairs that a subsample keeps are allowance pairs: the transfer graph has 74.8, 49.3 and 79.2 times as many edges as the allowance graph in the three subsamples. If the time of a solve followed $|E|$ alone, the time ratio would follow these edge ratios; it is 74.2, 60.2 and 108.0, that is 0.99, 1.22 and 1.36 times the edge ratio, with 48, 26 and 62 iterations for EndorseRank against 94,

Table 4.4: Runtime of EndorseRank (ER) and AWP on subsamples of the matched cohort and on the whole contract graph. The numbered rows are deterministic SHA256 subsamples of the cohort (seed `contract-benchmark-v1`); each keeps the pairs incident to its n sampled contracts, so $|V|$ and $|E|$ of both solver graphs are reported per row. AWP/ER time and AWP/ER $|E|$ are the ratios of the two runtimes and of the two edge counts in the same row. Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up. [a]The measurement of Table 4.3 on the whole contract graph, which holds every pair between contracts; $16{,}324$ allowance pairs and $1{,}063{,}593$ transfer pairs are incident to the full cohort of $n = 6{,}443$ contracts, and the graph of those pairs has no timing of its own.

n	ER (s)	AWP (s)	AWP/ER time	AWP/ER $\lvert E\rvert$	ER $\lvert V\rvert$	ER $\lvert E\rvert$	AWP $\lvert V\rvert$	AWP $\lvert E\rvert$
1,000	0.001	0.095	74.2×	74.8×	1,293	1,608	78,869	120,252
2,000	0.002	0.134	60.2×	49.3×	3,972	4,784	129,533	235,864
5,000	0.006	0.689	108.0×	79.2×	8,720	11,240	426,557	889,662
Whole graph[a]	0.101	2.821	27.9×	22.4×	109,381	114,997	1,298,649	2,580,381

90 and 104 for AWP. The iterations each walk needs on each subgraph, and not the edges alone, set the ratio. The last row, the measurement of Table 4.3, holds every pair between contracts and not only the pairs incident to the cohort, so the step from the third row to the last is a change of graph more than of n. The rows show how runtime grows with the edges up to the size of the whole graph and not beyond it (Figure 4.1).

4.3 Same-window alignment (Claim C5)

Claim C5 asks what each score agrees with in the window it is computed from. Table 4.5 reports the family-mean Kendall τ of every score on the transfer, allowance and Sybil-stability families; the proxies are computed from the events between contracts (Section 3.10). Apart from the secondary part of rule A, fixed in the analysis plan (Section 3.8) and applied here as a re-analysis, the comparisons below belong to no rule.

The scores divide by the edges they walk on. The two that walk on transfers alone, AWP and C-PR ($\lambda = 0$), reach 0.755 ($[0.748, 0.762]$) and 0.804 ($[0.799, 0.809]$) on the transfer family and 0.227 ($[0.206, 0.245]$) and 0.238 ($[0.218, 0.258]$) on the allowance family. The two that walk on allowances alone, EndorseRank and C-PR ($\lambda = 1$), reach 0.839 ($[0.828, 0.849]$) and 0.838

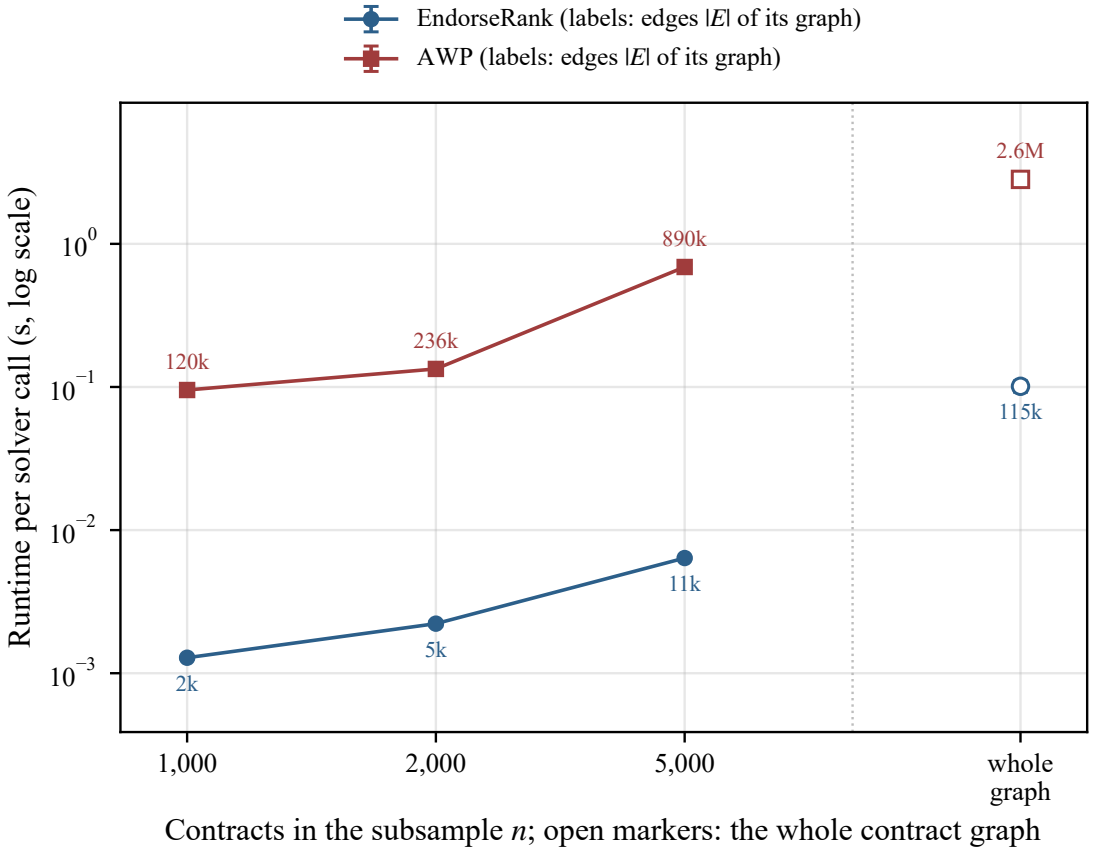


Figure 4.1: Wall-clock time of one solver call of EndorseRank and AWP for each row of Table 4.4, on a log scale: the three subsamples of the matched cohort (filled markers, joined by lines) and the whole contract graph (open markers). The label beside each point gives the edge count $|E|$ of that graph, and error bars are the standard deviations over five timed runs.

($[0.827, 0.850]$) on the allowance family and 0.297 ($[0.280, 0.317]$) and 0.298 ($[0.281, 0.318]$) on the transfer family; S-PR and the interior mixtures of C-PR lie between the two layers. On the Sybil-stability family the scores that walk on transfers lead, AWP with 0.643 ($[0.635, 0.651]$), while EndorseRank stays at 0.220 ($[0.206, 0.237]$). Part of this agreement is built in: the main proxies of each family come from the same events as one of the layers, in-degree and in-value from the transfers and in-approve degree and value from the capped latest allowances, and a PageRank smooths the in-degree of its own graph. Agreement with the proxies of a score's own edge type is therefore an intended-construct check, not an external test.

Table 4.5: Same-window family-mean Kendall τ with bootstrap intervals for every score of this study (matched cohort, $n = 6{,}443$ contracts). The scores are EndorseRank and AWP, the coupled operator C-PR at five values of λ and the endorsement-seeded walk S-PR; $\lambda = 1$ is the allowance layer walked alone (latest allowances between contracts in USD, each counted up to its owner's balance and capped) and $\lambda = 0$ the transfer layer walked alone (USD amounts discounted by the decay of AWP); both restart uniformly. The raw degrees have no row because they are proxies of the transfer and allowance families; EndorseRank with activity restarts is in Table 4.10. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Method	Transfer	Allowance	Sybil
EndorseRank	0.297 [0.280, 0.317]	0.839 [0.828, 0.849]	0.220 [0.206, 0.237]
AWP	0.755 [0.748, 0.762]	0.227 [0.206, 0.245]	0.643 [0.635, 0.651]
C-PR ($\lambda = 1$)	0.298 [0.281, 0.318]	0.838 [0.827, 0.850]	0.222 [0.207, 0.238]
C-PR ($\lambda = 0$)	0.804 [0.799, 0.809]	0.238 [0.218, 0.258]	0.637 [0.628, 0.646]
C-PR ($\lambda = 0.25$)	0.742 [0.734, 0.751]	0.424 [0.411, 0.437]	0.577 [0.567, 0.588]
C-PR ($\lambda = 0.5$)	0.738 [0.729, 0.747]	0.433 [0.419, 0.445]	0.574 [0.564, 0.585]
C-PR ($\lambda = 0.75$)	0.734 [0.725, 0.743]	0.438 [0.424, 0.450]	0.572 [0.561, 0.583]
S-PR	0.782 [0.773, 0.791]	0.371 [0.350, 0.391]	0.584 [0.575, 0.593]

On the allowance side the reach of the layer builds in a second part. EndorseRank sets only the 926 cohort contracts with an in-edge in its graph above the rest: it gives 317 others, which are in the graph only as owners, the score of a node without in-edges, and the remaining $5{,}200$ zero (Section 4.4). The in-approve degree and value of a contract that receives no capped allowance are zero. Most of the pairs that EndorseRank and these proxies order alike therefore set a contract that the layer reaches as a spender against one that it does not, and the coefficient of 0.839 says mainly that score and count agree on that boundary.

Appendix 8.5.2 breaks the family means of EndorseRank and AWP into their proxies. AWP follows the transfer in-degree (0.764, $[0.757, 0.772]$) and the in-value (0.746, $[0.738, 0.754]$) about equally, and EndorseRank the in-approve degree (0.847, $[0.835, 0.857]$) and the in-approve value (0.831, $[0.820, 0.841]$). C-PR ($\lambda = 1$) walks the allowance layer with capped amounts, also restarts uniformly, and stays within 0.002 of EndorseRank on every family.

The first half of Claim C5 is supported in a narrow sense: every score that walks on one edge type agrees most with the proxies of that edge type, the mixtures of C-PR lie between their two layers

on every family, and the agreement is partly built in, by the shared events and by the reach of each layer. Section 4.8.1 locates C-PR within the window.

4.4 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 concern whether EndorseRank and AWP put the same contracts in different orders. The inter-method Kendall τ_b between the two score vectors is 0.275, with interval $[0.258, 0.294]$ (last row of Table 4.6), so the interval excludes both zero and one. Of the $20{,}752{,}903$ contract pairs, only 32.3% are ordered by both scores, and 26.7% of those pairs are discordant.

Ties explain the first of these shares (Figure 4.2). EndorseRank gives 801 distinct scores and ties 65.4% of the contract pairs; AWP gives $3{,}982$ and ties 13.7%. Under each score the largest tie block is formed by the contracts that score zero, $5{,}200$ under EndorseRank and $2{,}384$ under AWP: the $5{,}193$ and $2{,}384$ cohort contracts outside each layer and, under EndorseRank, 7 whose only pairs carry amounts too small for the value weight to differ from zero. Under EndorseRank 317 further contracts share the score $c = 8.45 \times 10^{-6}$ of a node without in-edges. In Figure 4.3, among the contracts that EndorseRank ranks above c, the points spread over most of AWP's range.

The zero blocks come from one rule of this pipeline: a contract missing from a graph scores zero (Section 3.6). The analysis plan fixed the alternative, keeping the missing contracts as isolated nodes, as a sensitivity analysis, and Table 4.6 gives it. Under uniform restarts an isolated node scores like any node without in-edges, so the $5{,}193$ contracts outside the allowance layer tie with the 317. EndorseRank's allowance coefficient then rises from 0.839 to 0.956 ($[0.952, 0.960]$), because the score now ties the contracts that the allowance proxies tie at zero, and its transfer coefficient falls from 0.297 to 0.259 ($[0.239, 0.278]$). The agreement between the two scores falls from 0.275 to 0.227 ($[0.207, 0.245]$): the zero rule sets the 317 above the $5{,}193$, and AWP on balance orders those pairs the same way. AWP restarts only at contracts that sent a transfer, so an isolated contract keeps the score zero and AWP's coefficients are the same under both rules.

H1 is supported: the two scores order the cohort differently, but the support is narrower than the

Table 4.6: Same-window family-mean Kendall τ with contracts missing from a graph scored zero or kept as isolated nodes (matched cohort, $n = 6{,}443$ contracts). The main analysis scores a contract that is missing from a method's graph zero; keeping it as an isolated node is the sensitivity analysis fixed in the analysis plan. The rule concerns 5,193 contracts for EndorseRank and 2,384 for AWP (2,173 are missing from both graphs). Under EndorseRank's uniform restarts an isolated node gets the score of a node without in-edges. AWP restarts only at contracts that sent a transfer, so an isolated contract keeps the score zero and AWP's coefficients are the same under both rules. The last row is the agreement between the two scores. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Missing contracts scored zero		Missing contracts as isolated nodes	
Family	EndorseRank	AWP	EndorseRank	AWP
Transfer	0.297 [0.280, 0.317]	0.755 [0.748, 0.762]	0.259 [0.239, 0.278]	0.755 [0.748, 0.762]
Allowance	0.839 [0.828, 0.849]	0.227 [0.206, 0.245]	0.956 [0.952, 0.960]	0.227 [0.206, 0.245]
Sybil stability	0.220 [0.206, 0.237]	0.643 [0.635, 0.651]	0.171 [0.155, 0.187]	0.643 [0.635, 0.651]
EndorseRank vs. AWP	0.275 [0.258, 0.294]		0.227 [0.207, 0.245]	

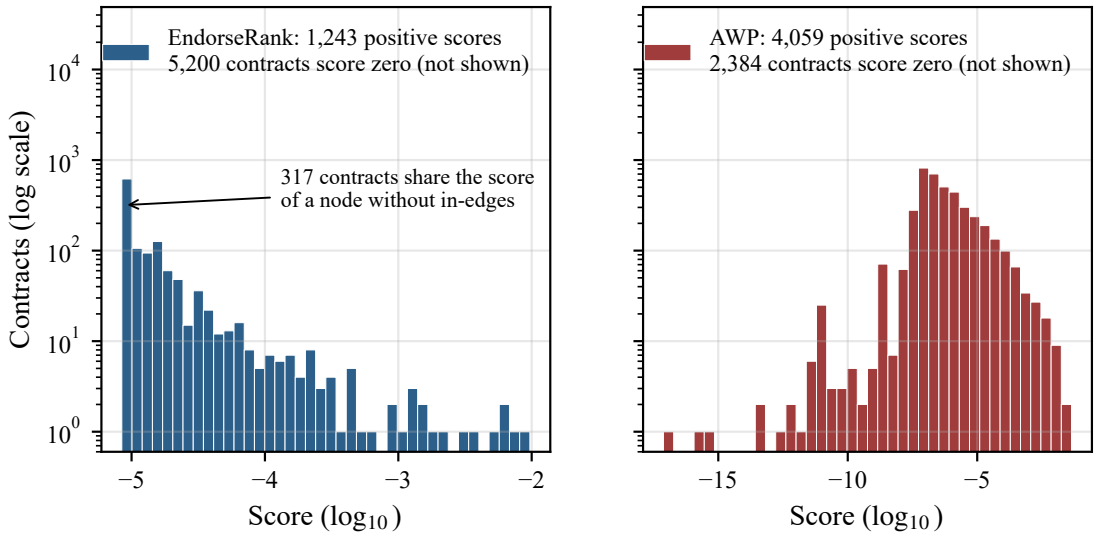


Figure 4.2: Marginal EndorseRank and AWP score distributions on the matched cohort ($n = 6{,}443$ contracts), drawn on a $\log_{10}$ scale for contracts with a positive score; counts are on a log scale. Zero-score contracts, which no edge of positive weight in the respective graph touches, are not drawn: 5,200 for EndorseRank and 2,384 for AWP. Under EndorseRank 317 contracts share the score of a node without in-edges.

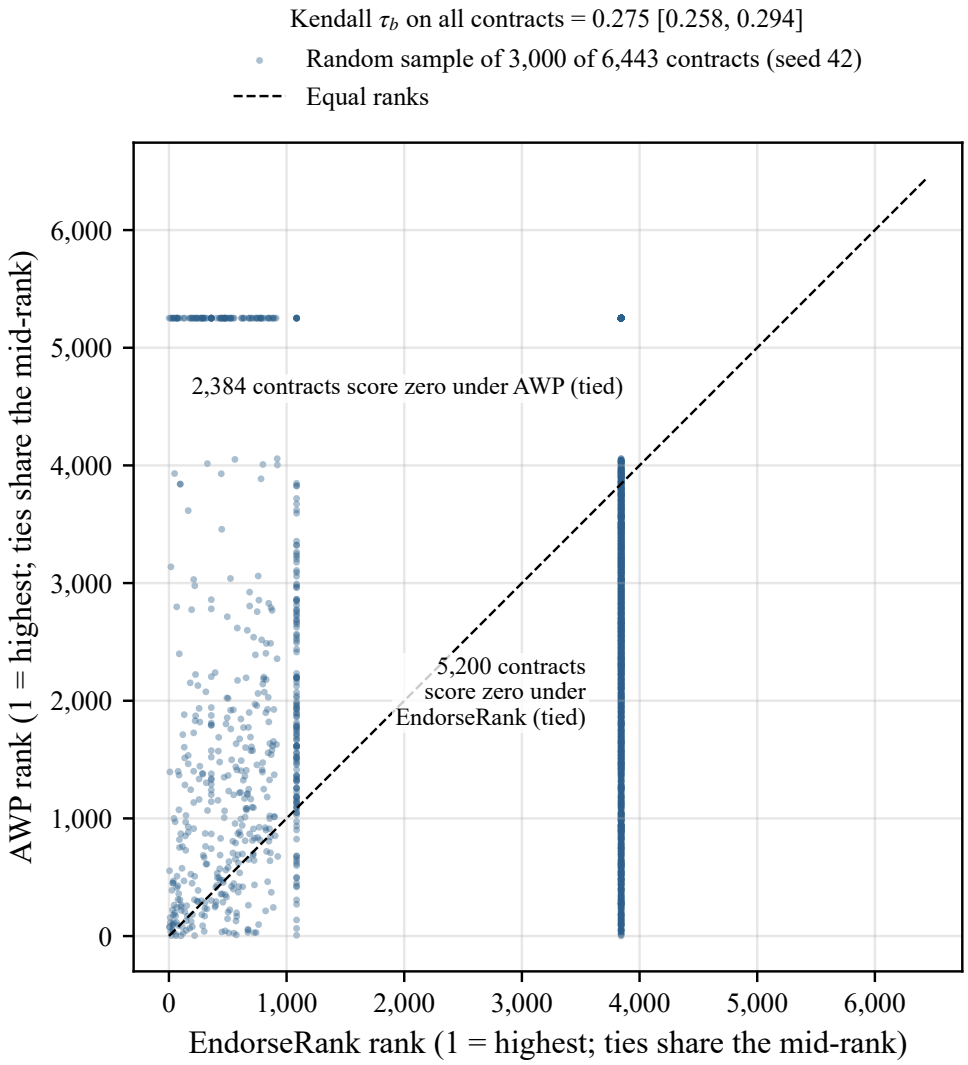


Figure 4.3: Rank–rank scatter of EndorseRank against AWP on the matched cohort (random sample of 3,000 of the 6,443 contracts, seed 42). Tied scores share their average rank, as in Kendall's τ_b. The right-hand column holds the sampled contracts among the 5,200 that score zero under EndorseRank, the column to its left those among the 317 that share the score of a node without in-edges, and the row at the top those among the 2,384 that score zero under AWP. A point on the dashed identity line has the same rank in both orderings; the legend gives τ_b on all contracts with its 95% interval.

coefficient suggests. Two thirds of the pairs are not ordered by both scores, because the allowance layer between contracts reaches a fifth of the cohort, and the rule for missing contracts, which leaves AWP's coefficients as they are, moves EndorseRank's. This divergence, the second half of Claim C5, is also the premise of the coupled walk (Section 4.8).

4.5 Robustness and sensitivity checks

Three checks ask whether the results of EndorseRank and AWP hinge on one damping value, one slope of the value weight or one sample definition; the third varies the sample itself, so its coefficients are expected to move, and it is not evidence of robustness. Two more test EndorseRank's restart rule and the balance cap.

4.5.1 Damping

Table 4.7 recomputes the family means at $d \in \{0.75, 0.85, 0.95\}$ on the same graphs.

Table 4.7: Robustness: family-mean Kendall τ of EndorseRank (ER) and AWP under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort ($n = 6{,}443$ contracts); all. = allowance, tr. = transfer, Syb. = Sybil stability. Runtimes are means of five timed solver calls on the full graphs; the $d = 0.85$ runtimes are the measurements of Table 4.3, reused, not timed again (Table 8.1 lists every run). CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)
0.75	0.839 [0.828, 0.849]	0.224 [0.204, 0.244]	0.297 [0.280, 0.317]	0.750 [0.742, 0.757]	0.220 [0.206, 0.237]	0.643 [0.634, 0.651]	0.086	1.755
0.85	0.839 [0.828, 0.849]	0.227 [0.206, 0.245]	0.297 [0.280, 0.317]	0.755 [0.748, 0.762]	0.220 [0.206, 0.237]	0.643 [0.635, 0.651]	0.101	2.821
0.95	0.839 [0.828, 0.849]	0.228 [0.207, 0.246]	0.297 [0.280, 0.317]	0.761 [0.754, 0.768]	0.220 [0.206, 0.237]	0.641 [0.633, 0.650]	0.208	7.182

For EndorseRank no family mean moves at the third decimal across the three values. For AWP the transfer coefficient rises from 0.750 ($[0.742, 0.757]$) to 0.761 ($[0.754, 0.768]$) as d rises, and its other two family means move by at most 0.004. Neither score's pattern is an artifact of the conventional $d = 0.85$. EndorseRank's stability is largely structural. Its two tie blocks do not depend on d, and for a spender whose owners receive no approvals EndorseRank equals $c\,(1 + d\,\delta(v))$, with $\delta(v)$ the

owner-normalized in-strength (Section 5.3.2), so the order among such spenders does not depend on d either. At T_{obs} the formula holds within 1% for all 651 such spenders among the 926 cohort contracts with in-edges, and for 33.8% of the other 275. Runtime grows with the iterations, which rise with d from 32 to 176 for EndorseRank and from 57 to 294 for AWP, the last just inside the cap of 300; the time ratio is 20.5, 27.9 and 34.5 at an edge ratio of 22.4. Figure 4.4 plots the family means.

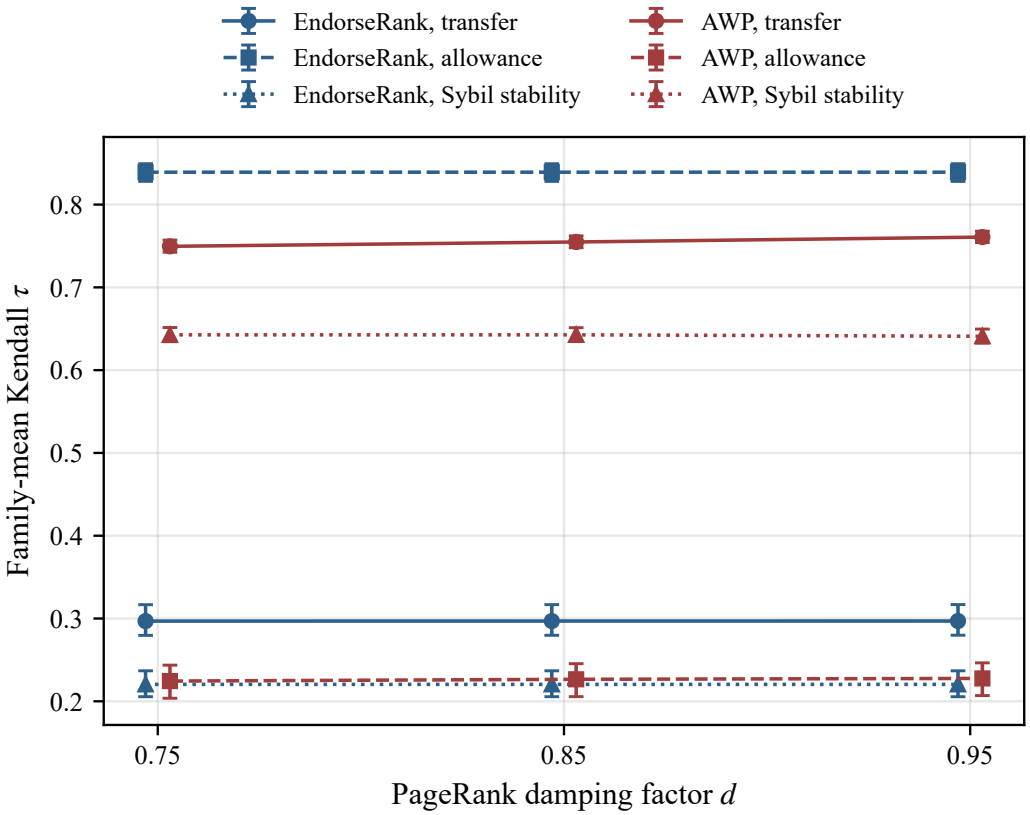


Figure 4.4: Damping sensitivity of the family-mean Kendall τ of EndorseRank and AWP on the transfer, allowance and Sybil-stability families across $d \in \{0.75, 0.85, 0.95\}$, with 95% bootstrap intervals (values from Table 4.7; matched cohort, $n = 6{,}443$ contracts).

4.5.2 Value-slope sensitivity

Table 4.8 recomputes EndorseRank and AWP at T_{obs} with the slope b of the value weight divided and multiplied by ten (Section 3.13.3), on the same graphs and proxies.

Table 4.8: Robustness: family-mean Kendall τ on the matched cohort ($n = 6{,}443$ contracts) when the slope b of the value weight V is divided or multiplied by ten, next to the main analysis. The main slopes put $V = 1/2$ at the median amount, \$160 for transfers and \$0.071 for allowances; $b/10$ moves that point to ten times the median and $10b$ to a tenth of it. The rank agreement τ of each variant with the main score of the same method (EndorseRank and AWP) is 0.999 and 0.986 at $b/10$, and 0.998 and 0.984 at $10b$. No intervals were computed for the variants.

Method	Transfer			Allowance			Sybil		
	$b/10$	b	$10b$	$b/10$	b	$10b$	$b/10$	b	$10b$
EndorseRank	0.297	0.297	0.297	0.839	0.839	0.838	0.220	0.220	0.221
AWP	0.754	0.755	0.756	0.227	0.227	0.225	0.642	0.643	0.643

Under either slope no family mean moves by more than 0.002, and the pattern stays: EndorseRank leads on the allowance family, and AWP on the transfer and Sybil-stability families. Each variant orders the contracts almost as the main score of its method does (τ of 0.999 and 0.998 for EndorseRank and of 0.986 and 0.984 for AWP).

4.5.3 Sample-definition sensitivity

Table 4.9 recomputes the family means on the subsamples of the runtime scaling and on the full cohort, each scored on the pairs incident to its contracts.

Table 4.9: Sensitivity of the family means to the evaluation set: family-mean Kendall τ of EndorseRank (ER) and AWP on deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`, full cohort $n = 6{,}443$ contracts), each row scored on the pairs incident to its contracts, whose numbers the columns $|E_A|$ (allowance) and $|E_T|$ (transfer) give; all. = allowance, tr. = transfer, Syb. = Sybil stability. Rows are evaluated on different contract sets and on other graphs than the whole contract graph, so their τ values are not comparable with Table 4.5; they show how much the statistic moves with the evaluation population. No intervals were computed for the rows. Runtimes are the measurements of Table 4.4; the graph of the pairs incident to the full cohort has no timing of its own (–).

n	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	$\lvert E_A\rvert$	$\lvert E_T\rvert$	ER (s)	AWP (s)
1,000	0.860	0.246	0.322	0.771	0.230	0.666	1,611	120,296	0.001	0.095
2,000	0.853	0.248	0.324	0.762	0.237	0.657	4,788	235,934	0.002	0.134
5,000	0.839	0.235	0.303	0.760	0.226	0.648	11,249	889,794	0.006	0.689
6,443	0.839	0.226	0.297	0.753	0.220	0.643	16,324	1,063,593	–	–

The coefficients are lower at the full cohort than at $n = 1{,}000$: EndorseRank's allowance coefficient

from 0.860 to 0.839 and its transfer coefficient from 0.322 to 0.297, AWP's transfer coefficient from 0.771 to 0.753 and its Sybil-stability coefficient from 0.666 to 0.643. None of these shifts is evidence about the scores. A stage cuts the paths through other contracts, so each score comes closer to a count of the sampled contract's own edges, which is what the proxies count. The last stage holds the whole cohort with the pairs incident to it, and its family means lie within 0.002 of the main analysis, which scores the cohort on the whole contract graph. The other rows, a sensitivity analysis of the sample definition (Section 3.13.4), must not be compared with Table 4.5.

4.5.4 Restarts of EndorseRank

EndorseRank restarts the walk uniformly, while AWP restarts it at contracts in proportion to their recent sending activity (Section 3.6.2). The analysis plan fixed uniform restarts for EndorseRank and named the variant as a score to report beside it, without fixing any comparison between the two. Table 4.10 scores both versions, with AWP beside them.

Table 4.10: EndorseRank with uniform restarts and with restarts weighted by approving activity, AWP's rule applied to the allowance graph, next to AWP: family-mean Kendall τ on the matched cohort ($n = 6{,}443$ contracts) and τ with the two spender labels in W0 ($n = 7{,}864$ spender contracts, scores frozen at 31 March 2026) and W1 ($n = 7{,}915$, frozen at 30 June 2026). Both EndorseRank rows weight each latest allowance by $\sigma(\Delta t)V(z)$. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

		Same window, matched cohort			W0, spenders		W1, spenders	
Method	Restarts	Transfer	Allowance	Sybil	New approval pairs	New senders	New approval pairs	New senders
EndorseRank	uniform	0.297 [0.280, 0.317]	0.839 [0.828, 0.849]	0.220 [0.206, 0.237]	0.168 [0.145, 0.193]	0.084 [0.062, 0.105]	0.147 [0.125, 0.170]	0.064 [0.041, 0.085]
EndorseRank	by approving activity	0.299 [0.281, 0.318]	0.799 [0.782, 0.813]	0.225 [0.210, 0.242]	0.163 [0.142, 0.188]	0.089 [0.069, 0.111]	0.148 [0.126, 0.171]	0.072 [0.049, 0.093]
AWP	by sending activity	0.755 [0.748, 0.762]	0.227 [0.206, 0.245]	0.643 [0.635, 0.651]	0.098 [0.079, 0.119]	0.330 [0.313, 0.347]	0.064 [0.042, 0.084]	0.314 [0.295, 0.332]

With AWP's rule the restart mass goes to the owner contracts in proportion to their approving activity. In the same window EndorseRank's allowance coefficient then falls from 0.839 to 0.799 ($[0.782, 0.813]$), and its transfer coefficient hardly moves (0.299, $[0.281, 0.318]$). Out of window the variant changes little. On new approval pairs it is slightly below the uniform version in W0 (-0.004, $[-0.008, -0.001]$) and level with it in W1 ($+0.001$, $[-0.002, 0.005]$); on new senders it is slightly above it in both windows ($+0.005$, $[0.002, 0.009]$, and $+0.008$, $[0.004, 0.012]$). The W1

contrasts are in Table 4.18; all were computed after the labels were known. Neither version ranks new approval pairs above the in-approve degree (Section 4.7), and Section 4.8.4 sets the variant beside C-PR.

4.5.5 Balance-cap sensitivity

Table 4.11 repeats the out-of-window comparisons on an allowance layer without the cap, in which every positive latest allowance between contracts is an edge at its own value (Section 3.13.5); that layer has $275{,}917$ pairs at t_1 and $302{,}559$ at T_{obs}. It decides no rule.

Table 4.11: Sensitivity to the balance cap: Kendall τ on the two spender labels with the allowance layer capped at the owner's balance (the main analysis) and uncapped, where every positive latest allowance between contracts counts at its own value. Each cell reads capped / uncapped. Without the cap the allowance layer has 275,917 pairs at t_1 against 17,940 with it, and 302,559 against 115,006 at T_{obs}. AWP reads no allowance, so its two values are equal. In the lower block the left cell is the contrast of the main analysis and the right cell the same contrast on the uncapped layer. The uncapped layer decides no rule. Intervals: 95% paired bootstrap (400 resamples, seed 42).

	W0 spenders		W1 spenders	
Score	New approval pairs	New transfer senders	New approval pairs	New transfer senders
EndorseRank	0.168 / 0.088	0.084 / 0.185	0.147 / 0.078	0.064 / 0.177
C-PR ($\lambda = 1$)	0.167 / 0.086	0.084 / 0.183	0.147 / 0.077	0.064 / 0.181
C-PR ($\lambda = 0.5$)	0.142 / 0.150	0.254 / 0.248	0.104 / 0.118	0.236 / 0.234
AWP	0.098 / 0.098	0.330 / 0.330	0.064 / 0.064	0.314 / 0.314
In-approve degree at the freeze	0.204 / 0.105	0.091 / 0.196	0.181 / 0.102	0.072 / 0.189
Union in-degree at the freeze	0.126 / 0.146	0.306 / 0.286	0.093 / 0.118	0.292 / 0.274
Contrasts ($\Delta\tau$ [95% CI]), capped / uncapped				
W0: C-PR minus C-PR ($\lambda = 1$), new approval pairs	-0.025 [-0.045, -0.004]		$+0.064$ [0.051, 0.076]	
W0: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.014 [-0.019, -0.008]		-0.019 [-0.027, -0.012]	
W1: C-PR minus C-PR ($\lambda = 0$), new approval pairs	$+0.074$ [0.060, 0.087]		$+0.088$ [0.074, 0.103]	
W1: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.016 [-0.022, -0.012]		-0.018 [-0.028, -0.011]	

Without the cap the allowance side has lower coefficients on later approvals and higher ones on later inflow. EndorseRank ranks new approval pairs at 0.088 ($[0.067, 0.107]$) in W0 and 0.078 ($[0.058, 0.099]$) in W1, about half its coefficients with the cap, and new senders at 0.185 ($[0.166, 0.202]$) and 0.177 ($[0.156, 0.197]$), more than twice those with the cap; the in-approve degree moves likewise. No paired contrast between the two layers was computed. C-PR moves little. The contrasts of the rules keep their sign except the contrast with the allowance layer walked alone on new approval pairs, which turns positive in W0 ($+0.064$, $[0.051, 0.076]$, against -0.025

with the cap) and in W1 ($+0.041$, $[0.031, 0.053]$, against -0.043), because the uncapped layer ranks them so much lower. On the uncapped layer C-PR is level with the union in-degree on new approval pairs ($+0.004$, $[-0.004, 0.012]$, in W0 and $+0.000$, $[-0.007, 0.008]$, in W1).

4.6 Discovery order (RQ2)

RQ2 asks whether a crawl ordered by C-PR on the contracts read so far reaches the contracts with the most user-account demand sooner than a breadth-first crawl and than a crawl ordered by the number of contract backlinks. The discovery-order test (Section 3.7) repeats the crawl-ordering experiment of Cho et al. (1998) on the contract graph. It was defined after the registered results were known and decides no rule of the analysis plan or the registration; its own decision rule is read in one run, at T_{obs} in the primary reveal mode. In that mode, `both`, a visit reveals a contract's in-edges and out-edges, and the universe $\mathcal{U}$ is the weakly connected component that holds the giant strongly connected component, $1{,}169{,}044$ contracts. The protocol's rule made it primary because the other universe, the giant strongly connected component with the contracts reachable from it (SCC$\cup$OUT), holds fewer than half as many contracts in the structure pilot at t_1 (Table 8.7 in Appendix 8.4.4). The pilot also raised the protocol's stop flag on a thin allowance layer (Section 3.7); what the allowance layer adds is read from the contrasts outside the decision. Importance is the demand of user accounts, which no ordering reads: the dollars they sent to a contract in the window (I_T) and the balance-capped dollars they had authorized it to spend at the freeze (I_A). The universe holds 96.5% of I_T and 93.0% of I_A (Table 8.8) and $16{,}776$ of the capped allowance pairs, so the pairs of the largest owner lie outside it. Its hot sets hold $13{,}141$ contracts by I_T and $11{,}662$ by I_A; each ordering crawls from 20 seeds, and $\text{AUC}_{10\%}$ measures how early it visits the hot contracts.

Table 4.12 gives the two decision contrasts. D_1, C-PR$'$ minus the breadth-first crawl, is $+0.0520$ ($[+0.0485, +0.0558]$) on I_T and $+0.0061$ ($[+0.0014, +0.0111]$) on I_A, both intervals above zero. D_2, C-PR$'$ minus the crawl ordered by contract backlinks, is -0.0051 ($[-0.0087, -0.0010]$) on I_T and -0.0571 ($[-0.0621, -0.0519]$) on I_A, both intervals below zero. The verdict is V2 on both

importances, with the sentence that an interval below zero adds, and the statement of the test reads: "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most user-account inflow and those with the most user-authorised value sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks. A crawl ordered by the number of contract backlinks reached them sooner." The answer to RQ2 is therefore no, and the lead over the breadth-first crawl is small on I_A.

Table 4.12: Discovery-order test: the two decision contrasts on each importance, $D_1 = \mathrm{AUC}_{10\%}(\text{C-PR}') - \mathrm{AUC}_{10\%}(\text{BFS})$ and $D_2 = \mathrm{AUC}_{10\%}(\text{C-PR}') - \mathrm{AUC}_{10\%}(\mathrm{IB}'_U)$, with 95% percentile intervals of a paired bootstrap over the hot contracts (400 resamples, seed 42) and the shares of resamples above 0 for D_1 and D_2. The decision is read at T_{obs} in the primary mode only. Its verdict on an importance is V1 when both intervals lie above 0, V2 when the interval of D_1 lies above 0 and that of D_2 does not, and V3 when the interval of D_1 does not lie above 0; an interval of D_2 below 0 adds that the crawl ordered by contract backlinks was sooner. The answer to the question of the test is the weaker of the two verdicts, so it is yes only when both are V1. The other three runs carry no verdict: for each importance the contrasts whose interval lies above 0 and those whose interval lies below 0 are listed. The two freezes share nine months of their score windows and are not independent. The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

Importance	$\lvert H_q \rvert$	D_1	95% interval	D_2	95% interval	Share above 0 (D_1 / D_2)	Reading
T_{obs}, `both` mode: the decision							
I_T	13,141	+0.0520	[+0.0485, +0.0558]	-0.0051	[-0.0087, -0.0010]	1.000 / 0.005	V2: sooner than BFS, not sooner than IB'_U; IB'_U sooner
I_A	11,662	+0.0061	[+0.0014, +0.0111]	-0.0571	[-0.0621, -0.0519]	0.998 / 0.000	V2: sooner than BFS, not sooner than IB'_U; IB'_U sooner
T_{obs}, `out` mode: no verdict							
I_T	5,087	+0.2650	[+0.2540, +0.2743]	+0.0803	[+0.0690, +0.0920]	1.000 / 1.000	above 0: D_1, D_2; below 0: none
I_A	5,088	+0.1143	[+0.1053, +0.1237]	+0.0436	[+0.0347, +0.0522]	1.000 / 1.000	above 0: D_1, D_2; below 0: none
t_1, `both` mode: no verdict							
I_T	12,697	+0.0574	[+0.0530, +0.0611]	-0.0158	[-0.0196, -0.0121]	1.000 / 0.000	above 0: D_1; below 0: D_2
I_A	8,136	+0.0230	[+0.0164, +0.0291]	-0.1010	[-0.1083, -0.0933]	1.000 / 0.000	above 0: D_1; below 0: D_2
t_1, `out` mode: no verdict							
I_T	3,324	+0.3414	[+0.3282, +0.3547]	+0.1222	[+0.1079, +0.1361]	1.000 / 1.000	above 0: D_1, D_2; below 0: none
I_A	3,324	+0.1705	[+0.1564, +0.1822]	+0.0739	[+0.0607, +0.0848]	1.000 / 1.000	above 0: D_1, D_2; below 0: none

Table 4.13 gives the level of every ordering and Figure 4.5 the curves. In the deciding run the breadth-first crawl has an $\mathrm{AUC}_{10\%}$ of 0.0565 on I_T and 0.0882 on I_A, C-PR′ 0.1085 and 0.0943, and IB'_U 0.1136 and 0.1514. AWP′ has the highest value of the crawls on both, 0.1465 and 0.1582, apart from the diagnostic δ'. Every crawl stays far from the ideal order, which reaches 0.9438 and 0.9501 (the random curve 0.0500). The partial view is not the only cause: ordered with the full graph in view, C-PR reaches 0.1061 and 0.1157 and the union in-degree 0.1593 and 0.1994.

Table 4.13: Discovery-order test: $\mathrm{AUC}_{10\%}$ of every ordering on the hot sets of the two importances, user-account inflow I_T and user-authorized value I_A, in the four runs. $\mathrm{AUC}_{10\%}$ is the mean over the hot contracts of $\max(0, 1 - t_h/\lceil 0.1|\mathcal{U}|\rceil)$, where $\mathcal{U}$ is the universe of the run and t_h the visit at which the crawl reaches hot contract h, averaged over 20 seeds from the giant strongly connected component; a crawl recomputes its ordering each time its visits have grown by the share $\rho = 1/50$. The first two columns (T_{obs}, `both` mode) are the run that the decision reads (Table 4.12). A primed name is the count or walk computed on the contracts read so far. δ' is a diagnostic outside every comparison. The references are not crawls: the orders by C-PR and by union in-degree of the full graph, the ideal order (hot contracts first) and the random curve. The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

	T_{obs}, `both`		T_{obs}, `out`		t_1, `both`		t_1, `out`	
Ordering	I_T	I_A	I_T	I_A	I_T	I_A	I_T	I_A
Breadth-first (BFS)	0.0565	0.0882	0.1179	0.0595	0.0543	0.1088	0.1269	0.0595
Random	0.0609	0.0725	0.1085	0.0621	0.0543	0.0930	0.1169	0.0703
IB'_A (allowance backlinks)	0.0626	0.0881	0.1438	0.0818	0.0539	0.1094	0.1750	0.1076
IB'_T (transfer backlinks)	0.1135	0.1466	0.3023	0.1233	0.1107	0.2001	0.3460	0.1478
IB'_U (contract backlinks)	0.1136	0.1514	0.3026	0.1302	0.1276	0.2328	0.3461	0.1561
EndorseRank′	0.0621	0.0869	0.1404	0.0802	0.0526	0.1062	0.1733	0.1068
EndorseRank-activity′	0.0612	0.1042	0.1404	0.0802	0.0587	0.1426	0.1734	0.1066
AWP′	0.1465	0.1582	0.4446	0.1753	0.1524	0.2220	0.4766	0.2231
C-PR′ ($\lambda = 1$)	0.0617	0.0864	0.1398	0.0789	0.0526	0.1057	0.1728	0.1063
C-PR′ ($\lambda = 0$)	0.1061	0.0806	0.4133	0.1584	0.1088	0.1097	0.4818	0.2020
C-PR′ ($\lambda = 0.25$)	0.1081	0.0930	0.4129	0.1835	0.1110	0.1297	0.4817	0.2349
C-PR′ ($\lambda = 0.5$)	0.1085	0.0943	0.3829	0.1738	0.1118	0.1319	0.4683	0.2300
C-PR′ ($\lambda = 0.75$)	0.1091	0.0952	0.3744	0.1608	0.1121	0.1323	0.4562	0.2197
S-PR′	0.1069	0.0861	0.4190	0.1805	0.1095	0.1170	0.4874	0.2360
δ' (diagnostic, outside the comparisons)	0.1643	0.1942	0.5357	0.2317	0.1775	0.2778	0.5795	0.2937
Reference: C-PR of the full graph	0.1061	0.1157	0.2449	0.1279	0.1165	0.1646	0.3239	0.1773
Reference: union in-degree of the full graph	0.1593	0.1994	0.4600	0.2142	0.1757	0.3151	0.5826	0.2846
Reference: ideal order	0.9438	0.9501	0.9500	0.9500	0.9375	0.9600	0.9500	0.9500
Reference: random curve	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500

Table 4.14 gives the contrasts outside the decision in the order the protocol fixes. First, each walk is set against the count it smooths. AWP′ is the one single-layer walk above its count on both importances ($+0.0329$, $[+0.0298, +0.0368]$, and $+0.0116$, $[+0.0071, +0.0162]$, over IB'_T). The walks on the allowance layer alone lie at or below IB'_A, except EndorseRank-activity′ on I_A; C-PR′ ($\lambda = 0$) lies below IB'_T; and C-PR′ lies above IB'_A and below IB'_T. Second, among the walks, the allowance layer adds a small, resolved amount to C-PR′ over its transfer layer walked alone

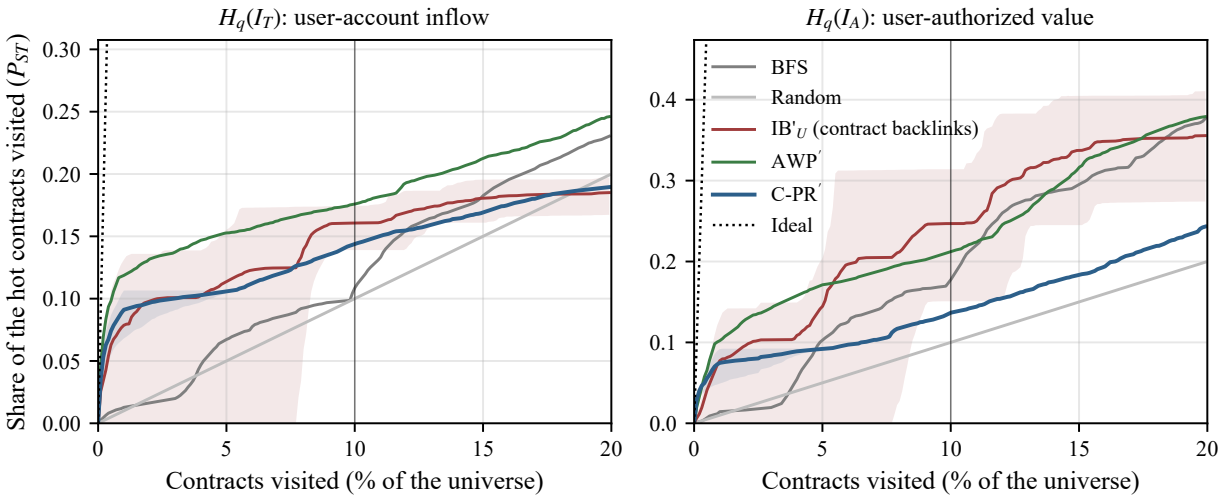


Figure 4.5: Discovery-order test at T_{obs} in the primary mode: the share of the hot contracts visited, P_{ST}, against the contracts visited as a percentage of the universe, for the hot sets of user-account inflow I_T (left) and of user-authorized value I_A (right). The curves are means over the 20 seeds for the breadth-first and random orderings, the backlink count IB'_U, AWP′ and C-PR′; the shaded bands give the minimum and maximum over the seeds for IB'_U and C-PR′, the dotted line is the ideal order, and the vertical line marks 10% of the universe, the range of $\text{AUC}_{10\%}$. The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

($+0.0024$, $[+0.0016, +0.0033]$, and $+0.0137$, $[+0.0117, +0.0156]$), and C-PR′ lies below AWP′ (-0.0379, $[-0.0412, -0.0347]$, and -0.0639, $[-0.0683, -0.0591]$). Third, $\lambda = 0.25$ and 0.75 give the same reading. Fourth, on hot sets defined on the edges that the orderings read, each ordering finds its own target, an intended-construct check: on the hot set of the union in-degree the backlink crawl is far sooner than C-PR′ (-0.3031, $[-0.3106, -0.2958]$), and on the hot set of full-graph C-PR, C-PR′ is far sooner ($+0.2606$, $[+0.2544, +0.2663]$). The first of these repeats the comparison of Cho et al. (1998), in which the PageRank ordering was sooner; here it goes the other way.

Fifth, the sensitivities change single cells and not the reading that C-PR′ does not beat the backlink count (Table 8.9). No row puts the interval of D_2 above zero. D_1 on I_A, the smallest decision contrast, does not hold throughout: it lies below zero with the area taken over 20% of the universe and under a schedule that recomputes four times as often, and its interval contains zero without the hubs and under a bootstrap over the seeds. The other reveal mode belongs to these sensitivities and carries no verdict. In the `out` mode, where a visit reveals out-edges only and the universe is

Table 4.14: Discovery-order test at T_{obs} in the primary mode: the contrasts reported outside the decision, in the order fixed in the protocol. Each cell is the difference of $\text{AUC}_{10\%}$ between two orderings with its 95% percentile intervals of a paired bootstrap over the hot contracts (400 resamples, seed 42); an interval above 0 means the first ordering reached the hot contracts sooner. Block 1 sets each walk beside the count it smooths; block 2 compares the walks, first C-PR′ with its transfer layer walked alone, C-PR′ ($\lambda = 0$), and with AWP′, which shows what the allowance layer adds. In block 4 the hot sets are defined on the edges that the orderings read, the union in-degree IB_U and C-PR of the full graph (IR), so agreement there is an intended-construct check; C-PR′ minus IB'_U on $H_q(\text{IB}_U)$ is the contrast that repeats the comparison of Cho et al. The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

Contrast	On $H_q(I_T)$	On $H_q(I_A)$
1. Each walk′ against the count it smooths		
EndorseRank′ − IB'_A	-0.0006 [-0.0007, -0.0004]	-0.0012 [-0.0014, -0.0010]
EndorseRank-activity′ − IB'_A	-0.0014 [-0.0030, +0.0001]	+0.0161 [+0.0136, +0.0185]
C-PR′ ($\lambda = 1$) − IB'_A	-0.0009 [-0.0010, -0.0007]	-0.0017 [-0.0019, -0.0015]
AWP′ − IB'_T	+0.0329 [+0.0298, +0.0368]	+0.0116 [+0.0071, +0.0162]
C-PR′ ($\lambda = 0$) − IB'_T	-0.0074 [-0.0109, -0.0031]	-0.0660 [-0.0709, -0.0609]
C-PR′ − IB'_A	+0.0459 [+0.0427, +0.0492]	+0.0062 [+0.0023, +0.0106]
C-PR′ − IB'_T	-0.0050 [-0.0086, -0.0008]	-0.0523 [-0.0575, -0.0468]
2. Among the walks		
C-PR′ − C-PR′ ($\lambda = 0$)	+0.0024 [+0.0016, +0.0033]	+0.0137 [+0.0117, +0.0156]
C-PR′ − AWP′	-0.0379 [-0.0412, -0.0347]	-0.0639 [-0.0683, -0.0591]
C-PR′ − C-PR′ ($\lambda = 1$)	+0.0468 [+0.0436, +0.0501]	+0.0078 [+0.0040, +0.0123]
C-PR′ − EndorseRank′	+0.0465 [+0.0433, +0.0498]	+0.0074 [+0.0035, +0.0117]
S-PR′ − C-PR′ ($\lambda = 0$)	+0.0008 [+0.0006, +0.0011]	+0.0054 [+0.0044, +0.0066]
3. Other values of λ		
C-PR′ ($\lambda = 0.25$) − BFS	+0.0515 [+0.0480, +0.0555]	+0.0049 [+0.0002, +0.0100]
C-PR′ ($\lambda = 0.25$) − IB'_U	-0.0055 [-0.0091, -0.0016]	-0.0584 [-0.0633, -0.0531]
C-PR′ ($\lambda = 0.25$) − C-PR′ ($\lambda = 0.5$)	-0.0005 [-0.0008, -0.0001]	-0.0012 [-0.0017, -0.0008]
C-PR′ ($\lambda = 0.75$) − BFS	+0.0526 [+0.0491, +0.0563]	+0.0070 [+0.0024, +0.0121]
C-PR′ ($\lambda = 0.75$) − IB'_U	-0.0045 [-0.0080, -0.0005]	-0.0562 [-0.0610, -0.0508]
C-PR′ ($\lambda = 0.75$) − C-PR′ ($\lambda = 0.5$)	+0.0006 [+0.0002, +0.0010]	+0.0009 [+0.0005, +0.0014]
4. Hot sets in the manner of Cho et al. (an intended-construct check)		
C-PR′ − BFS on $H_q(\text{IB}_U)$	+0.0176 [+0.0119, +0.0238]	
C-PR′ − IB'_U on $H_q(\text{IB}_U)$	-0.3031 [-0.3106, -0.2958]	
C-PR′ − BFS on $H_q(\text{IR})$	+0.3084 [+0.3024, +0.3143]	
C-PR′ − IB'_U on $H_q(\text{IR})$	+0.2606 [+0.2544, +0.2663]	

SCC ∪ OUT, all four contrasts lie above zero at both freezes (Table 4.12): at T_{obs} D_2 is $+0.0803$ ($[+0.0690, +0.0920]$) on I_T and $+0.0436$ ($[+0.0347, +0.0522]$) on I_A, and at t_1 it is larger. The result therefore depends on what a visit reveals (Figure 8.1). Sixth, at t_1 in the primary mode the intervals of D_1 lie above zero and those of D_2 below zero on both importances, as at T_{obs}; the two freezes share nine months of their score windows and are not independent.

The diagnostics show how the crawls differ (Table 8.10). The count orderings decide nearly every pick by a tie (0.9998 to 0.9999 of the picks), as do the walks on the allowance layer alone, and the order of discovery breaks a tie, as in a breadth-first crawl. C-PR′ decides 0.535 of its picks by a tie in the primary mode and 0.023 in the `out` mode, AWP′ 0.252 and 0.005. Under every walk and count, 0.84 to 0.91 of the first tenth of the visits go to contracts with no user-account demand.

4.7 Temporal holdout of the single-layer walks (RQ3)

The same-window check cannot separate a score from its own construct, and the discovery-order test reads importance in the score window; the out-of-window evaluation tests whether a score frozen at a cut-off predicts what happens after it. Following Section 3.8, scores on the W0 spender cohort are frozen at t_1, and two labels are counted for each contract between 1 April and 30 June 2026: new owner–spender approval pairs and new transfer senders. Table 4.15 sets every score against both labels, with four raw counts as baselines: the in-approve degree and value, the transfer in-degree and the union in-degree. Table 4.16 gives the paired contrasts: the two of rule A, fixed in the analysis plan, and those computed after the labels were known. This section reads each walk against the count it smooths; rule A and the other rows of C-PR and S-PR are taken up in Section 4.8.

Two facts frame every out-of-window number. The labels count every account: 90.4% of the $748{,}258$ new approval pairs of W0 come from user accounts, and 86.3% of its $651{,}474$ new senders are user accounts (in W1 87.9% and 82.3%; Table 8.6 in Appendix 8.4.3), and no score reads an edge of a user account. And 27.5% of the W0 spenders and 27.2% of the W1 spenders lie outside

the union graph and score zero under every score and count (Table 4.2). Both labels are also sparse: of the 7,864 spenders of W0, 1,725 gained a new approval pair and 1,416 a new sender. Each label is one kind of event carried into the future, so these results show whether a score predicts more of the activity it summarizes and not whether a contract is safe.

Table 4.15: Temporal holdout W0 on the spender cohort ($n = 7{,}864$ contracts): Kendall τ between scores frozen at 31 March 2026 on the contract graph and two labels counted on the contract between 1 April 2026 and 30 June 2026: the number of new owner–spender approval pairs and the number of accounts that sent a transfer to the contract for the first time. The cohort keeps the 7,864 contracts among the 7,962 spenders of the labels; 97 user accounts and 1 system address are left out. 1,725 contracts received at least one new pair and 1,416 at least one new sender. 5,700 of the cohort (72.5%) lie in the union graph; a contract outside a graph scores zero. Baseline rows are raw counts on the contract graph at t_1: the in-approve degree and value and the transfer in-degree that the single-layer walks smooth, and the union in-degree, the number of distinct contracts with an edge to the contract in either layer. C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$) are the allowance and transfer layers of the coupled operator walked alone. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.204 [0.181, 0.228]	0.091 [0.066, 0.113]
In-approve value at t_1 (baseline)	0.189 [0.166, 0.213]	0.081 [0.057, 0.103]
Transfer in-degree at t_1 (baseline)	0.070 [0.050, 0.091]	0.318 [0.298, 0.336]
Union in-degree at t_1 (baseline)	0.126 [0.108, 0.146]	0.306 [0.287, 0.324]
EndorseRank	0.168 [0.145, 0.193]	0.084 [0.062, 0.105]
AWP	0.098 [0.079, 0.119]	0.330 [0.313, 0.347]
C-PR ($\lambda = 1$)	0.167 [0.145, 0.192]	0.084 [0.062, 0.105]
C-PR ($\lambda = 0$)	0.070 [0.052, 0.091]	0.268 [0.250, 0.285]
C-PR ($\lambda = 0.25$)	0.141 [0.123, 0.161]	0.256 [0.238, 0.273]
C-PR ($\lambda = 0.5$)	0.142 [0.124, 0.162]	0.254 [0.235, 0.270]
C-PR ($\lambda = 0.75$)	0.142 [0.124, 0.162]	0.251 [0.232, 0.267]
S-PR	0.091 [0.074, 0.113]	0.264 [0.246, 0.283]

The first finding answers the second part of RQ3 for the single-layer walks, and it differs by layer. On the allowance side the count wins. The in-approve degree at t_1 ranks new approval pairs at $\tau = 0.204$ ($[0.181, 0.228]$) and EndorseRank at 0.168 ($[0.145, 0.193]$), a paired difference of -0.036 ($[-0.046, -0.027]$); C-PR ($\lambda = 1$) lies as far below the count (-0.037, $[-0.046, -0.028]$), and so does EndorseRank with activity restarts (-0.041, $[-0.053, -0.029]$). On the transfer side AWP ranks new transfer senders at 0.330 ($[0.313, 0.347]$), slightly above the transfer in-degree at 0.318

Table 4.16: Paired bootstrap differences on the spender holdout W0 ($n = 7{,}864$ contracts; 400 resamples shared by a and b, seed 42). The first block holds the two contrasts of the primary part of rule A, fixed in the analysis plan before the study extracted any data and applied here to the scores of the contract graph (C-PR against C-PR ($\lambda = 1$) on new approval pairs and against C-PR ($\lambda = 0$) on new transfer senders). The second block was computed after the labels were known: the increment of C-PR ($\lambda = 1$), the allowance layer walked alone, over its raw degree at t_1, the same comparisons for S-PR, and the increments of EndorseRank and AWP over their raw degrees; C-PR's other contrasts are in Table 4.22. The third block sets C-PR beside the union in-degree. Primary part of rule A (C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders): no worse fails, better fails, so the rule is not met. The differences between EndorseRank and AWP are in Table 8.15.

Contrast (a minus b)	Label	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Rule A, primary part, fixed in the plan						
C-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.142	0.167	-0.025	[-0.045, -0.004]	0.015
C-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.254	0.268	-0.014	[-0.019, -0.008]	0.003
Computed after the labels were known						
C-PR ($\lambda = 1$) minus t_1 in-approve degree	new approval pairs	0.167	0.204	-0.037	[-0.046, -0.028]	0.000
S-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.091	0.167	-0.077	[-0.098, -0.055]	0.000
S-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.264	0.268	-0.003	[-0.013, 0.008]	0.285
EndorseRank minus t_1 in-approve degree	new approval pairs	0.168	0.204	-0.036	[-0.046, -0.027]	0.000
AWP minus t_1 in-degree	new transfer senders	0.330	0.318	+0.012	[0.003, 0.023]	0.998
C-PR against the union in-degree, the baseline of the contract graph						
C-PR minus t_1 union in-degree	new approval pairs	0.142	0.126	+0.016	[0.007, 0.024]	1.000
C-PR minus t_1 union in-degree	new transfer senders	0.254	0.306	-0.053	[-0.061, -0.043]	0.000

($[0.298, 0.336]$): the paired difference is $+0.012$ ($[0.003, 0.023]$). The transfer layer walked with uniform restarts lies lower, C-PR ($\lambda = 0$) at 0.268 ($[0.250, 0.285]$). These degree contrasts were computed after the labels were known. The coupled walk smooths the union in-degree, the count of the edges of both layers, and against it the result is split: C-PR ranks new approval pairs higher ($+0.016$, $[0.007, 0.024]$) and new senders lower (-0.053, $[-0.061, -0.043]$).

AWP's lead holds among the W0 spenders inside the union graph ($+0.021$, $[0.005, 0.038]$) but not among the contracts that the count can rank by their inflow. Table 4.17 restricts the holdout to the 4,569 spenders that another contract had paid before the freeze. There AWP ranks new senders below the transfer in-degree (0.411, $[0.389, 0.429]$, against 0.446, $[0.423, 0.467]$; -0.035, $[-0.049, -0.022]$), and EndorseRank stays below the in-approve degree on new approval pairs (-0.045, $[-0.058, -0.033]$). The two results for AWP are compatible: the transfer in-degree is zero for the other 3,295 spenders, whereas AWP, which restarts at senders, is zero only for the

2,421 without a transfer pair and sets the remaining 874 above them. On new approval pairs AWP reaches 0.238 ($[0.219, 0.260]$) on the subset, above the point estimates of EndorseRank (0.167, $[0.139, 0.197]$) and of the in-approve degree (0.212, $[0.183, 0.242]$), so the lead of the allowance side on this label rests on the spenders outside the subset. The restriction was chosen after the results were known. No paired contrast was computed for the three estimates on the subset, for the 3,295 alone, or for C-PR ($\lambda = 0$) against its count.

Table 4.17: Spender holdout W0 restricted to the 4,569 of the 7,864 spender contracts that had received a transfer from another contract before the freeze ($n = 4{,}569$). The other 3,295 have no inbound transfer edge from another contract, and 2,421 spenders score zero under AWP. The restriction was chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42). [a]Computed afterwards from the saved scores, with the same resamples; its contrasts with AWP and the in-approve degree on this subset are in Table 4.22.

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
EndorseRank	0.167 [0.139, 0.197]	0.068 [0.040, 0.094]
AWP	0.238 [0.219, 0.260]	0.411 [0.389, 0.429]
C-PR ($\lambda = 0.5$)[a]	0.264 [0.244, 0.286]	0.327 [0.306, 0.347]
In-approve degree at t_1	0.212 [0.183, 0.242]	0.104 [0.076, 0.133]
Transfer in-degree at t_1	0.220 [0.196, 0.243]	0.446 [0.423, 0.467]
EndorseRank minus t_1 in-approve degree (new approval pairs)	$\Delta\tau = -0.045$ [-0.058, -0.033]	
AWP minus t_1 in-degree (new transfer senders)	$\Delta\tau = -0.035$ [-0.049, -0.022]	

Read across the whole spender cohort, each label is ranked best by the scores and counts of its own edge type. On new approval pairs the in-approve degree leads, followed by the in-approve value, EndorseRank and C-PR ($\lambda = 1$), then the C-PR mixtures and the union in-degree, and the transfer side stays lowest (AWP 0.098, $[0.079, 0.119]$). On new transfer senders AWP, the transfer in-degree and the union in-degree lead, and the allowance side stays below 0.10 (EndorseRank 0.084, $[0.062, 0.105]$). The reach of the layers is part of this. EndorseRank scores $5{,}966$ of the $7{,}864$ spenders zero and AWP $2{,}421$; the spenders without a transfer pair gained new approval pairs about as often as the rest (23.4% against 21.3%) and new senders far less often (7.9% against 22.5%), so AWP's zeros cost it little on the first label and help it on the second. Within a three-month window a spender gains more new owners and senders the more it already has, as preferential attachment would predict (Barabási & Albert, 1999), and a local count captures this directly.

Two other candidate labels are reported without being offered as support. Revocation counts, which for some tokens include allowances spent in full (Section 3.8), and the drain heuristic both rise with every score, most steeply with the raw in-approve degree ($\tau = 0.190$, $[0.161, 0.220]$, and 0.213, $[0.188, 0.238]$). For an endorsement reading the sign is wrong, since a trusted spender should lose approvals less often; a spender with many owners sees more of everything owners do, so these labels cannot separate trust from use.

4.7.1 The single-layer walks in the registered window W1

The registered window W1 (Section 3.8.1) freezes the scores at T_{obs}, 30 June 2026, and counts the same two labels from July to September 2026; rule B, which concerns C-PR, is reported in Section 4.8.3. Of the 7,915 spender contracts, 1,491 gained a new approval pair in the label window and 1,369 a new sender. Table 4.18 gives EndorseRank and its variant with activity restarts, which the registration does not name, beside AWP, C-PR and the two degrees. None of its rows enters a rule, and its contrasts were computed after the labels were known.

The W0 pattern recurs in this window, on an allowance layer of another shape (Section 4.1). The in-approve degree at the freeze again ranks new approval pairs above EndorseRank (0.181, $[0.158, 0.208]$, against 0.147, $[0.125, 0.170]$; -0.034, $[-0.043, -0.025]$), and with activity restarts EndorseRank lies as far below it (-0.033, $[-0.044, -0.022]$). AWP again ranks new senders slightly above the transfer in-degree (0.314, $[0.295, 0.332]$, against 0.302, $[0.278, 0.323]$; $+0.011$, $[0.001, 0.022]$); the restriction to the spenders that another contract had paid was not computed for this contrast in W1. C-PR again lies above the union in-degree on new approval pairs ($+0.011$, $[0.003, 0.019]$) and below it on new senders (-0.056, $[-0.065, -0.047]$; Table 4.22).

4.8 The coupled walk C-PR (RQ4, RQ5)

C-PR, the method of this thesis, walks both US-dollar layers and mixes their transition rows at each node with weight λ (Section 3.6.3). The analysis plan fixed one decision rule, and its object is C-PR:

Table 4.18: EndorseRank and its variant with activity restarts on the spender cohort and labels of window W1 ($n = 7{,}915$ contracts). Neither these rows nor the contrasts below them belong to the two applied criteria of rule B, and the contrasts were not named in the registration. The contrasts of EndorseRank with activity restarts were computed afterwards from the saved W1 scores. The AWP, C-PR and degree rows repeat Table 4.20, and every contrast uses the same contract resamples; the differences between EndorseRank and AWP are in Table 8.15, and those of C-PR in Table 4.22. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at 30 June 2026	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at the freeze (baseline)	0.181 [0.158, 0.208]	0.072 [0.046, 0.096]
Transfer in-degree at the freeze (baseline)	0.031 [0.008, 0.052]	0.302 [0.278, 0.323]
EndorseRank	0.147 [0.125, 0.170]	0.064 [0.041, 0.085]
EndorseRank, restarts by approving activity	0.148 [0.126, 0.171]	0.072 [0.049, 0.093]
AWP	0.064 [0.042, 0.084]	0.314 [0.295, 0.332]
C-PR ($\lambda = 0.5$)	0.104 [0.083, 0.126]	0.236 [0.217, 0.255]
EndorseRank minus in-approve degree, new approval pairs	$\Delta\tau = -0.034$ [-0.043, -0.025]	
AWP minus transfer in-degree, new transfer senders	$\Delta\tau = +0.011$ [0.001, 0.022]	
EndorseRank (activity restarts) minus in-approve degree, new approval pairs	$\Delta\tau = -0.033$ [-0.044, -0.022]	
EndorseRank (activity restarts) minus EndorseRank, new approval pairs	$\Delta\tau = +0.001$ [-0.002, 0.005]	
EndorseRank (activity restarts) minus EndorseRank, new transfer senders	$\Delta\tau = +0.008$ [0.004, 0.012]	

rule A asks the coupled walk to beat each of its layers walked alone (RQ4). The registration of W1 added rule B, which asks it to improve on its transfer layer on new data (RQ5). Both rules compare C-PR with its own layers, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), because only that contrast changes the coupling and nothing else; C-PR also differs from EndorseRank and AWP in its amount weights and its uniform restarts. Both are applied to the scores of the contract graph as a re-analysis. The contrasts of C-PR with every other score and count, at the end of the section, decide no rule.

4.8.1 The coupled walk within the window

Table 4.19 gives the paired contrasts in this window of C-PR and its ablation S-PR, which keeps the transfer layer but restarts from the allowance walk (Section 3.6.3): first the eight of Section 3.11 against the two layers walked alone, then six against EndorseRank and AWP, two of them computed afterwards from the saved scores. RQ4 and RQ5 are decided out of window; this window only locates the hybrids.

C-PR lies between its two layers on every family and stays nearer to its transfer layer. At the primary setting it reaches 0.738 ($[0.729, 0.747]$) on the transfer family, below C-PR ($\lambda = 0$) by

Table 4.19: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the two layers they are built from, the allowance layer (C-PR ($\lambda = 1$)) and the transfer layer (C-PR ($\lambda = 0$)), and of C-PR against EndorseRank and AWP (matched cohort, $n = 6{,}443$ contracts). Family-mean τ; the same contract resamples are used for a and b. The contrasts with the layers are a stricter check beside the secondary part of rule A, which compares C-PR's point estimate with the better layer's interval in Table 4.5. The secondary part of rule A, fixed in the analysis plan and applied here to the scores of the contract graph, is not met: C-PR's point estimate lies outside the interval of C-PR ($\lambda = 0$) on the transfer family and outside the interval of C-PR ($\lambda = 1$) on the allowance family, and C-PR does not exceed both layers on the Sybil-stability family. An interval below 0 means a is below b beyond sampling variation; one above 0 means it exceeds it. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42). [a]Computed afterwards from the saved scores, with the same resamples: C-PR against EndorseRank on the transfer family and against AWP on the allowance family, the family that each single-layer walk follows least.

Contrast (a minus b)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Against the two layers walked alone					
C-PR transfer minus C-PR ($\lambda = 0$) transfer	0.738	0.804	-0.066	[-0.074, -0.058]	0.000
C-PR allowance minus C-PR ($\lambda = 1$) allowance	0.433	0.838	-0.406	[-0.418, -0.393]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.574	0.637	-0.063	[-0.071, -0.053]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.574	0.222	+0.352	[0.340, 0.365]	1.000
S-PR transfer minus C-PR ($\lambda = 0$) transfer	0.782	0.804	-0.022	[-0.030, -0.013]	0.000
S-PR allowance minus C-PR ($\lambda = 1$) allowance	0.371	0.838	-0.467	[-0.488, -0.446]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.584	0.637	-0.053	[-0.058, -0.048]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.584	0.222	+0.362	[0.346, 0.379]	1.000
Against EndorseRank and AWP					
C-PR transfer minus AWP transfer	0.738	0.755	-0.017	[-0.028, -0.006]	0.000
C-PR allowance minus EndorseRank allowance	0.433	0.839	-0.406	[-0.419, -0.393]	0.000
C-PR sybil-stability minus AWP sybil-stability	0.574	0.643	-0.069	[-0.077, -0.059]	0.000
C-PR sybil-stability minus EndorseRank sybil-stability	0.574	0.220	+0.353	[0.341, 0.367]	1.000
C-PR transfer minus EndorseRank transfer[a]	0.738	0.297	+0.441	[0.423, 0.457]	1.000
C-PR allowance minus AWP allowance[a]	0.433	0.227	+0.206	[0.188, 0.224]	1.000

$\Delta\tau = -0.066$ ($[-0.074, -0.058]$), and 0.433 ($[0.419, 0.445]$) on the allowance family, below C-PR ($\lambda = 1$) by $\Delta\tau = -0.406$ ($[-0.418, -0.393]$). The gap on the allowance family is in part one of reach. C-PR gives a positive score to the $4{,}270$ cohort contracts in the union graph, more than four times the 926 that the allowance layer reaches as spenders, and so orders by their transfers more than three thousand contracts that the allowance proxies tie at zero, which τ_b counts against it. AWP ranks the transfer family better than C-PR (-0.017, $[-0.028, -0.006]$), and EndorseRank the allowance family (-0.406, $[-0.419, -0.393]$). On the family that each single-layer walk follows

least, C-PR leads that walk: it ranks the transfer family 0.441 above EndorseRank ($[0.423, 0.457]$) and the allowance family 0.206 above AWP ($[0.188, 0.224]$). A node with out-edges in only one layer follows that layer whatever λ is (Equation (3.9)); λ acts only at the nodes with out-edges in both layers, and at T_{obs} only $9{,}459$ contracts are owners in the allowance layer.

On every family S-PR falls below the layer that leads that family. Restarting the transfer walk from the allowance walk lowers the transfer coefficient to 0.782 ($[0.773, 0.791]$; $\Delta\tau = -0.022$, $[-0.030, -0.013]$, against C-PR ($\lambda = 0$)) and raises the allowance coefficient only as far as 0.371 ($[0.350, 0.391]$; $\Delta\tau = -0.467$, $[-0.488, -0.446]$, against C-PR ($\lambda = 1$)). A restart vector changes where the walk lands after each teleport, and it carries only part of the allowance signal into the transfer walk.

As fixed in the analysis plan, the secondary part of rule A asks C-PR's point estimate to lie within the 95% interval of the better layer on the transfer and allowance families, and C-PR to exceed both layers on Sybil-adjusted stability. At $\lambda = 0.5$, C-PR fails all three parts, as in the registered analysis (Appendix 8.6). Its transfer coefficient (0.738) lies below the interval $[0.799, 0.809]$ of C-PR ($\lambda = 0$), its allowance coefficient (0.433) below the interval $[0.827, 0.850]$ of C-PR ($\lambda = 1$), and on Sybil-adjusted stability it exceeds the allowance layer ($\Delta\tau = +0.352$, $[0.340, 0.365]$) and not the transfer layer (-0.063, $[-0.071, -0.053]$). The paired contrasts of Table 4.19, reported alongside the rule as a stricter check, point the same way: the mixture gives up part of each layer's agreement with its own family and exceeds the better layer on no family.

4.8.2 Rule A in the window W0

Rule A is not met out of window either. The rule asked C-PR to do no worse than C-PR ($\lambda = 1$) on new approval pairs and to beat C-PR ($\lambda = 0$) on new transfer senders in W0 (first block of Table 4.16). At $\lambda = 0.5$, C-PR lies below the allowance layer on new approval pairs ($\Delta\tau = -0.025$, interval $[-0.045, -0.004]$), so the first condition fails, and below the transfer layer on new transfer senders ($\Delta\tau = -0.014$, interval $[-0.019, -0.008]$), so the second fails as well. In the registered analysis the rule was not met either: its primary part failed through its second condition alone, and

its secondary part failed (Table 8.16).

Three comparisons outside the rule qualify the first failure. Against its transfer layer walked alone C-PR ranks new approval pairs higher, 0.142 ($[0.124, 0.162]$) against 0.070 ($[0.052, 0.091]$), a paired difference of $+0.071$ ($[0.059, 0.084]$); this contrast, computed after the labels were known, is the one that the registration of W1 made its criterion F1. Inside the union graph the contrast with the allowance layer does not exclude zero ($+0.020$, $[-0.002, 0.043]$; Table 4.23): on the whole cohort the contrast also reflects that C-PR gives a positive score to the $5{,}700$ spenders of the union graph where its allowance layer reaches $1{,}901$. And against the uncapped allowance layer walked alone C-PR ranks new approval pairs higher ($+0.064$; Section 4.5.5). None of the three decides the rule. The second failure does not depend on these choices: C-PR lies below its transfer layer on new senders inside the union graph (-0.016, $[-0.025, -0.007]$) and with the uncapped layer (-0.019, $[-0.027, -0.012]$). Across the three values of λ the coefficients of C-PR differ by at most 0.005 on either label in either window (Tables 4.15 and 4.20), and λ was not tuned.

The seeded ablation shows where the allowance signal has to sit. S-PR is level with the transfer layer on new senders ($\Delta\tau = -0.003$, interval $[-0.013, 0.008]$) and falls below the allowance layer on new approval pairs ($\Delta\tau = -0.077$, $[-0.098, -0.055]$), which it ranks at 0.091 ($[0.074, 0.113]$) against 0.070 for the transfer layer; C-PR ranks them 0.051 above S-PR ($[0.038, 0.064]$). When transfer edges decide where the walk goes, a restart distribution derived from allowances changes little. The allowance signal reaches the ranking when allowance edges carry the walk.

4.8.3 Rule B in the registered window W1

The registration of Section 3.8.1 was committed to the authors' Git repository after the W0 results of the registered analysis were known and before the study extracted the July–September logs. It fixed rule B with the margin $\delta = 0.01$, the half-width of the W0 interval of the F2 contrast in the registered analysis (0.0040), rounded up. Of rule B the thesis applies two of its registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds. F1 asks C-PR to rank new approval pairs above its transfer layer walked alone, with an interval above zero, and F2

asks it not to fall below that layer on new transfer senders by more than δ. Both are applied to the scores of the contract graph as a re-analysis, beside F6, the primary part of rule A on this window, and the spender parts of the two registered sensitivity analyses. Table 4.20 gives every score on the two labels of the spender cohort of Section 4.7.1, and Table 4.21 the contrasts with their verdicts.

Table 4.20: Window W1: Kendall τ between scores frozen at 30 June 2026 on the contract graph and the two spender labels, counted from 1 July 2026 to 30 September 2026. The spenders are the $n = 7{,}915$ contracts among the 7,995 spenders of the labels (79 user accounts and 1 system address are left out); 1,491 of them gained a new approval pair and 1,369 a new sender, and 5,759 (72.8%) lie in the union graph. C-PR ($\lambda = 0$) is the comparator of the two applied criteria of rule B and AWP that of the first sensitivity analysis (Table 4.21); EndorseRank is in Table 4.18. The baselines are raw counts on the contract graph at the freeze. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at 30 June 2026	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at the freeze (baseline)	0.181 [0.158, 0.208]	0.072 [0.046, 0.096]
In-approve value at the freeze (baseline)	0.163 [0.141, 0.188]	0.062 [0.037, 0.084]
Transfer in-degree at the freeze (baseline)	0.031 [0.008, 0.052]	0.302 [0.278, 0.323]
Union in-degree at the freeze (baseline)	0.093 [0.071, 0.114]	0.292 [0.270, 0.312]
AWP	0.064 [0.042, 0.084]	0.314 [0.295, 0.332]
C-PR ($\lambda = 1$)	0.147 [0.124, 0.170]	0.064 [0.041, 0.084]
C-PR ($\lambda = 0$)	0.030 [0.008, 0.050]	0.253 [0.232, 0.272]
C-PR ($\lambda = 0.25$)	0.103 [0.081, 0.124]	0.239 [0.219, 0.258]
C-PR ($\lambda = 0.5$)	0.104 [0.083, 0.126]	0.236 [0.217, 0.255]
C-PR ($\lambda = 0.75$)	0.104 [0.082, 0.126]	0.234 [0.215, 0.252]
S-PR	0.055 [0.034, 0.073]	0.242 [0.222, 0.261]

Rule B is not met. F1 holds: C-PR ranks the spenders that later gained new approval pairs at $\tau = 0.104$ ($[0.083, 0.126]$) against 0.030 ($[0.008, 0.050]$) for its transfer layer walked alone, a difference of $+0.074$ (interval $[0.060, 0.087]$), so the W0 gain recurs. F2 fails. On new transfer senders C-PR reaches 0.236 ($[0.217, 0.255]$) against 0.253 ($[0.232, 0.272]$); the difference is -0.016 ($[-0.022, -0.012]$), and the whole interval lies below the margin of -0.01. The thesis therefore makes no claim that C-PR improves on a walk over the transfers alone. The registered analysis gave the same reading of both criteria (Table 8.16). F1 shows a gain over the transfer layer and not over the approval count: the in-approve degree at the freeze ranks new approval pairs at 0.181, and C-PR lies below it in both windows (Section 4.8.4).

The results reported beside the two criteria agree. F6 fails in both of its parts: C-PR lies below its

Table 4.21: Window W1: the paired contrasts F1 and F2, two of the registered criteria of rule B, with their verdicts on the scores of the contract graph, and the spender parts of the two registered sensitivity analyses. Under the registration the rule holds only if each of its criteria holds; this analysis applies F1 and F2 only, so its reading is "not met" when one of them fails and "both applied criteria hold" when both hold. The criteria and the margin are those of the registration of W1; they are applied here as a re-analysis after the registered results were known. $\Delta\tau = \tau(a) - \tau(b)$, with C-PR ($\lambda = 0.5$) as a. Superiority holds when the interval lies above zero; non-inferiority holds when its lower end lies above $-\delta = -0.01$; F6a, the first part of rule A, holds when the interval contains zero or lies above it. Every contrast is on the spenders ($n = 7{,}915$ contracts) of Table 4.20. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Contrast	Label	Test	$\tau(a)$	$\tau(b)$	$\Delta\tau$ [95% CI]	Verdict
F1 and F2, two of rule B's registered criteria: C-PR ($\lambda = 0.5$) against its transfer layer walked alone, C-PR ($\lambda = 0$)							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.104	0.030	+0.074 [0.060, 0.087]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.236	0.253	-0.016 [-0.022, -0.012]	fails
Reading of F1 and F2							not met
Reported beside them							
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.104	0.147	-0.043 [-0.061, -0.024]	fails
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.236	0.253	-0.016 [-0.022, -0.012]	fails
Sensitivity analysis 1: AWP as the comparator							
F1	C-PR vs. AWP	new approval pairs	superiority	0.104	0.064	+0.040 [0.026, 0.055]	holds
F2	C-PR vs. AWP	new transfer senders	non-inferiority, -0.01	0.236	0.314	-0.077 [-0.086, -0.069]	fails
Reading of F1 and F2							not met
Sensitivity analysis 2: contracts outside a graph kept as isolated nodes							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.111	0.038	+0.073 [0.059, 0.086]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.238	0.253	-0.016 [-0.022, -0.011]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.111	0.173	-0.062 [-0.081, -0.041]	fails
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.238	0.253	-0.016 [-0.022, -0.011]	fails
Reading of F1 and F2							not met

allowance layer on new approval pairs (-0.043, $[-0.061, -0.024]$) and below its transfer layer on new senders. With AWP as the comparator F1 still holds and F2 fails by more ($+0.040$ and -0.077; Section 4.8.4). With the contracts outside a graph kept as isolated nodes every verdict stays as it was ($+0.073$, $[0.059, 0.086]$, and -0.016, $[-0.022, -0.011]$). Inside the union graph F1's contrast is larger ($+0.118$, $[0.101, 0.138]$) and F2's about the same (-0.020, $[-0.027, -0.011]$), and the contrast with the allowance layer does not exclude zero ($+0.007$, $[-0.017, 0.029]$; Table 4.23).

4.8.4 C-PR beside every other score

Table 4.22 sets C-PR against every other score of the study and the four raw counts, on the spender labels of both windows, and Table 4.23 inside the union graph. Only the cells marked in the first

table belong to a rule. The contrasts with the union in-degree belong to the definition of the contract-graph analysis, which was made after the registered results, and so the labels, were known; like every other contrast of the table outside the rules, they decide no rule.

Table 4.22: Paired contrasts of C-PR ($\lambda = 0.5$) with every other score of this study and with the four raw counts of the contract graph, on the spender labels of W0 ($n = 7{,}864$ contracts) and W1 ($n = 7{,}915$). Each cell is $\Delta\tau = \tau_{\text{C-PR}} - \tau_b$ over its 95% interval; within a cohort every cell uses the same resamples. Cells marked ^{A} are the contrasts of rule A, fixed in the analysis plan; ^{B} marks F1, F2 and F6a of the registration of W1 and ^{S} the contrasts of its first sensitivity analysis, with AWP as the comparator. These rules were fixed for the raw-unit analysis and are applied here to the scores of the contract graph. Cells marked ^{U}, against the union in-degree, are part of the definition of the contract-graph analysis, defined after the registered results were known and so with the labels known. Like every cell without a mark, also computed after the labels were known, they belong to no rule. The lower block repeats the contrasts with AWP and the in-approve degree on the spenders that had received a transfer from another contract before the freeze, the spenders AWP can order; that restriction was also chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	W0 spenders		W1 spenders	
C-PR minus	New approval pairs	New transfer senders	New approval pairs	New transfer senders
C-PR ($\lambda = 1$)	-0.025^{A} [-0.045, -0.004]	+0.170 [0.148, 0.191]	-0.043^{B} [-0.061, -0.024]	+0.173 [0.152, 0.194]
C-PR ($\lambda = 0$)	+0.071 [0.059, 0.084]	-0.014^{A} [-0.019, -0.008]	+0.074^{B} [0.060, 0.087]	-0.016^{B} [-0.022, -0.012]
EndorseRank	-0.026 [-0.046, -0.004]	+0.170 [0.148, 0.191]	-0.043 [-0.062, -0.025]	+0.172 [0.152, 0.193]
EndorseRank (activity restarts)	-0.022 [-0.042, -0.000]	+0.164 [0.142, 0.187]	-0.044 [-0.063, -0.025]	+0.164 [0.144, 0.186]
AWP	+0.044 [0.029, 0.058]	-0.077 [-0.087, -0.068]	+0.040^{S} [0.026, 0.055]	-0.077^{S} [-0.086, -0.069]
S-PR	+0.051 [0.038, 0.064]	-0.011 [-0.021, -0.000]	+0.049 [0.036, 0.063]	-0.006 [-0.015, 0.002]
In-approve degree at the freeze	-0.062 [-0.082, -0.037]	+0.163 [0.142, 0.187]	-0.077 [-0.097, -0.057]	+0.164 [0.141, 0.188]
In-approve value at the freeze	-0.048 [-0.068, -0.024]	+0.173 [0.152, 0.195]	-0.059 [-0.079, -0.039]	+0.174 [0.151, 0.197]
Transfer in-degree at the freeze	+0.072 [0.059, 0.086]	-0.064 [-0.075, -0.054]	+0.072 [0.058, 0.087]	-0.066 [-0.075, -0.057]
Union in-degree at the freeze	+0.016^{U} [0.007, 0.024]	-0.053^{U} [-0.061, -0.043]	+0.011^{U} [0.003, 0.019]	-0.056^{U} [-0.065, -0.047]
Spenders that had received a transfer from another contract before the freeze (W0 $n = 4{,}569$, W1 $n = 4{,}642$)				
AWP	+0.026 [0.013, 0.041]	-0.084 [-0.097, -0.069]	+0.007 [-0.008, 0.021]	-0.094 [-0.108, -0.079]
In-approve degree at the freeze	+0.051 [0.025, 0.076]	—	+0.053 [0.025, 0.080]	—

Against EndorseRank, the walk over allowances alone, C-PR ranks new approval pairs lower in

Table 4.23: The main out-of-window contrasts on the whole spender cohorts and on their members inside the union graph only (W0: 5,700 of 7,864 contracts; W1: 5,759 of 7,915). A spender outside the union graph ends no pair between contracts and scores zero under every score and baseline, so on the whole cohort a contrast also reflects how the two scores order the members with a pair against those without one. The right column decides no rule. $\Delta\tau$ with 95% paired bootstrap interval (400 resamples, seed 42).

Window	Contrast (a minus b)	Label	Whole cohort	Inside the union graph
W0	C-PR minus C-PR ($\lambda = 1$)	new approval pairs	-0.025 [-0.045, -0.004]	$+0.020$ [-0.002, 0.043]
W0	C-PR minus C-PR ($\lambda = 0$)	new transfer senders	-0.014 [-0.019, -0.008]	-0.016 [-0.025, -0.007]
W0	C-PR minus union in-degree	new approval pairs	$+0.016$ [0.007, 0.024]	$+0.035$ [0.022, 0.049]
W0	C-PR minus union in-degree	new transfer senders	-0.053 [-0.061, -0.043]	-0.089 [-0.103, -0.075]
W1	C-PR minus C-PR ($\lambda = 0$)	new approval pairs	$+0.074$ [0.060, 0.087]	$+0.118$ [0.101, 0.138]
W1	C-PR minus C-PR ($\lambda = 0$)	new transfer senders	-0.016 [-0.022, -0.012]	-0.020 [-0.027, -0.011]
W1	C-PR minus C-PR ($\lambda = 1$)	new approval pairs	-0.043 [-0.061, -0.024]	$+0.007$ [-0.017, 0.029]
W1	C-PR minus union in-degree	new approval pairs	$+0.011$ [0.003, 0.019]	$+0.027$ [0.014, 0.041]
W1	C-PR minus union in-degree	new transfer senders	-0.056 [-0.065, -0.047]	-0.093 [-0.107, -0.077]

both windows (-0.026, $[-0.046, -0.004]$, in W0 and -0.043, $[-0.062, -0.025]$, in W1) and new senders higher ($+0.170$, $[0.148, 0.191]$, and $+0.172$, $[0.152, 0.193]$). The variant of EndorseRank with activity restarts gives the same picture: C-PR lies below it on new approval pairs (-0.022, $[-0.042, -0.0004]$, in W0, the upper end just below zero, and -0.044, $[-0.063, -0.025]$, in W1) and above it on new senders.

Against AWP, the walk over transfers alone, the signs are reversed: C-PR ranks new approval pairs higher ($+0.044$, $[0.029, 0.058]$, and $+0.040$, $[0.026, 0.055]$) and new senders lower (-0.077, $[-0.087, -0.068]$, and -0.077, $[-0.086, -0.069]$). Among the spenders that another contract had paid before the freeze its lead on approvals shrinks to $+0.026$ ($[0.013, 0.041]$) in W0 and is not resolved in W1 ($+0.007$, $[-0.008, 0.021]$), while AWP ranks new senders higher by more (-0.084, $[-0.097, -0.069]$, and -0.094, $[-0.108, -0.079]$) (lower block of Table 4.22; this restriction, too, was chosen after the results were known). On that subset C-PR ranks new approval pairs above the in-approve degree ($+0.051$, $[0.025, 0.076]$, and $+0.053$, $[0.025, 0.080]$).

Against the counts, C-PR does not rank either label above the count of the same edge type. On new approval pairs it lies below the in-approve degree in both windows (-0.062, $[-0.082, -0.037]$, and -0.077, $[-0.097, -0.057]$) and below the in-approve value; on new senders it lies below

the transfer in-degree (-0.064, $[-0.075, -0.054]$, and -0.066, $[-0.075, -0.057]$). It lies above each of these counts on the other label. Against the union in-degree, the count of the edges that C-PR walks, the result is split (Section 4.7), with the same directions and larger gaps inside the union graph: $+0.035$ ($[0.022, 0.049]$) and $+0.027$ ($[0.014, 0.041]$) on new approval pairs, -0.089 ($[-0.103, -0.075]$) and -0.093 ($[-0.107, -0.077]$) on new senders. Against its ablation S-PR it ranks new approval pairs higher ($+0.051$, $[0.038, 0.064]$, and $+0.049$, $[0.036, 0.063]$); on new senders it is slightly lower in W0 (-0.011, $[-0.021, -0.0004]$, the upper end just below zero) and level in W1 (-0.006, $[-0.015, 0.002]$). C-PR thus stands between the two sides on each label, and no score or count of this study has the highest coefficient on both: the in-approve degree has it on new approval pairs and AWP on new senders.

4.9 Claims (C1–C6)

The chapter's results support the claims below, cross-referenced to the hypothesis and research question columns of Table 1.2. C1 rests on the discovery-order test, which was defined after the registered results were known and decides no rule of the analysis plan or the registration. The verdicts of rules A and B in C3 are a re-analysis on the scores of the contract graph; the registered analysis gave the same readings (Appendix 8.6). C2 and C4 rest on contrasts outside every rule. The direct comparisons between EndorseRank and AWP are in Appendix 8.5 and carry no claim of the thesis.

C1. In the run that decides, at T_{obs} in the primary mode, a crawl ordered by C-PR on the contracts read so far reached the contracts with the most user-account demand sooner than a breadth-first crawl ($D_1 = +0.0520$, $[+0.0485, +0.0558]$, on I_T and $+0.0061$, $[+0.0014, +0.0111]$, on I_A) but not sooner than a crawl ordered by the number of contract backlinks, which was sooner ($D_2 = -0.0051$, $[-0.0087, -0.0010]$, and -0.0571, $[-0.0621, -0.0519]$). The verdict is V2 on both importances, so the answer to RQ2 is no (Table 4.12). AWP′ has the highest $\text{AUC}_{10\%}$ of the crawls in that run, apart from the diagnostic δ', every crawl stays far from the

ideal order, and the allowance layer adds a small, resolved amount to C-PR′ over its transfer layer (Tables 4.13 and 4.14). The lead over breadth-first order on I_A does not hold in every sensitivity. In the `out` mode, a sensitivity without a verdict, all four contrasts lie above zero at both freezes, so the result depends on what a visit reveals.

C2. Out of window EndorseRank does not beat the count it smooths, and AWP beats its count by a small margin that does not hold among the contracts with inflow. EndorseRank ranks new approval pairs below the in-approve degree in both windows (W0 -0.036, $[-0.046, -0.027]$; W1 -0.034, $[-0.043, -0.025]$), and so does its variant with activity restarts. AWP ranks new transfer senders slightly above the transfer in-degree on the whole cohorts (W0 $+0.012$, $[0.003, 0.023]$; W1 $+0.011$, $[0.001, 0.022]$) and below it among the W0 spenders that another contract had paid (-0.035, $[-0.049, -0.022]$) (Tables 4.16, 4.18 and 4.17). Against the union in-degree, the count of the edges it walks, C-PR ranks new approval pairs higher and new senders lower in both windows (Table 4.22). The labels count mostly user accounts, which no score reads, and about 27% of each spender cohort lies outside the union graph and scores zero.

C3. Under its rules, C-PR, a fixed-λ mixture of two dollar-weighted layers with uniform restarts, does not beat its two layers walked alone. Rule A is not met: in W0 C-PR lies below its allowance layer on new approval pairs (-0.025, $[-0.045, -0.004]$) and below its transfer layer on new senders (-0.014, $[-0.019, -0.008]$), and its same-window part fails as well (Tables 4.16 and 4.19). Rule B is not met by its two applied criteria (Table 4.21): F1 holds ($+0.074$, $[0.060, 0.087]$), so the gain over the transfer layer on new approval pairs recurs in W1, and F2 fails (-0.016, $[-0.022, -0.012]$, against the margin of -0.01). Inside the union graph and with the uncapped allowance layer the contrast with the allowance layer reads differently and the contrast with the transfer layer does not (Tables 4.23 and 4.11); neither decides a rule.

C4. Beside the rules, C-PR stands between the two sides on each label in both windows (Table 4.22). Against EndorseRank it ranks new approval pairs lower and new senders higher

(in W0 -0.026, $[-0.046, -0.004]$, and $+0.170$, $[0.148, 0.191]$); against AWP it ranks new approval pairs higher and new senders lower (in W0 $+0.044$, $[0.029, 0.058]$, and -0.077, $[-0.087, -0.068]$). It ranks new approval pairs above S-PR and below the in-approve degree and value, and new senders below the transfer in-degree.

C5. In the same window each single-layer walk agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family (0.839, $[0.828, 0.849]$) and AWP with the transfer family (0.755, $[0.748, 0.762]$), and the mixtures of C-PR lie between their two layers (Table 4.5). These are intended-construct checks, and part of the agreement is built in. The two walks order the cohort differently: their agreement is $\tau_b = 0.275$ ($[0.258, 0.294]$), only 32.3% of the contract pairs are ordered by both scores, because $5{,}200$ of the $6{,}443$ cohort contracts score zero under EndorseRank, and 26.7% of the pairs that both order are discordant. With the contracts outside a graph kept as isolated nodes EndorseRank's allowance coefficient rises to 0.956 ($[0.952, 0.960]$), and AWP's coefficients do not change (Table 4.6, Figure 4.3).

C6. The cost of each score's solver call grows with the size of the layers it walks and with the iterations its walk needs. The allowance graph has 4.5% of the transfer graph's edges, and AWP takes 27.9 times as long as EndorseRank at an edge ratio of 22.4 (Table 4.3). On the subsamples the time ratio is 0.99 to 1.36 times the edge ratio, and the rows do not test edge counts beyond those of the contract graph (Table 4.4). The low cost belongs to the smaller graph and is no advantage of the algorithm: C-PR walks both layers and takes 1.09 times as long as AWP.

5 Discussion

Chapter 4 reported the measurements on Arbitrum One. Here each research question and objective of Chapter 1 is held to the criterion stated there, and every interpretation points to the table of Chapter 4 it rests on. After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. It is the main analysis, and none of its results is a registered result (Section 3.5.3). Unless noted otherwise, same-window statements concern the matched cohort of contracts and out-of-window statements the spender cohorts.

5.1 Findings by research question

5.1.1 RQ1: The two single-layer walks and their cost

RQ1 asks whether EndorseRank and AWP order the same contracts differently, whether each agrees most with current checks of its own edge type, and what each walk, the coupled one included, costs. They order the contracts differently: the interval of their agreement excludes both zero and one, whether a contract missing from a graph is scored zero or kept as an isolated node (Table 4.6). The coefficient compares fewer contracts than the cohort holds: four in five cohort contracts grant or receive no capped allowance between contracts and tie at zero under EndorseRank (Table 4.2), so both walks order only about a third of the contract pairs, and about a quarter of those in opposite directions (Section 4.4).

The divergence follows from how the graphs are built. The allowance graph records the latest

allowances between contracts that the owner's balance backs and is far smaller than the transfer graph, which records every transfer (T. Do et al., 2023), and the two methods restart differently (Table 4.3). H1 is supported, largely by construction. One score cannot stand in for the other, and the divergence is also the premise of C-PR: if the two walks ordered the contracts alike, one walk over both could add little to either.

In the same window each score follows the edges it walks on: the walks over transfers agree most with the transfer family, the walks over allowances with the allowance family, and every interval on a score's own family excludes zero (Table 4.5). Circularity bounds H2: every proxy is computed from the same events between contracts as the matching graph, so the agreement is an intended-construct check in the sense of Cronbach and Meehl (1955) and not an external test. The rule for missing contracts also moves EndorseRank's coefficients: as isolated nodes, the contracts outside the allowance graph share the score of a node without in-edges, which raises its agreement with the allowance family and lowers it with the other two (Table 4.6). In the narrow sense of agreement with the checks of its own edges, H2 is supported, largely by construction (paired differences in Appendix 8.5).

The cost of each walk follows the size of its graph. EndorseRank is the cheapest score in time, memory and iterations; C-PR, on the union of both layers, takes about one tenth longer than AWP, and S-PR about as long as AWP (Table 4.3). AWP's time ratio to EndorseRank lies above the edge ratio at $d = 0.85$ and 0.95 and below it at $d = 0.75$ (Table 4.7), so the cost grows with the size of each graph (Langville & Meyer, 2006) and with the iterations its walk needs, which the benchmark does not separate; the speed belongs to the smaller graph, not to the algorithm. The scaling table qualifies this: its subsample rows keep only the pairs incident to the sampled contracts and its last row is the whole contract graph, so the rows are not sizes of one growing graph (Table 4.4). The damping and value-slope checks leave the alignment pattern as it was (Tables 4.7 and 4.8). H3 is supported, again largely by construction. H1–H3 are thus descriptive checks of the instruments; the tests whose outcome was open are RQ2–RQ5.

5.1.2 RQ2: Discovery order

RQ2 asks whether a crawl ordered by C-PR on the contracts read so far reaches the contracts with the most user-account demand sooner than breadth-first and backlink-count crawls. The discovery-order test was defined after the registered results were known and decides no rule of the plan or the registration. By the decision rule of its protocol (Section 3.7), the answer is yes only if, on each importance, the intervals of D_1 (C-PR′ minus breadth-first) and of D_2 (C-PR′ minus the backlink count) lie above zero at T_{obs} in the primary reveal mode, in which a visit reveals a contract's in-edges and out-edges. On both importances the interval of D_1 lies above zero and that of D_2 below zero (Table 4.12), and the verdict reads: "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most user-account inflow and those with the most user-authorised value sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks. A crawl ordered by the number of contract backlinks reached them sooner." The answer to RQ2 is no.

C-PR′ is sooner than breadth-first order on both importances, but on user-authorized value the margin is small and does not hold in every sensitivity, and in no row does C-PR′ lead the backlink count (Table 8.9). The one single-layer walk that does better than its count is AWP′: it reaches the hot contracts of both importances sooner than the transfer backlink count, has the highest $\text{AUC}_{10\%}$ of the compared crawls and leads C-PR′ on both importances, whereas C-PR's transfer layer walked alone, the same edges with other weights and restarts, lies below that count (Tables 4.13 and 4.14). Every crawl stays far from the ideal order, and under every ordering most of the first tenth of the visits goes to contracts with no user-account demand (Table 8.10).

Two facts bound this reading. First, the result depends on what a visit reveals. In the `out` mode, which reveals out-edges only and whose universe is the giant strongly connected component with the contracts reachable from it, all four contrasts lie above zero at both freezes; at t_1 in the primary mode they repeat the signs of the decision (Table 4.12). The `out` mode carries no verdict: the

protocol makes the mode on the weakly connected component primary when the other universe holds fewer than half of its contracts, as the structure pilot found (Table 8.7). The modes differ in the universe, in what the crawl sees of a frontier contract and in how often C-PR′ picks by a tie (Table 8.10); which difference reverses the comparison with the backlink count was not tested. Second, the allowance layer between contracts is thin: a capped allowance in-edge reaches about half a percent of the contracts of SCC ∪ OUT at t_1 in the structure pilot (Section 3.7). In the deciding run C-PR′ is therefore close to a transfer walk: its allowance layer raises the area by a small, resolved amount on both importances (Table 4.14), and in the `out` mode at T_{obs} it lowers the area on user-account inflow (Appendix 8.4.4).

5.1.3 RQ3: Out-of-window prediction and the raw counts

RQ3 has two clauses. The first asks whether frozen scores predict the new owner–spender approval pairs and the new transfer senders that the spender contracts gain in the following three months. Every score predicts both labels in both windows, and the edge type decides how well (Tables 4.15, 4.20 and 4.18). On new approval pairs the in-approve degree leads and the scores built on transfers alone rank lowest; on new senders AWP and the transfer in-degree lead and the scores built on allowances alone rank lowest. Two facts frame every coefficient: more than four in five of the new pairs and senders come from user accounts, which no score reads (Table 8.6), and about 27% of each spender cohort lies outside the union graph and scores zero (Table 4.2).

The second clause asks whether any walk beats the raw count of the edges it smooths; the answer differs by walk. EndorseRank ranks new approval pairs below the in-approve degree in both windows, with either restart rule (Tables 4.16 and 4.18; Section 4.5.4). AWP ranks new senders above the transfer in-degree in both windows, with intervals just above zero, but among the W0 spenders that another contract had paid before the freeze, which the count can order, it ranks them below the count (Table 4.17), a restriction chosen after the results were known and not computed for this contrast in W1. C-PR ranks new approval pairs above the union in-degree, the count of the edges it walks, and new senders below it, with larger gaps inside the union graph (Table 4.23); it lies be-

low the in-approve degree on new approval pairs and below the transfer in-degree on new senders (Table 4.22). So of the single-layer walks AWP alone gains on its count, by one or two hundredths of τ on the whole cohorts and inside the union graph, and C-PR gains on the count of its own edges on one label and loses on the other. Known results lead one to expect the count to be hard to beat: in growing networks new links attach preferentially to nodes that already have many (Barabási & Albert, 1999), which makes degree a standard link-prediction baseline (Liben-Nowell & Kleinberg, 2007), also on Ethereum transaction networks (D. Lin et al., 2022), and where degree correlations are weak PageRank is on average proportional to in-degree (Fortunato et al., 2008). For most spenders EndorseRank is a reweighted count of the contracts that authorize them (Section 5.3.2). All of these contrasts were computed after the labels were known.

5.1.4 RQ4: The coupled walk against its layers

RQ4 and RQ5 concern the method of the thesis, and for C-PR, the coupling the plan fixed, both get a negative answer under their own rules, applied to the scores of the contract graph as a re-analysis; the registered analysis reached the same verdicts (Appendix 8.6). RQ4 asks whether one walk over both layers ranks contracts better than either layer walked alone. Under rule A, fixed in the analysis plan, it does not. On W0 C-PR ranks new approval pairs below its allowance layer and new senders below its transfer layer, so both conditions of the primary part fail (Table 4.16), and in the same window the secondary part fails on all three families (Section 4.8.1). The plan fixed the rule and λ before the study extracted any log, so λ could not have been tuned to it. The result limits an obvious next step: a convex mixture of the two edge types at the transition level ends between its layers on both labels (Table 4.22).

Two results outside the rule qualify the loss to the allowance layer. Three in four spenders score zero under that layer, and among the spenders inside the union graph the contrast of C-PR with it on new approval pairs does not exclude zero in either window (Tables 4.2 and 4.23). And when every positive allowance counts at its own value, the allowance layer walked alone ranks new approval pairs much lower and C-PR lies above it in W0 (Table 4.11). Neither result decides the rule, and

the loss to the transfer layer on new senders stays in both.

The ablation and the layer weights explain part of this answer: S-PR, which restarts the transfer walk from the allowance walk, ranks new senders level with the transfer layer and new approval pairs far below the allowance layer (Table 4.16), and C-PR ranks new approval pairs above it in both windows, so the allowance signal survives better where allowance edges carry part of the walk than where they only seed its restarts. The layers of C-PR carry dollar amounts and restart uniformly, whereas AWP bounds the weight of each transfer and restarts at active senders; walked alone, C-PR's transfer layer already ranks new senders below AWP, without a paired contrast (Tables 4.15 and 4.20), so part of C-PR's gap to AWP comes from the weights and restarts, not from the coupling.

5.1.5 RQ5: Allowance edges added to a transfer walk

RQ5 asks whether adding the allowance layer improves a transfer walk on new data, gaining on later approvals without losing on later inflow. Rule B was registered with W1 before the study extracted the July–September logs. The thesis applies two of its registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds. On W1 F1 holds, as the same contrast did on W0 (Table 4.22): C-PR ranks new approval pairs above its transfer layer walked alone. F2 fails: on new senders C-PR lies below that layer by more than the registered margin of 0.01, with its whole interval beyond the margin (Table 4.21). Rule B is not met. The two registered sensitivity analyses give the same reading, and so do the spenders inside the union graph and the uncapped allowance layer, which decide no rule (Tables 4.23 and 4.11).

Beside the rules, the contrasts of Table 4.22 set C-PR against every other score and show losses on both labels: on new approval pairs C-PR lies below EndorseRank with either restart rule, its own allowance layer and the in-approve degree and value in both windows, and on new senders below AWP and the transfer and union in-degrees, and level with or slightly below S-PR. Among the spenders that another contract had paid before the freeze, C-PR ranks new approval pairs above the in-approve degree in both windows, and its lead over AWP there is smaller in W0 and not resolved in W1. What the allowance layer adds to a transfer walk out of window is therefore information

about later approvals, bought with a small, resolved loss on later inflow, and a walk or a count over allowances alone carries that information better. These contrasts were computed after the labels were known and decide nothing.

5.2 Achievement of the research objectives

Chapter 1 attached a measurement and a criterion to each of its five objectives. Table 5.1 checks them against the evidence.

5.3 Theoretical implications

5.3.1 Propagation semantics beyond transfers

PageRank was first formulated for hyperlinks, read as a citation network (Page et al., 1999), and Cho et al. (1998) found that a crawler ordering the pages it has seen by PageRank reaches important pages sooner than one ordering them by backlink count or breadth-first (Section 2.6). H.-D. Do and Do (2023) carried the citation reading over to Ethereum transactions, and AWP (T. Do et al., 2023) added value and age weights and restarts at active senders. EndorseRank applies it to ERC-20 allowances, where an owner that lets a spender act for it issues an endorsement of spending authority (Vogelsteller & Buterin, 2015), and C-PR lets one walk follow both relations. The RQ1 divergence shows that changing what the edges mean, and where the walk restarts, changes the ranking even on the same nodes, so graph-based reputation is a family of methods, each valid relative to the relationships its edges encode.

For C-PR that result does not carry over to contracts in the deciding run: on the hot set of the contract backlink count, the comparison that repeats the one of Cho et al. (1998), the count ordering is far sooner than C-PR′, and on user-account demand it is sooner as well (Table 4.14). The out-of-window results add a qualification that the propagation metaphor hides: the edge type decided which label a score could anticipate, but of the single-layer walks only AWP added to the count of its own edges, and not among the spenders that another contract had paid (Section 5.1.3). On these

Table 5.1: Research objectives, the criterion stated in Chapter 1, and the outcome.

Obj.	Criterion	Evidence	Outcome
O1	95% interval of the τ between EndorseRank and AWP contains neither 0 nor 1; a score agrees with a construct when its interval on that family excludes 0, and agreement with the construct of a score's own edges is an intended-construct check; the runtime ratio is given alongside the edge ratio, under a single fixed protocol	Table 4.6, last row: the interval lies strictly inside $(0, 1)$. Table 4.5: every walk over transfers agrees with the transfer family and both walks over allowances alone with the allowance family, each interval excluding 0. Tables 4.3 and 4.7: time ratio beside edge ratio at each damping value	Met (largely by construction)
O2	Decision rule of the discovery-order protocol, read at T_{obs} in the primary mode: on each importance the verdict is V1 when the 95% intervals of D_1 (C-PR′ minus BFS) and D_2 (C-PR′ minus IB'_U) both lie above 0, V2 when only that of D_1 does and V3 otherwise; the answer is the weaker of the two verdicts	Table 4.12: on both importances the intervals of D_1 lie above 0 and those of D_2 below 0. Verdict V2 on both	Not met
O3	A score predicts a label when its interval excludes 0; a walk beats a count only when the interval of the paired difference lies above 0	Tables 4.15, 4.20 and 4.18: every score predicts both labels in W0 and W1. Tables 4.16, 4.17 and 4.18: AWP lies above the transfer in-degree on the whole cohorts and below it among the W0 spenders another contract had paid; EndorseRank lies below the in-approve degree with either restart rule. Table 4.22: C-PR lies above the union in-degree on new approval pairs and below it on new senders. All computed after the labels were known	First clause met; second met by AWP on the whole cohorts only, and by C-PR against the union in-degree on new approval pairs only
O4	Rule A of the analysis plan: on W0, C-PR does no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new senders; in the same window, its point estimate lies within the leading layer's interval on the transfer and allowance families, and it exceeds both layers on Sybil stability	Scores of the contract graph, re-analysis. Table 4.16: both conditions of the primary part fail. Same window: C-PR's point estimates lie outside the leading layer's interval on both families (Table 4.5), and Sybil stability lies between the layers (Table 4.19); see Sections 4.8.1 and 4.8.2	Not met, as in the registered analysis (Table 8.16)
O5	Two of the registered criteria of rule B (registration of W1): on W1, C-PR beats C-PR ($\lambda = 0$) on new approval pairs (F1) and is not worse by more than 0.01 on new senders (F2: lower 95% bound above -0.01). Under the registration the rule holds only if each of its criteria holds	Scores of the contract graph, re-analysis. Table 4.21: F1 holds; the interval of F2 lies wholly below -0.01, so F2 fails. The two registered sensitivity analyses read the same	Rule B is not met; F1 held and F2 failed in the registered analysis as well (Table 8.16)

data the choice of edge decides which contracts a score can order, and a count of that edge orders them about as well as a walk over it or better.

5.3.2 Depth-one spenders: what a walk adds to a count

The allowance graph keeps the latest allowances between contracts that stay positive under the balance cap (Section 3.6). Most of its nodes are pure owners, with out-edges only, or pure spenders, with in-edges only, but a contract can be both, so the graph also has paths of more than one step. For a spender whose owners receive no allowances, a walk with uniform restarts has a closed form.

Proposition. Let PageRank with transition matrix P be computed by Equation (3.1) with damping d and uniform restarts, so that dangling mass is also spread uniformly. Then every node without in-edges has the same score c, and every node v whose in-neighbors all lack in-edges has

$$r_v = c\,\big(1 + d\,\delta(v)\big), \qquad \delta(v) = \sum_u P_{uv}, \tag{5.1}$$

the sum of the transition probabilities into v. In EndorseRank $P_{uv} = w(u,v)/o(u)$, so $\delta(v)$ is the owner-normalized in-strength of v: each owner's endorsement is divided over all the spenders it approves. In C-PR P_{uv} is the coupled transition of Equation (3.9), and an in-neighbor without in-edges is one that has none in either layer.

Derivation. At the fixed point, $r_v = (1 - d + d\,r_D)/|V| + d\sum_u r_u\,P_{uv}$, where r_D is the mass held by the dangling nodes. A node without in-edges receives only the first term, so every such node has the score $c = (1 - d + d\,r_D)/|V|$, the constant c of Section 5.6.3. If every in-neighbor u of v is such a node, substituting $r_u = c$ gives Equation (5.1).

On EndorseRank's graph at T_{obs}, 651 of the 926 cohort contracts that are spenders have depth one, and EndorseRank lies within 1% of Equation (5.1) for each of them; of the deeper spenders, which have an owner that itself receives allowances, 33.8% do. Over all 926 the share is 80.3%, and over the 1,362 W0 spenders of the graph at t_1 it is 68.4%. Over the cohort spenders EndorseRank's Kendall τ with δ is 0.923, a point estimate. These shares were not computed for the union graph

of C-PR.

Two consequences follow. First, for a depth-one spender EndorseRank only reweights the count of the contracts that authorize it, counting an owner that authorizes many spenders for less at each; besides the time and value weights, that normalization is all that separates it from the count, which ranks new approval pairs better (Section 5.1.3). For these spenders EndorseRank's negative answer to the second clause of RQ3 thus has an analytic basis: there was no propagated information for the walk to add, and with activity restarts the variant is likewise a one-step sum over the owners, weighted by their restart mass. Second, for most deeper spenders the score departs from the closed form, so multi-hop propagation is present in this graph, but over all spenders EndorseRank did not beat the raw count out of window, and the deeper spenders were not tested apart. The edges are read from the whole chain, so the short paths are no artifact of a sample. The one-step form reappears in the discovery-order test: the diagnostic δ', the normalized in-strength on the visited part with C-PR's mixed weights, has a higher $\mathrm{AUC}_{10\%}$ than every walk and every count ordering in all four runs (Table 4.13); it stands outside the comparisons of the protocol and has no paired contrast.

5.3.3 Multi-construct domain alignment

Each score aligns with the construct of the edges it walks on, the mixtures falling between their layers. Classical construct-validity theory would predict this pattern for instruments that measure different constructs (Cronbach & Meehl, 1955), and the integrated view of Messick (1995) goes further, counting the consequences of using a score as part of its validity. The same theory says what such a pattern cannot settle: the instrument that comes out ahead on a check is the one built for that check, so ordering the instruments by their coefficients does not show which is the better measure of reputation. The importance measures of the discovery-order test come closest to a criterion that no instrument was built for, since no ordering reads an edge of a user account; they are still sums of transfers and allowances, so they test whether a position among contracts agrees with demand from users, not whether either makes a contract worth relying on.

5.3.4 Reputation under cheap pseudonyms

Identities are cheap to create on public ledgers (Nakamoto, 2008), and under Ethereum's account model a new address needs no registration step (Wood, 2025). Any reputation system that rewards inbound transfers or allowances is therefore exposed to Sybil strategies, in which an attacker creates edges among addresses it controls (Douceur, 2002). A graph whose nodes are contracts changes the price of an identity and not the principle: a contract costs a deployment and a user account does not, the study does not measure who controls any account, and one actor can deploy many contracts. Section 5.6 formalizes the cost of a farm of contracts and shows that on this study's allowance graph its gain is close to the upper bound.

5.4 Practical implications and deployment scenarios

A wallet interface, a block explorer, an auditor or an integrator faces very many contracts and has to choose which to look at first. The results do not single out one ordering for that choice: in the deciding discovery-order run a crawl ordered by the contract backlink count reached the contracts with the most user-account demand sooner than C-PR′, and out of window no walk clearly improved on the leading count of a label.

5.4.1 The coupled walk and combined use

Folding the two edge types into one walk did not pay against the published transfer walk on these data: C-PR costs slightly more than AWP (Table 4.3), trails it on new senders in both windows and on both importances in the deciding discovery-order run, and leads it out of window only on new approval pairs (Sections 5.1.2 and 5.1.5). An operator free to choose should keep the signals apart, using a count or a walk over allowances where authorization matters and AWP or the transfer count where flow matters. For an operator tied to a single transfer walk weighted by dollar amounts, adding allowance edges bought information about later approvals that held on new data, at a loss on inflow larger than the registered margin; rule B is not met.

5.4.2 Transfer-hub and flow monitoring

A protocol that describes current flow can read it from in-degree and in-value directly. AWP (T. Do et al., 2023) had the highest coefficient of the walks on later new senders (Tables 4.15 and 4.20) and ordered the crawl with the highest area in the deciding discovery-order run (Table 4.13), the best record of the walks in this study. Its margin over the transfer in-degree out of window is about a hundredth of τ and reverses in W0 among the spenders that the count can order (Table 4.17), so little is lost by counting.

5.4.3 Authorization-aware and low-latency refresh

Where explicit spending authorization matters most, there is reason to use allowance edges. The cheapest form is the spender's in-approve degree among contracts, which needs no solver and ranked new approval pairs above EndorseRank, with either restart rule, and above C-PR in both windows (Tables 4.16, 4.18 and 4.22). EndorseRank, its propagated version, solves in a fraction of AWP's time (Table 4.3), but for most spenders it reflects the authorizing contracts only through how many spenders each authorizes (Section 5.3.2), and it is open to inflation by closed farms (Section 5.6). Any allowance score between contracts covers little: one cohort contract in five and one spender in four grant or receive one (Table 4.2). And the balance cap matters: when every positive allowance counts at its own value, the in-approve degree and EndorseRank have about half the coefficient on new approval pairs (Table 4.11; point estimates), so a tool that reads allowances to anticipate authorization should count each one up to the balance behind it.

5.4.4 Industry adoption scenarios

- Wallet interfaces and security tooling. A user asked to approve a spender wants to know whether it is widely authorized. A badge that shows how many contracts with a balance have authorized a spender, or its EndorseRank, makes that delegation explicit (Lundfall, 2020; Vogelsteller & Buterin, 2015); it conveys the position of a graph node and is neither a security audit nor a guarantee against malicious contracts, and it is empty for the spenders, at least three

in four, that no contract with a balance authorizes.

- Block explorers, auditors and integrators. A service that labels or indexes contracts, a reviewer choosing which contracts to read first and an aggregator choosing what to integrate cannot treat millions of contracts alike and need an order of work. In the deciding run the contract backlink count, which needs no solver, reached the contracts with the most user-account demand sooner than C-PR′, and most early visits of every ordering went to contracts without such demand (Section 5.1.2). An ordering by position among contracts is a first filter at best and says nothing about the quality of a contract's code.
- Programs that rank or reward contracts. Sybil clusters routinely abuse token distributions and governance eligibility (Douceur, 2002; Luo et al., 2025), and a listing, grant or incentive program that rewards contracts by rank invites the same response. EndorseRank as defined here offers little protection: a closed farm collects about six times its population share and can concentrate it on one target, so four deployed contracts lift a target into the cohort's top hundredth and about 375 to its top, and the bound holds for C-PR as well (Section 5.6).
- Lending protocols. A ranking of the contracts that users rely on could later serve as one input to credit decisions about contracts, for instance when a protocol weighs lending to a vault or a strategy contract. This thesis does not test that use: no score was compared with a default, repayment or loss outcome.

5.4.5 Standardized scores as shared infrastructure

A score shared by several parties is useful only if each can recompute it and see how it might be manipulated. C-PR, EndorseRank and AWP can be recomputed with the open pipeline from public logs, public prices and the code at each address (Appendix 8.1), and Section 5.6 documents how they can be manipulated.

5.5 Societal and practical impact

On a rollup with millions of contracts, finding out which of them others rely on is expensive, and the capacity to do it is unevenly spread: wealthy actors may afford proprietary risk tools, while retail participants work from anecdotal evidence or skip vetting altogether. An ordering built from public transfers and allowances between contracts, cheap to recompute (Table 4.3), lowers the barrier to a first look at the contracts that ask for permissions. An allowance that stays in force is one route by which tokens are drained (Section 2.18), but no score is shown to anticipate drains: on W0 the drain heuristic and the revocations rose with every score, the transfer-based ones included, so they track volume and are not offered as evidence (Section 4.7).

A score built from public events and public prices with versioned parameters supplies the explanatory information that regulators suggest should be required when DeFi protocols are analyzed (Zetzsche et al., 2020), and it brings the duties of Section 5.7.

5.6 Sybil attack cost model on allowance graphs

A Sybil attack exploits cheap pseudonyms to create a false impression of trustworthiness (Douceur, 2002). On a transfer graph, an actor can inflate in-degree and in-flow by moving assets among accounts it controls, at the price of transaction fees and possible inventory loss (Cong et al., 2023). An allowance edge records an authorization, not a transfer, and exposes a different attack surface. This section formulates a simple cost model for inflating a walk over allowance edges with fake allowances, computed on EndorseRank's graph of contracts at T_{obs}; its bound holds for C-PR as well (Section 5.6.3). It builds on the optimal link farms of Gyöngyi and Garcia-Molina (2005) and on the result of Cheng and Friedman (2005) that symmetric functions such as PageRank are not Sybil-proof (Section 2.7), and adds the costs of a farm whose every address must be a contract that holds a balance and the size of the gain on this study's graph. Its quantities decide no rule.

5.6.1 Attack setup

Let the attacker control a set of contracts $\mathcal{S}$, and write $\mathcal{H} = V \setminus \mathcal{S}$ for the honest contracts in the endorsement graph $G_{\text{approve}} = (V, E, w)$. Every node is a contract, so the attacker has to deploy each member of $\mathcal{S}$; a user account, with or without code, is no node (Section 3.5.1). An edge (u, v) of weight $w_{uv} > 0$ records that u has authorized v to spend its tokens and held a balance of the token at the freeze. The weight is that of Equation (3.7). With damping parameter $d \in (0, 1)$, EndorseRank finds the PageRank vector $\mathbf{r}$ on G_{approve} as the solution of (Page et al., 1999):

$$\mathbf{r} = d\, P^{\top}\mathbf{r} + \frac{1-d}{|V|}\mathbf{1}, \tag{5.2}$$

where P is the transition matrix of Equation (2.1). A node with out-edges splits its row in proportion to its outgoing allowance weights. A dangling node, one without out-edges, has every entry of its row equal to $1/|V|$, so its mass is spread uniformly over all nodes (Langville & Meyer, 2006). The fixed point equals that of the iteration in Equation (3.1), which the implementation runs and which feeds dangling mass back through the uniform teleportation vector. The attacker's objective is to maximize the mean EndorseRank of a target set $\mathcal{G} \subseteq \mathcal{S}$,

$$J(\mathcal{G}) = \frac{1}{|\mathcal{G}|}\sum_{i \in \mathcal{G}} r_i, \tag{5.3}$$

subject to a budget constraint on the cost of the attack, which is defined next.

5.6.2 Cost components

The attacker deploys the contracts of $\mathcal{S}$, creates a set of edges $E_{\mathcal{S}}$, all with sources in $\mathcal{S}$, and may also obtain a set $E_{\mathcal{H}\to\mathcal{S}}$ of edges from honest owner contracts into $\mathcal{S}$. Costs are counted over q score windows, the number of score refreshes for which the targets are meant to hold their rank.

1. Deployment cost. Each farm contract is created once. Under the fee schedule of the EVM a creation costs at least 32,000 gas from a factory contract and at least 53,000 gas by a trans-

action of its own, plus 200 gas for each byte of code (Wood, 2025); this is execution gas, to which a rollup adds the fee for posting the transaction's data to its base layer. With c_{dep} the fee of one deployment, $C_{\text{dep}} = |\mathcal{S}|\, c_{\text{dep}}$.

2. Gas cost. Write c_{gas} for the expected fee of one allowance update. EndorseRank reads the `Approval` events of the twelve months before each anchor, so an approval enters only the graphs whose window holds it; inside a closed farm its time weight does not matter, since row normalization removes it. A closed farm therefore has to emit its approvals again at least once every twelve months, and over score refreshes that span y such periods $C_{\text{gas}} = y\,|E_{\mathcal{S}}|\, c_{\text{gas}}$.
3. Balance cost. An allowance counts only up to its owner's balance of the token at the freeze, so every farm contract that grants one must hold one of the five tokens then (Section 3.5.2). The amount is almost free: any balance above the dust level of 10^{-6} token makes the allowance an edge, a balance of one dollar already gives the full value weight ($V = 1.000$, since m_A is 0.071 dollars), and after row normalization a single out-edge gives $P_{uv} = 1$ whatever its weight. The tokens stay under the attacker's control, since the spenders are its own contracts, and are needed only at the freeze. With θ the balance of each farm contract in dollars and κ the cost of holding one dollar across one freeze, $C_{\text{bal}} = q\,|\mathcal{S}|\,\theta\,\kappa$.
4. Revocation and detection cost. A farm of fresh contracts that authorize only one another might be identifiable. Let c_{rev} be the expected cost per window of detection and of the revocations the attacker issues to avoid it; over q windows, $C_{\text{rev}} = q\, c_{\text{rev}}$.
5. Edges from honest owners. The attacker can also get a contract it does not control to authorize one of its own; the model prices such an edge without asking how it is obtained. An edge (h, a), with $h \in \mathcal{H}$ and $a \in \mathcal{S}$, stays in the graph for twelve months after h last sets it, while h holds a balance of the token, and it loses share of h's row as it ages, so keeping it effective costs, per window,

$$c_{\text{in}}(h, a) = c_{\text{gas}} + p_h + \rho_h, \tag{5.4}$$

where p_h is the payment to whoever controls the honest owner and ρ_h is a risk premium

against the allowance being spent by the attacker.

Over the q windows, the total cost of the attack is

$$C = C_{\text{dep}} + C_{\text{gas}} + C_{\text{bal}} + C_{\text{rev}} + q \sum_{(h,a)\in E_{\mathcal{H}\to\mathcal{S}}} c_{\text{in}}(h, a). \tag{5.5}$$

For a closed farm, $E_{\mathcal{H}\to\mathcal{S}}$ is empty, so $C = |\mathcal{S}|\, c_{\text{dep}} + y\,|E_{\mathcal{S}}|\, c_{\text{gas}} + q\,|\mathcal{S}|\,\theta\,\kappa + q\, c_{\text{rev}}$.

5.6.3 Influence and a cost-effectiveness bound

By the usual sensitivity results for PageRank, an edge moves **r** in proportion to the stationary mass at its source node and to its normalized outgoing weight (Langville & Meyer, 2006). Consider first a closed farm: every node of $\mathcal{S}$ has at least one out-edge, all of these edges end in $\mathcal{S}$, and no honest owner authorizes a contract of $\mathcal{S}$. Let D be the set of dangling nodes, all of which then lie in $\mathcal{H}$, and let $r_D = \sum_{j\in D} r_j$ be their stationary mass. By Equation (5.2), a node $i \in \mathcal{S}$ receives $d \sum_{j\in\mathcal{S}} r_j P_{ji}$ from the farm, $(1-d)/|V|$ from teleportation and $d\, r_D/|V|$ from the dangling rows. The rows of the farm's nodes sum to one inside $\mathcal{S}$, so summing over $\mathcal{S}$ gives $\sum_{i\in\mathcal{S}} r_i = d \sum_{i\in\mathcal{S}} r_i + |\mathcal{S}|\,(1 - d + d\, r_D)/|V|$, that is,

$$\sum_{i\in\mathcal{S}} r_i = \frac{|\mathcal{S}|}{|V|} \cdot \frac{1 - d + d\, r_D}{1 - d}. \tag{5.6}$$

The first factor is the farm's population share, and the second says how many times that share the farm collects. The second factor equals one only when $r_D = 0$, that is, on a graph without dangling nodes; otherwise the dangling rows pass part of the honest mass to the farm. Since the dangling nodes lie outside $\mathcal{S}$, $r_D \le 1 - \sum_{i\in\mathcal{S}} r_i$, and substituting this into Equation (5.6) gives a bound that holds on any graph:

$$\sum_{i\in\mathcal{S}} r_i \le \frac{|\mathcal{S}|}{(1 - d)\,|V| + d\,|\mathcal{S}|}. \tag{5.7}$$

For a farm that is small against $|V|$, the bound is close to $1/(1 - d)$ times the population share. On the EndorseRank graph of contracts at T_{obs}, with $109{,}381$ nodes, the pipeline's own PageRank code

gives $r_D = 0.911$, 0.911 and 0.910 at $d = 0.75$, 0.85 and 0.95: the $9{,}452$ owners are the only nodes with out-edges, so nine nodes in ten are dangling. A closed farm on this graph therefore collects 3.73, 6.16 and 18.30 times its population share, close to the bounds of 4, 6.67 and 20. A direct check with the same code agrees: a ring of 20 contracts, each authorizing the next, added to the graph collects 6.16 times its population share at $d = 0.85$.

Equation (5.7) limits the farm's total mass, not the score of one target, and the attacker can concentrate that mass. Take a star in which f farm contracts each authorize the target t and t authorizes each of them back. Every farm contract a has t as its only out-neighbor, so $P_{at} = 1$, and the row of t sums to one over the farm. Write $c = (1-d+d\,r_D)/|V|$ for the mass that each node receives from teleportation and the dangling rows. The balance equations $r_a = c + d\,r_t P_{ta}$ and $r_t = c + d\sum_a r_a$ then give

$$r_t = \frac{c\,(1+df)}{1-d^2}.$$

At $d = 0.85$, $c = 8.45 \times 10^{-6}$, which is the common score of every contract of the graph that receives no allowance. With $f = 10$ (eleven deployments and twenty approvals), the target reaches $r_t = 2.89 \times 10^{-4}$ on the graph with the star added, above the 99th percentile of the cohort contracts but only 3.0% of the highest EndorseRank among them (0.0097). The score grows linearly in f: each further farm contract adds $d\,c/(1-d^2)$ to r_t and costs one deployment and two approvals. Four in five cohort contracts score zero, so any contract that enters the graph ranks above them. By the formula the smallest farm, a single contract, already lifts the target past the cohort's 90th percentile (the target then scores $c(1+d)/(1-d^2)$, about $6.7c$), three farm contracts pass its 99th and about 374 reach its highest score, that is, 375 deployments and 748 approvals, since the highest-ranked cohort contract holds about 1% of the graph's stationary mass. This study did not measure c_{dep}, c_{gas} or κ, so these counts are not turned into a price. They describe the allowance layer of T_{obs}, in which one owner contract holds 85% of the pairs (Section 5.8.2); the model was not computed at t_1.

Honest in-edges are therefore not needed to reach the upper part of the ranking, but they add to the gain. When the farm has no out-edges to honest contracts, an edge (h, a) obtained from an honest

owner h adds, to first order, $d\,r_h\,P_{ha}/(1-d)$ to the farm's mass, where P_{ha} is the share of h's row that the new edge takes. A fresh allowance from h that a balance of a dollar or more backs, with $V(z)$ close to one, takes about $\sigma(0)/(o(h)+\sigma(0))$ of h's row; for the median owner in the graph that share is 0.687. Such an allowance puts the owner's balance of the token at the attacker's disposal, so the risk premium ρ_h in Equation (5.4) grows with what the owner holds, while the weight of the edge stops growing at about a dollar. The attack's cost-effectiveness is $\eta = \Delta J(\mathcal{G})/C$, the gain in the objective of Equation (5.3) over the cost of Equation (5.5).

Equations (5.6) and (5.7) show that EndorseRank, as defined here, is open to inflation by closed farms of contracts. Under uniform teleportation and dangling redistribution, a closed cluster is not held to its population share, and on this graph it collects nearly the most the bound allows. The coupled walk is open to the same farm: the derivation of Equation (5.7) uses only these two uniform rules, which C-PR shares; a farm contract whose only out-edges are allowances follows them with weight one whatever λ is (Equation (3.9)), and a single out-edge takes the whole row whatever its amount, so neither the cap U nor the balance cap protects C-PR. The farm factors and star sizes above were computed on the allowance graph only; on the union graph of C-PR the dangling mass, and with it c, differ and were not computed. The control that does bound closed clusters changes the walk itself: if teleportation and dangling redistribution go only to a vetted seed set, as in TrustRank (Gyöngyi et al., 2004), a cluster outside that set gains mass only through edges from honest owners, which the attacker has to obtain at the price of Equation (5.4). AWP's restart rule gives the transfer side no protection either: a farm whose contracts send small amounts to one another raises their activeness and so draws restart mass to itself, and EndorseRank with activity restarts inherits the same exposure. The Sybil-adjusted stability proxies (Section 3.10.3) do not flag a farm of fresh contracts, and an adversarial assessment is left to future work.

5.6.4 A threshold on the balance

The balance cap makes a farm hold tokens but asks for almost none. A stricter graph would keep only the allowance pairs whose capped value is at least θ dollars, so that every farm contract must

hold θ at the freeze; Table 5.2 repeats the model on such graphs.

Table 5.2: The Sybil cost model on the EndorseRank graph of contracts at T_{obs}, and its sensitivity to a balance threshold θ: the graph keeps only the allowance pairs whose capped value is at least θ, so that every farm contract must hold θ of the token at the freeze. Columns: the pairs kept and their share, the owners and the nodes of the graph, the rank agreement τ of the matched cohort's scores with those of the main analysis, the score of the target of a star farm of 10 contracts as a share of the highest EndorseRank in the cohort, and the farm contracts that the closed form needs to reach that highest score. Every farm address is a contract: a creation costs at least 32,000 gas from a factory and 53,000 gas by its own transaction, plus 200 gas per byte of code (execution gas of the EVM fee schedule, without the fee for posting data to L1). The value weight of an allowance backed by one dollar is $V = 1.000$, so a larger balance buys no further weight. The quantities decide no rule.

Threshold	Pairs kept	Share	Owners	Nodes	τ with main	Star target / highest	Farm contracts to the top
None (main analysis)	114,997	100%	9,452	109,381	1.000	0.030	374
$\theta = \$1$	4,247	3.7%	2,390	3,394	0.709	0.092	119
$\theta = \$10$	2,167	1.9%	1,000	1,769	0.589	0.325	33
$\theta = \$100$	992	0.9%	500	837	0.464	1.054	9

The threshold raises the capital that a farm must hold and shrinks the honest graph faster, so the reach of the same farm grows. Most capped allowances between contracts are worth less than a dollar: at $\theta = 1$ dollar 3.7% of the pairs stay and the common score c rises about twentyfold, and a star of 10 then reaches 9.2% of the highest cohort score, against 3.0% without a threshold, and 119 farm contracts reach the top, against 374. At 100 dollars 0.9% of the pairs stay, and the star of 10, whose eleven contracts hold $1{,}100$ dollars between them, already exceeds the highest cohort score. The honest ranking changes as well, as its falling agreement with the main analysis shows, and the capital is held, not spent, and only at the freeze. A requirement that protects the ranking would have to bound the mass that flows into a cluster, as a vetted seed set does; pricing the members of a cluster does not.

5.7 Regulatory and ethical considerations

On-chain reputation metrics raise questions of transparency as well as equity. Legal scholarship has shown that decentralized finance markets remain bound by disclosure and consumer protection rules even though settlement happens on-chain (Zetzsche et al., 2020). Packin and Lev-Aretz (2024)

warn that decentralized scoring may perpetuate black-box harms when models cannot be interpreted or contested, or when they are linked to identity databases without due process. C-PR, EndorseRank and AWP are computed deterministically from public data with open-source algorithms, so anyone can recalculate them (Appendix 8.1). That openness partly answers the black-box criticism. It leaves the distributional concern untouched, because contracts with little activity, new ones among them, will score lower whatever their code, and a contract that only user accounts pay or authorize has no position in the graph.

Ethically, these scores should never be used as security ratings, as measures of the ability to repay, or as evidence that the contracts that authorize a spender belong to distinct parties (Douceur, 2002); they are rank-based indicators of a contract's position in the defined graph. A deployment that grants or withholds a service by score thresholds should document the graph definitions behind the scores, the timeframe, and the limitations, above all the exposure to Sybil attacks.

Privacy on an open blockchain is limited by the ledger's very nature (Nakamoto, 2008), and the scores use public data only. Further attempts to uncover the identity behind each address are discouraged, and the scores should not be treated as permanent, fingerprint-like identifiers. This study scores contract addresses alone and makes no attempt to link any address to a real-world individual (Section 3.17).

5.8 Threats to validity

5.8.1 Construct validity

The threats below concern whether the scores and checks measure what they are read as measuring.

- The scores are not ratings of default risk: no default, repayment or loss label enters the analysis.
- Same-window checks rest on local allowance and transfer statistics between contracts, and own-construct agreement is partly mechanical: EndorseRank smooths spender in-approve degree over the whole graph, and AWP does the same for incoming transfers weighted by their

value (Table 8.11); EndorseRank's coefficients also depend on the rule for missing contracts (Table 4.6).

- Importance in the discovery-order test. User-account inflow and user-authorized value are measured on-chain, on five tokens, from user accounts only. They record dollars that user accounts moved or authorized, not whether a contract is needed, safe or valuable, and no label of importance from outside the chain was used. Beside the 16.61 billion dollars of inflow to typed contracts at T_{obs}, user accounts sent 1.68 billion to addresses that were not typed because only user accounts transact with them (Section 3.15); a contract among these has no edge to another contract and lies outside every crawl. Transfer volume can be inflated by wash trading (Victor & Weintraud, 2021); the cap per account bounds one account and not one actor.
- Out-of-window construct. A later approval pair is still an endorsement event and a later sender a flow event, so neither is an independent construct, and as counts of arrivals they favor any score that tracks the current count of counterparties (Section 5.1.3). They count mostly user accounts, which no score reads (Table 8.6), and the part that comes from contracts was not evaluated on its own.
- Account types. Every node is an address typed as a contract by its code at one block, read after both label windows (Section 3.5.1); a contract that destroyed itself before EIP-6780 (Ballet et al., 2023) is classed as a user account. Safes and ERC-4337 accounts are user accounts by rule and by listed code families, and kinds of smart account that the rule misses stay classed as contracts: after the lists, $115{,}084$ typed contracts still carry a `validateUserOp` selector and no paymaster selector, and the contracts that services create for each user, order or payment are not picked out (Appendix 8.4.3). Users may control such contracts, so some edges of the graph are in effect edges of user accounts.
- Value weights. EndorseRank and AWP pass each US-dollar amount through $V(z)$ with a slope set so that the median amount weighs one half (Section 3.5.3), and the slopes were

checked only in the same window (Table 4.8). Under the balance cap the median positive allowance is 0.071 dollars, so V is at its ceiling for any allowance backed by a dollar or more and separates very small balances from the rest and little else.

5.8.2 Internal validity

The threats below concern whether the comparisons are free of choices that could have produced their results.

- Shared score windows. Every method and proxy is computed over the twelve months before its freeze, and market regimes peculiar to those months may push alignment up or down. An allowance last set before the window began is not an edge even while it is in force, and an allowance is an edge only if its owner holds a balance at the freeze, so the allowance layer is a snapshot of one instant, and its size differs severalfold between the two freezes (Table 4.1). No check varies the calendar window.
- One owner contract at T_{obs}. One owner contract holds 85% of the capped allowance pairs at T_{obs}; at t_1 the largest owner holds 256 pairs (Table 4.1). The same-window results on the allowance side, the W1 scores and the quantities of the Sybil model are computed on that layer, so one contract's allowances weigh on all of them; most of its pairs lie outside the universe of the discovery-order test (Section 4.6), and the thesis does not identify the contract. W0, scored at t_1, has no such owner, and every contrast of Table 4.22 has the same sign in W0 as in W1.
- The thin allowance layer. Between contracts, capped allowances are few beside transfers: about one pair for every twenty-two transfer pairs at T_{obs} and far fewer at t_1 (Table 4.1). The coupled walk is therefore close to its transfer layer, and every comparison of C-PR with that layer, in the discovery-order test above all, measures a small perturbation.
- Cohort selection and coverage. The matched cohort is the $6{,}443$ contracts that received non-zero approvals of one of the five tokens from at least three owners in the twelve months before

T_{obs}; the rule counts every approving account, and results may not carry over to other contracts. About a third of the matched cohort and 27% of each spender cohort end no pair between contracts and score zero under every score and baseline (Table 4.2), so on the whole cohorts a contrast also reflects how two scores order the members with a pair against those without one. Among the members inside the union graph the main contrasts keep their direction, except that of C-PR with its allowance layer on new approval pairs, which does not exclude zero there (Table 4.23); the sample-definition stages show how the coefficients move with the evaluation population (Table 4.9).

- Reveal mode of the discovery-order test. The decision is read in the mode the protocol made primary, and in the other mode D_2 has the opposite sign at both freezes (Section 5.1.2); the test does not say which mode is closer to how its users meet contracts. Its intervals resample the hot contracts for twenty fixed seeds, and under a bootstrap over the seeds D_2 on user-account inflow and D_1 on user-authorized value are not resolved (Table 8.9). The two freezes share nine months of their score windows and are not independent.
- Order of analysis. The cohort rule, scores, labels, statistics and rule A were fixed in the analysis plan before any log of the study was extracted, and rule B with its margin and its two sensitivity analyses was registered after the W0 results were known and before the study extracted the July–September logs (Section 3.8.1). The main analysis, with the definitions listed at the start of this chapter, was defined after the registered results were known, so it is post hoc: its verdicts under rule A and under F1 and F2 are a re-analysis, not registered tests, and the labels of both windows were known when it was defined. The registered analysis reached the same verdicts, although some contrasts differ, among them the sign of C-PR against its allowance layer on new approval pairs in W0, which lay above zero there (Table 8.16).
- Comparators and baseline fidelity. The rules compare C-PR with its own layers walked alone, the only contrast that changes the coupling and nothing else; its contrasts with EndorseRank and AWP mix the coupling with their weights and restarts (Table 4.22). AWP follows the published method in its edge weights and restarts (Section 3.6.1), but the paper gives no

parameter values, so k, t_0 and b are this study's choices, and here a sender's activeness counts only its transfers of five tokens to contracts.

- Dependence, balances and timing. Contracts that share owners or a protocol are resampled as independent, and the post hoc contrasts are not adjusted for multiplicity (Section 3.11); the five tokens were chosen by their rows among the events of the plan's cohort up to T_{obs}, which include the label window of W0. Balances are reconstructed from `Transfer` logs: in a check, 99.5% of $1{,}500$ sampled balances equaled `balanceOf` at the block of T_{obs} (Section 3.5.2). Runtimes are wall-clock times from one machine (Appendix 8.1).

5.8.3 External validity

The threats below concern how far the results carry beyond the data studied.

- Single chain and window. The evidence comes from Arbitrum One alone, from contracts that at least three owners approved on one of five tokens, over the twelve months before each of two freezes three months apart and two three-month label windows. Extending the findings to other rollups, Layer-1 chains or later months would need the same tests there, since DeFi venues differ in protocol design (Werner et al., 2022) and in market structure (Schär, 2021).
- Five tokens and edges between contracts. A transfer or an allowance of another token, a payment in native ether and any event with a user account at one end are not edges, so a contract that users rely on directly, without another contract paying or authorizing it on these tokens, has no position in the graph (Table 8.8).
- Path depth. Few contracts both grant and receive a capped allowance, so the findings about propagation on the allowance layer do not carry over to graphs with longer endorsement paths (Section 5.3.2).

5.8.4 Adversarial validity

The threats below concern what an attacker could do to the scores.

- Sybil and wash trading. Rankings may be manipulated through allowance or transfer patterns among contracts that one actor deploys (Douceur, 2002); the Sybil-adjusted proxies only partly address this, and the cost model of Section 5.6 is analytical, not an empirical red-team evaluation.
- Strategic behavior. Once a score built on allowances is live, rational actors may deploy contracts, fund them at the freeze and grant allowances to farm rank, and the account-type rule is public, so a contract can be written to fall on either side of it. An observational τ estimated before deployment may overstate how well the ranking resists such behavior.

5.9 Limitations

On this sample, the strengths of C-PR are a crawl that reaches the contracts with the most user-account demand sooner than breadth-first order, a gain from its allowance layer on later approval pairs and, by a small amount, in that crawl, and a ranking of later approval pairs above AWP and of later senders above every allowance-side score. Its weaknesses are a crawl that is no sooner than the backlink count and later than AWP's, the failure of rule A and of F2, rankings of later senders below AWP and the transfer and union in-degrees and of later approval pairs below EndorseRank, its own allowance layer and the in-approve degree, a Sybil-stability coefficient below AWP's and the highest cost of the walks.

The coupled operator was tried at three fixed values of λ, with uniform teleportation and a per-node convex mixture at the transition level, on two layers weighted by dollar amounts, with each allowance counted up to its owner's balance and up to U. Layers weighted as in EndorseRank and AWP, node-dependent or learned weights, mixtures of the stationary vectors in place of the transition rows, and other Markov chains are not evaluated, so the answers to RQ2, RQ4 and RQ5 apply only to that family of methods on this dataset. EndorseRank, the allowance reference, orders one cohort contract in five, ranks later approval pairs below the raw in-approve degree and, like C-PR, is exposed to closed farms. None of the scores says anything about contracts that few owners

approve or about user accounts, which are not scored (Section 3.2).

The data add four limits. The graphs hold five tokens only (Table 8.3), so a contract that owners approve only on other tokens is outside the cohort. Every price comes from one public service, the daily series of DefiLlama (DefiLlama, n.d.), so an error in it on some date would pass into every amount of that day (Appendix 8.4.2). The account type was read once, from code at one block, and the detection of smart accounts is incomplete (Table 8.4). And importance is measured on-chain from user accounts only (Section 5.8.1).

None of the scores draws on off-chain evidence about who controls an address; their only off-chain inputs are the token list and the daily price series. Unlike attestation tooling such as Human Passport (Human Passport, n.d.), they give no guarantee that two highly ranked contracts are controlled by different actors, so they should be read as orderings of an already admitted population and not as a Sybil defense.

For this cohort of contracts and time frame, neither C-PR nor EndorseRank is a rating of default risk, a Sybil-resistant identity mechanism or a security audit. On these data counts were hard to beat: the backlink count in discovery and the in-approve degree on later approvals, with AWP the one single-layer walk that added to its count. Chapter 6 restates what has been shown, what that knowledge is worth and what it would take to extend these bounds.

6 Conclusion and Future Work

Chapter 5 interpreted the findings and judged whether the study met its objectives.

6.1 Answers to the research questions

The thesis built a coupled PageRank (C-PR), one walk over the ERC-20 allowance and transfer edges between Arbitrum One contracts, and set it beside EndorseRank, Adaptive Weighted PageRank (AWP) (T. Do et al., 2023), the ablation S-PR and the raw counts in a discovery-order test at two freezes and in the holdout windows W0 (April to June 2026) and W1 (July to September 2026). After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. The verdicts below under rules A and B are a re-analysis, and the registered analysis gave the same readings (Appendix 8.6).

RQ1 asked whether EndorseRank and AWP order the same contracts differently, whether each agrees most with current checks of its own edge type, and what each walk costs. H1 is supported, largely by construction: most cohort contracts hold no capped allowance from another contract and score zero under EndorseRank, so both walks order only about a third of the contract pairs (Section 4.4). Each walk agrees most with the proxies of its own edges (Table 4.5), which supports H2 only narrowly, since the proxies come from the same events as the graphs. Cost follows the size of the layers walked and the iterations needed, so EndorseRank, on the far smaller allowance layer, is cheapest and C-PR takes slightly longer than AWP (Table 4.3); H3 is supported, largely by construction.

RQ2 asked whether a crawl ordered by C-PR on the contracts read so far reaches the contracts with the most user-account demand sooner than breadth-first and backlink-count crawls. The answer is no. The test's decision rule, read at T_{obs} in the primary reveal mode, gives verdict V2 on both importances (Table 4.12): "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most user-account inflow and those with the most user-authorised value sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks. A crawl ordered by the number of contract backlinks reached them sooner." On user-authorized value its lead over breadth-first order is small and does not hold in every sensitivity (Table 8.9). Outside the decision (Table 4.14), AWP′ was the one single-layer walk sooner than its count on both importances, and the allowance layer added a small, resolved amount to C-PR′ over its transfer layer. In the other reveal mode, which shows out-edges only and carries no verdict, C-PR′ was sooner than both comparators at both freezes. Every crawl stayed far from the ideal order (Table 4.13). The test decides no rule of the plan or the registration.

RQ3 asked whether frozen scores predict later new approval pairs and new transfer senders, and whether any walk beats the raw count of the edges it smooths. The first part holds in both windows, and the edge type decides which label a score anticipates (Tables 4.15 and 4.20). On the second the counts mostly win. EndorseRank ranks later approval pairs below the in-approve degree in both windows with either restart rule (Section 4.7; Tables 4.16 and 4.18). AWP ranks later senders slightly above the transfer in-degree on the whole cohorts, but below it among the W0 spenders another contract had paid before the freeze (Table 4.17). C-PR ranks later approval pairs above the union in-degree, the count of its edges, and later senders below it, and lies below the in-approve degree and the transfer in-degree on their labels (Table 4.22). The labels count mostly user accounts, which no score reads, and more than a quarter of each spender cohort lies outside the union graph and scores zero (Tables 8.6 and 4.2).

RQ4 asked whether the coupled walk ranks contracts better than either of its layers walked alone, in and out of window. The answer is no. Rule A is not met in either part: in W0 C-PR ranks new

approval pairs below its allowance layer and new senders below its transfer layer (Table 4.16), and in the same window it lies outside the interval of the leading layer on the transfer and allowance families and does not exceed both layers on Sybil stability (Table 4.19). Among the spenders inside the union graph the contrast with the allowance layer does not exclude zero, a comparison that decides no rule (Table 4.23).

RQ5 asked whether adding the allowance layer improves a transfer walk on new data, gaining on later approvals without losing on later inflow. The answer is again no. The thesis applies two of rule B's registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds. F1 holds: in W1 C-PR ranks new approval pairs above its transfer layer walked alone. F2 fails: its loss on new senders exceeds the margin. Rule B is not met, and the two registered sensitivity analyses give the same reading (Table 4.21).

Claims C1–C6 (Section 4.9) state these findings formally.

The main research question asked whether one walk over both edge types finds the contracts that users rely on sooner, and predicts their later use better, than single-layer walks and raw counts. On these data it does not. Its crawl is sooner than breadth-first but later than the backlink-count and AWP′ crawls, it meets neither rule A nor rule B, and out of window it beats neither single-layer count on that count's own label. What the allowance layer adds is small and resolved: a little in discovery order, and a gain on later approval pairs paid for on later inflow.

6.2 Original contribution to knowledge

Read against the literature of Chapter 2, the five contributions of Section 1.9 establish the following. In the work reviewed, graph reputation on public ledgers ranked addresses by the transfers among all of them (T. Do et al., 2023), by risk propagation in transaction networks (D. Lin et al., 2025) or by explicit ratings (Bellini et al., 2020), and DeFi reputation tooling scored an address's holder from credentials (Human Passport, n.d.). None of it ranked contracts on the relations among contracts alone, used the ERC-20 allowance (Vogelsteller & Buterin, 2015) as the edge of a ranking

or coupled it with payments in one walk; approvals had been used to cluster the addresses of one entity (Victor, 2020) and studied as a security risk (Wang et al., 2022). None asked of a ledger ranking what the crawl-ordering experiment asked of web rankings (Cho et al., 1998), how soon it leads to what is important, and none set on-chain scores beside the raw counts they smooth, with confidence intervals, paired differences and temporal holdouts. This thesis supplies these.

The new knowledge has five parts. First, a ranking of contracts can be computed from the relations among contracts alone, and what it measures depends on its edges, which can be checked on one cohort with quantified uncertainty. Second, the coupled walk leads a crawl to the contracts users rely on sooner than breadth-first order but not sooner than the count of contract backlinks, a result that depends on what a visit reveals. Third, the coupled walk meets neither rule that sets it against its own layers. Fourth, the counts are hard to beat: C-PR ranks no label above the degree of that label's edge type, and of the single-layer walks only AWP leads its count, in the deciding discovery-order run on both importances and, slightly, on later inflow, though not among the spenders that another contract had paid. Fifth, an allowance between contracts is rarely backed: counted up to the owner's balance, most such allowances form no edge, and on the thin layer that remains a walk with uniform restarts is at depth one, up to a constant, a linear function of the transition mass into a spender (Section 5.3.2). The open pipeline, whose published summaries let others recompute every table, is part of the contribution, since automated scores that outsiders cannot inspect are a known concern (Packin & Lev-Aretz, 2024).

The scope is narrow. The results apply to the contract graph of the five tokens on Arbitrum One in these months, with addresses typed from code read at one block, to the cohorts drawn from it and to the coupling the plan fixed (a fixed λ, dollar-weighted layers, uniform restarts). The allowance layer between contracts is thin, and at T_{obs} one owner contract holds most of its pairs, so that contract's allowances weigh on the allowance-side results of that freeze (Table 4.1). No score reads a contract's code or says whether it is safe, and none resists Sybils: under uniform restarts a closed farm of contracts collects several times its population share (Section 5.6).

6.3 Future work

Each suggestion below answers a limitation of Chapter 5. Any new score, weighting or test it leads to should be fixed before it is scored, replicated on new data and checked against the counts it smooths.

- Credit decisions about contracts. A ranking that found the contracts users rely on could serve as one input to such decisions. The thesis does not test this use, and its results do not yet support it: the walk did not find those contracts sooner than a count, and the use would also need a contract outcome bearing on repayment, which this study lacked.
- Importance labels from outside the chain. The discovery-order test measures importance on chain, as the capped dollars user accounts pay to a contract or let it spend, and so depends on the rule that separates user accounts from contracts. Labels of which contracts matter, collected off chain by people who examine contracts, would test the orderings against a judgment sharing no data with them.
- Detection of smart accounts. The rule recognizes Safes and ERC-4337 accounts. The kinds it misses, and contracts that services create for a single user, order or payment, stay in the graphs as contracts (Appendix 8.4.3); a better rule would show how far they shape the transfer layer and the importance measures.
- A later freeze. The discovery-order test and the comparisons with the counts were defined after the registered results were known, and the two freezes share nine months of their score windows. Repeating the test on a later freeze, with its protocol and primary reveal mode fixed before that freeze's data exist, would give a registered answer to RQ2, and other rollups would show whether the answer extends beyond Arbitrum One.
- A coupled walk with activity restarts. C-PR differs from AWP in coupling, weights and restarts at once, and AWP′, which restarts by sending activity, was the soonest walk of the deciding run. A coupled walk whose layers carry the edges and weights of AWP and EndorseRank and whose restarts mix sending and approving activity (C-AWP) would set the

coupling alone against the published walk.

- Adversarial evaluation. The Sybil cost model is analytical, and no attack was carried out on chain. Restricting teleportation and dangling redistribution to a vetted seed set, as TrustRank does (Gyöngyi et al., 2004), would remove the uniform inflow that closed farms exploit, and red-teaming with farms of funded contracts would measure the realized cost of an attack.

6.4 Closing reflection

A rollup's contracts are many, and the relations among them are public. A transfer between two contracts records a payment, and an allowance, counted only up to what its owner holds, records a spending right the owner can back. PageRank will rank either graph, and one walk can follow both. Here that walk was not the best guide to the contracts that users rely on. A crawl ordered by the number of contracts that pay or approve a contract reached them sooner, and out of window each edge type's count ranked the later arrivals of that type above the coupled walk. For a wallet application, an explorer or an auditor choosing from public logs which contracts to examine first, the counts on the contract graph are the baseline a walk must beat: they cost least, and here they beat it. A walk earns its cost only where shown to beat its count, which AWP did only in part. The allowance between contracts proved a thin layer. Whether a ranking built from these relations finds what people who examine contracts would choose is a question for labels such people supply.

7 References

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the evaluation pipeline behind Chapter 4 and behind the registered analysis of Appendix 8.6.

8.1.1 Pipeline stages

The evaluation source is in the folder `experiments` of the public repository https://github.com/abitocodes/ERC20-Allowance-PageRank-Wallet-Reputation, and the paths below are relative to it. The evaluation has seven stages. The first four are those of the registered analysis: three run on Arbitrum One over the window of scan A, 1 October 2023 to 30 June 2026 (UTC, inclusive at both ends), and the fourth adds July to September 2026. The last three build and evaluate the contract graph of the main analysis.

1. Plan, cohort and graphs: Commit the analysis plan (`docs/analysis_plan.md`) and the configuration before any log of the study is extracted. Copy the logs of the window into a project table (scan A), list the spenders that at least three owners approved (`sql/02_spender_candidates.sql`), read their account types (`scripts/check_account_code.py`) and keep the contracts as the plan's cohort (`scripts/build_cohort.py`). Extract the ego events of the cohort (`sql/03_ego_events.sql`) and aggregate them into the pair tables of both graphs at T_{obs} and at t_1 (`sql/04_anchor_pairs.sql`).
2. Checks and alignment: Score every method at T_{obs} with the solver settings of Section 3.1, compute the checks of Section 3.10 and report Spearman ρ and Kendall τ with 95% paired-bootstrap intervals (400 resamples, seed 42) and the contrasts of Section 3.11 (`scripts/run_same_window.py`).

3. Temporal holdout W0: Score every method at t_1 and evaluate the spender cohort against the labels of April–June 2026, with the contrasts of rule A (`scripts/run_holdout.py`).
4. Registered window W1: Commit the registration (`docs/registration_w1.md`) with the status in the configuration set to registered. `scripts/extract_w1.py`, which will not run before that commit, runs scan B and the ego extraction of July–September and writes an extraction manifest; `scripts/run_registered.py` evaluates rule B.
5. Tokens, prices, cohort lists and labels: `scripts/usd_inputs.py` builds the token list and the daily prices, and `scripts/chainlink_check.py` compares them with the Chainlink feeds. `scripts/run_usd_pipeline.py` and `scripts/run_usd12m_pipeline.py` load both and, from the five-token events of the ego tables of stages 1 and 4, list for each freeze the contract spenders that at least three owners approved in its score window and count the labels of the spenders with a positive latest allowance at the freeze (`sql/21`, `sql/16`).
6. Contract graph: `scripts/run_contract12m_pipeline.py` runs the BigQuery jobs (`sql/24` to `sql/43`) and the RPC reads of the contract graph in a fixed order, in two groups of steps that it calls gate A and gate B. The steps of gate A are `check`, `subset`, `ends`, `activity`, `queue`, `typing`, `load`, `contracts`, `balances`, `anchors`, `audit` and `structure`: they read the five-token events of 1 April 2025 to 30 September 2026, type the accounts (`scripts/check_account_code.py`), keep the events between contracts, reconstruct and check the balances (`scripts/balance_audit.py`) and end with the structure pilot of Section 3.7. The steps of gate B are `constants`, `pairs`, `rest`, `checks` and `download`: they compute the capped allowances, the constants, the pair tables, the cohorts, the proxies, the label parts and the importance, check 64 invariants, each a count of violations that must be zero, and download the tables. The run manifest records every job, the result of every step and every decision by which the author let a step go on; the digest quotes these decisions.
7. Evaluation of the contract graph: With `CONTRACT_REP_VARIANT=contract12m` the evaluation scripts read the downloaded tables, in this order (all in `scripts/`): `run_same_window.`

py (with the damping sweep and the secondary part of rule A), run_discovery.py (the four runs of the discovery-order test), run_holdout.py (W0 and the primary part of rule A), run_registered.py (W1: F1, F2 and F6 with the two registered sensitivity analyses), run_robustness.py (value slope, sample definition and the uncapped allowance layer) and run_benchmark.py. Then run_supplementary.py computes, from the saved scores and with the same resamples, the contrasts that belong to no rule (Table 4.22), sybil_model.py the quantities of Section 5.6 and describe_data.py those of Appendix 8.4; publish_results.py writes the published summaries with their digest, and export_latex.py the tables in results/tables/ and the figures in Figures/generated/.

The timed runs were made with no other job of the pipeline running, on a machine with an Intel Core Ultra 9 185H processor and 64 GB of memory under Windows 11 (build 26200), with Python 3.11.9, NumPy 2.3.2 and SciPy 1.16.1.

The experiments folder holds the queries, scripts and tests, the versioned configuration (config/contract_reputation.yaml with the overlay config/overlay_contract12m.yaml), the plan, the registration and the four documents that define the main analysis (docs/: revision_price_weighting.md, revision_one_year_window.md, revision_contract_graph.md and discovery_protocol.md), the token list and daily prices with their Chainlink comparison (data/0-usd-token-list-and-prices/) and the cost ledger. The summaries behind the tables are those of commit 6b389df of the public repository, in data/3-published-results-for-thesis/revised-contract-12m/, and those of the registered analysis are in data/3-published-results-for-thesis/; the digest of the main analysis records the code revision at which it was written. Each of the two folders holds copies of the configuration, the plan and the registration, a digest results_digest.md, which names the summary file and key of every number, and SHA256SUMS.txt, so any number in a table can be checked against its summary without re-running the pipeline. The extracted tables and the per-contract scores, stored as parquet parts below 90 MB, are kept in a public Google Drive folder; the committed file drive-manifest.json gives its link and lists every part with its checksum.

Synthetic tests check the solver, the edge weights, the paired bootstrap, the US-dollar conversion, the balance cap, the account typing and the crawl offline. Real extraction needs Google Cloud credentials and a billing project.

8.1.2 BigQuery extraction design

The public table `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs` is read once for each window. Scan A copies the ERC-20 `Approval` and `Transfer` logs from 1 October 2023 to 30 June 2026 into one project table partitioned by month, and scan B does the same for July to September 2026. Every later query on these logs reads the project tables, so the chain-wide scan is paid for once per window. The registered analysis extracts from them the events with a contract of the plan's cohort at one end. The main analysis draws from them the events of the five tokens from 1 April 2025 to 30 September 2026 into one table, which its later jobs read: an edge needs a contract at both ends, so these events come from the whole chain and not from the neighborhood of a cohort. From the public data set it also reads the transaction senders among the addresses of these events, with two cross-checks of the typing, the last block at or before each freeze and the `Transfer` logs of the five tokens before 1 October 2023, with which the balances begin. Appendix 8.2 shows the queries.

Every job is listed in `data/bq_ledger.json` with its query, parameters, estimated and billed bytes and cost. The jobs of the study billed 8.78 TiB, or USD 54.88 at the on-demand rate of USD 6.25 per TiB, against the budget of USD 60 set in the configuration of the main analysis. The jobs of the registered analysis billed USD 29.71 and the 26 jobs of the contract graph USD 12.63; the ledger lists the other jobs one by one, those of stage 5 among them.

8.1.3 Account-type classification

Section 3.5.1 states the rule by which an address is a user account or a contract; this section documents the reads behind it. An address that sent a transaction is a user account without a read. `scripts/check_account_code.py` read every other address that can end an edge between contracts,

every member of the cohort and spender lists and two audit samples of 1,000 addresses each with `eth_getCode` at block 513,216,506, from the public endpoints `arb1.arbitrum.io` and `arbitrum-one.public.blastapi.io`. At that block the registered analysis had read the 29,022 candidates of the plan's cohort, of which 27,844 held contract code, 1,121 no code and 57 an EIP-7702 delegation. The block lies after the end of the data, so every address is typed by the code it held on that day.

No code, or a delegation designator (`0xef0100` followed by 20 bytes), means an externally owned account. For other code the script recognizes ERC-1167 clones (Murray et al., 2018), ERC-1967 proxies (Palladino et al., 2019) and Safe proxies (Safe Ecosystem Foundation, n.d.), reads their implementation and calls `getThreshold()`, `getOwners()` and `entryPoint()`: a Safe answers a threshold from 1 to 1,000 and a non-empty list of owners, and an ERC-4337 account (Buterin et al., 2021) answers one of the four canonical EntryPoint contracts (versions 0.6 to 0.9) and holds, in its code or its implementation's, a `validateUserOp` selector and no `validatePaymasterUserOp` selector. The calls are made per address for ERC-1967 and Safe proxies and for code that answers `getThreshold()`, and once per code hash otherwise. Table 8.4 in Appendix 8.4.3 gives the counts and the code families that complete the rule. No call was aborted at every attempt, and none of the 1,000 audited transaction senders held contract code.

8.1.4 Benchmark run logs

Table 8.1 lists the wall-clock seconds of every timed solver call of the benchmark, in the order the runs were made, with their mean and standard deviation. Each configuration had one untimed warm-up before its five timed runs, and each graph and setting was timed once: the $d = 0.85$ rows of Table 4.7 and the whole-graph row of Table 4.4 repeat the first block of this table.

8.1.5 Parameter summary

Table 8.2 collects the default parameters of the main analysis; every value is read from the versioned configuration or, for the tokens, prices and constants, from the inputs committed with the analysis

Table 8.1: Wall-clock seconds of every timed solver call of the benchmark, in the order the runs were made, with their mean and standard deviation. Each configuration had one untimed warm-up before its timed runs, and each graph and setting was timed once: the $d = 0.85$ rows of Table 4.7 and the whole-graph row of Table 4.4 are the first block, reused. The subsamples are those of Table 4.4.

Method	Run 1	Run 2	Run 3	Run 4	Run 5	Mean	SD
Whole contract graph, $d = 0.85$							
EndorseRank	0.101	0.112	0.096	0.107	0.089	0.101	0.009
AWP	2.964	2.792	2.852	2.514	2.984	2.821	0.189
C-PR ($\lambda = 0.5$)	2.954	2.963	3.394	3.095	2.993	3.080	0.184
S-PR	2.997	2.682	2.512	2.527	2.565	2.657	0.202
Whole contract graph, $d = 0.75$							
EndorseRank	0.082	0.088	0.093	0.085	0.081	0.086	0.005
AWP	2.146	1.687	1.613	1.562	1.769	1.755	0.232
Whole contract graph, $d = 0.95$							
EndorseRank	0.179	0.233	0.241	0.204	0.185	0.208	0.028
AWP	6.870	6.961	7.566	7.555	6.957	7.182	0.348
Subsample of the cohort, $n = 1{,}000$							
EndorseRank	0.001	0.001	0.001	0.001	0.001	0.001	0.000
AWP	0.096	0.096	0.095	0.095	0.095	0.095	0.000
Subsample of the cohort, $n = 2{,}000$							
EndorseRank	0.002	0.002	0.002	0.002	0.002	0.002	0.000
AWP	0.135	0.131	0.132	0.138	0.132	0.134	0.003
Subsample of the cohort, $n = 5{,}000$							
EndorseRank	0.007	0.006	0.006	0.006	0.006	0.006	0.000
AWP	0.693	0.735	0.698	0.657	0.663	0.689	0.031

(Section 3.5.3), and ρ, q and the primary reveal mode are those that the rules of the protocol set (Section 3.7).

8.2 SQL Templates

This appendix shows the two query patterns on which the contract graph rests, abridged; the full files, with those of the registered analysis, are in `sql/` of the experiments folder (Appendix 8.1.1). Names that start with `@` are query parameters.

Table 8.2: Default parameters used in the evaluation.

Parameter	Value	Role
PageRank damping d	0.85; sweep 0.75, 0.95	Teleport mixture
Convergence tolerance ε	10^{-8} in L_1 norm	Power-iteration stop
Max iterations $I_{\max}$	300	Power-iteration cap
Logistic k	0.01	Decay slope (AWP, EndorseRank, transfer layer of C-PR)
Logistic t_0	180 days	Decay midpoint (AWP, EndorseRank, transfer layer of C-PR)
Value transform b	$\ln 3/m$ per dollar, $m_T = 160$, $m_A = 0.071$	Value weight $V(z)$ of AWP (m_T) and EndorseRank (m_A)
Balance cap	allowance counted up to the owner's balance at the freeze	Every allowance weight
Allowance cap U	16,000 dollars	C-PR allowance layer; in-approve value
Cap per user account U_{user}	31,000 dollars	Importance I_T and I_A of the discovery-order test
Tokens and prices	WETH, USDC, USDT0, ARB, WBTC; DefiLlama daily close	Events and US-dollar amounts (Table 8.3)
Restart distribution	$\propto X_u$; uniform over the contracts of a graph	AWP; EndorseRank, C-PR
Coupling weight λ	0.5 primary; 0.25, 0.75	C-PR allowance-layer weight, fixed in the plan
S-PR restart vector	C-PR ($\lambda = 1$) on the transfer-layer nodes, renormalized	Endorsement-seeded teleportation
Score windows	2025-07-01 – 2026-06-30; 2025-04-01 – 2026-03-31	Same window and W1; W0. UTC inclusive; the freeze anchors the decay
Events of the contract graph	2025-04-01 – 2026-09-30	Both score windows and both label windows
Account-type block	513,216,506	Code and calls of the typing
Extraction (scan A)	2023-10-01 – 2026-06-30	Scored in full by the registered analysis
W0 freeze / labels	2026-03-31 / 2026-04-01 – 2026-06-30	Spender cohort; rule A
W1 freeze / labels	2026-06-30 / 2026-07-01 – 2026-09-30	Spender cohort; F1 and F2 of rule B, margin $\delta = 0.01$
Bootstrap resamples / seed	400 / 42	Paired bootstrap over contracts; over hot contracts in the crawls
Crawl seeds	20 contracts of the giant SCC	Starts of the crawls
Recompute share ρ	$1/50$; $1/200$ on five seeds	Schedule of the orderings; sensitivity
Hot share q / area share x	1%; 5% / 10%; 5%, 20%	Hot set $H_q(I)$ / AUC_x; sensitivities
Reveal mode	`both`; `out`	Primary; sensitivity
Timed runs / warm-up	5 / 1	Runtime protocol
Scaling seed	`contract-benchmark-v1`	Deterministic SHA256 subsamples
BigQuery budget	USD 60	Checked before every query

8.2.1 Event selection pattern

Scans A and B tag each log of the public table by the topic hash of its event and store its emitter, its two indexed addresses `a` and `b` and its 32-byte amount (`sql/01_materialize_logs.sql`). `sql/24_c12_logs5.sql` keeps the events of the five tokens from both scan tables, and `sql/31_c12_contracts.sql` those between two contracts. Their filters, abridged:

```
-- sql/24: the events of the five tokens (src: the rows of both scan tables)
FROM src AS s
WHERE s.emitter IN (SELECT token FROM `dissertation-bq.contract_rep.u_tokens`)
  AND (s.kind = 'A' OR LTRIM(TO_HEX(s.val), '0') != '')
-- sql/31: the events between contracts
FROM `dissertation-bq.contract_rep.c12_logs5` AS l
WHERE l.a IN (SELECT address FROM `dissertation-bq.contract_rep.c12_contracts`)
  AND l.b IN (SELECT address FROM `dissertation-bq.contract_rep.c12_contracts`);
```

For an approval (`kind` A), `a` is the owner and `b` the spender; for a transfer (T), `a` is the sender and `b` the recipient. Approvals of zero are kept, because an approval of zero sets an allowance to zero; a transfer is kept only when its value is not zero, and it is valued at its token's price on the UTC day

of its block. `c12_contracts` holds the addresses typed as contracts (Appendix 8.1.3).

8.2.2 Latest-allowance and pair-table pattern

`sql/36_c12_allow_triples.sql` reduces the approvals between contracts to the latest allowance of each token, owner and spender in the score window of a freeze and caps it at the owner's balance; `sql/37_c12_pairs.sql` sums the weights of each pair. Abridged, for T_{obs}:

```
latest_tobs AS (                                    -- sql/36; latest_t1 likewise
  SELECT token, owner, spender, value, block_timestamp FROM (
    SELECT token, a AS owner, b AS spender, value, block_timestamp,
           ROW_NUMBER() OVER (PARTITION BY token, a, b
                              ORDER BY block_number DESC, log_index DESC) AS rn
    FROM `dissertation-bq.contract_rep.c12_events_cc`
    WHERE kind = 'A' AND block_timestamp >= TIMESTAMP(@tobs_start)
      AND block_timestamp <= TIMESTAMP(@tobs))
  WHERE rn = 1 AND value > 0),
-- with the decimals, the anchor day's price_usd and the owner's balance bal:
LEAST(value / POW(10.0, decimals) * price_usd,
      IF(bal >= POW(10.0, decimals - CAST(@dust_digits AS INT64)), bal, 0.0)
        / POW(10.0, decimals) * price_usd) AS usd_eff
-- sql/37, per (tag, owner, spender) over the triples with usd_eff > 0:
SUM(sig(age_days) * vt(usd_eff, CAST(@b_allowance AS FLOAT64))) AS w_er,
SUM(LEAST(usd_eff, CAST(@cap_allowance AS FLOAT64))) AS w_raw,
-- and per (tag, sender, recipient) over the transfers of c12_events_cc:
-- w_awp = SUM(sig * vt(usd, @b_transfer)), w_raw = SUM(usd * sig)
```

The window function picks the latest approval of each token, owner and spender by block number and log index; it is kept only when its value is positive, so a revocation removes that token from the pair. `usd_eff` is the smaller of the allowance and the owner's balance, both at the price of the anchor's day, with a balance below 10^{-6} token counted as zero (Section 3.5.2). `w_er` and `w_awp` are the weights of EndorseRank and AWP and `w_raw` those of the layers of C-PR; `sig` and `vt` are the logistic decay of Equation (3.3) and the value transform of Equation (3.4), set to one from $bz = 40$ on, where it equals one in double precision. Each pair row carries the weights of every method, so all scores at one anchor come from the same rows. The same query writes the weights with the slope multiplied and divided by ten, for the check of Table 4.8, and the pairs without the balance cap, for that of Table 4.11.

8.3 Elementary Properties of PageRank and Rank Correlation

Chapters 2–4 rely on the elementary properties summarized here. The material is standard: the PageRank properties follow Page et al. (1999) and Langville and Meyer (2006), and the rank correlations Spearman (1904) and Kendall (1938).

8.3.1 Existence and uniqueness of PageRank

Take P to be a row-stochastic matrix on $|V|$ nodes, possibly after the dangling-node repair of Equation (2.1). For $d \in (0, 1)$ the Google matrix of Equation (2.2),

$$G_{\text{PR}} = dP^{\top} + \frac{1-d}{|V|}\mathbf{1}\mathbf{1}^{\top},$$

is positive and column-stochastic. By the Perron–Frobenius theorem for positive matrices, G_{PR} has a single eigenvalue of modulus one, and the corresponding eigenvector $\mathbf{r}$ is strictly positive; we can scale it so that $\sum_i r_i = 1$. It follows that, once the damping and the dangling-node rules are fixed, every directed graph has unique PageRank scores (Langville & Meyer, 2006).

Three restart vectors in this thesis are not uniform. AWP's, proportional to activeness, is zero at every contract that has sent nothing, that of EndorseRank with activity restarts is zero at every contract that has approved nothing, and S-PR's, the normalized score of the allowance walk, is zero at every node of the transfer layer that has no allowance score, so none of the three Google matrices is positive. The fixed point is still unique: for two probability vectors the restart term cancels in their difference, and the operator multiplies their $\|\cdot\|_1$ distance by at most d. In general it is not strictly positive, however, because a node with no restart mass that no path from a restart node reaches keeps a score of zero. Uniqueness means that, once its parameters are fixed, each method assigns exactly one score vector to a given edge set, so any difference between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ arises from their edges and restart distributions.

For fixed d the scores also depend continuously on the entries of P: a second row-stochastic matrix

$\tilde{P}$ on the same nodes moves them by at most $\frac{d}{1-d} \max_i \|P_{i\cdot} - \tilde{P}_{i\cdot}\|_1$ in the $\|\cdot\|_1$ norm (Langville & Meyer, 2006). A row of P is not continuous in the edge weights, however: it jumps between the uniform dangling row and a row concentrated on its out-neighbors when a node gains its first out-edge or loses its last one, however small the weight involved.

8.3.2 Sparsity and per-iteration cost

With $m = |E|$ nonzero edges and $n = |V|$ nodes, the sparse product $P^\top \mathbf{x}$ costs $O(m)$ arithmetic operations, each iteration also updates vectors of length n, and building the sparse operator from the edge list is $O(m)$, so power iteration over I iterations is $O(m + (m + n)I)$. Both scores run the same solver. In the measured calls an iteration of AWP takes about 15.9 times as long as one of EndorseRank, less than the edge ratio of 22.4, and AWP needs more iterations (Section 4.2), so neither m nor I alone sets the time ratio.

8.3.3 Properties of Spearman's ρ and Kendall's τ

Spearman's ρ and Kendall's τ_b lie in $[-1, 1]$, and the evaluation pipeline computes both with standard tie-aware implementations. Across the fourteen pairs of EndorseRank or AWP with a proxy (Tables 8.12 to 8.14) both coefficients are positive and ρ is the larger of the two. The denominator of τ_b in Equation (2.6) differs from n_0 only through ties, and among the scores of this thesis ties are frequent: a contract outside the graph of a score scores zero, so on the matched cohort EndorseRank ties 65.4% of the contract pairs and AWP 13.7% (Section 4.4), and the integer proxies tie many pairs as well. A contrast between two scores on one proxy therefore also reflects which contracts each score leaves at zero. Table 4.6 repeats the family means with the contracts outside a graph kept as isolated nodes, and Table 4.23 the main out-of-window contrasts on the members inside the union graph.

8.4 Data Description

This appendix describes the data behind Chapter 4: the events and the layers of the contract graph, the five tokens with their prices and allowance amounts, the account types, balances and labels of the contract graph, and the detail of the discovery-order test.

8.4.1 Extracted events and graphs

The events of the main analysis are the `Approval` logs of any value and the `Transfer` logs of non-zero value that the five tokens emitted from 1 April 2025 to 30 September 2026, the months that the two score windows and the two label windows cover; Section 4.1 gives their number. They form $73{,}058{,}133$ distinct ordered pairs of addresses. Only the rows with a contract at both ends enter a graph (Table 4.1). The node table numbers $1{,}403{,}959$ contracts over both freezes: every contract that ends a pair of either layer, with or without the balance cap, and every contract of the cohort and spender lists. A further $152{,}257{,}072$ rows of the two score windows lead from a user account to a contract or to an address that was not typed; no score reads them, and the importance measures of the discovery-order test are computed from them (Section 3.7). The registered analysis extracted the events of every ERC-20 token with a contract of the plan's cohort at one end, from 1 October 2023 to 30 June 2026: $389{,}866{,}882$ approval logs, zero-value ones included, and $1{,}538{,}404{,}067$ transfer logs of non-zero value.

8.4.2 Tokens, prices and allowance amounts

Section 3.5.3 gives the rule by which the five tokens were chosen. Table 8.3 gives for each of them the address, the decimals and the Chainlink feed, the positive latest allowances between contracts at T_{obs} before and after the balance cap, and the data behind the daily prices.

Each of the five tokens has a DefiLlama price under its own Arbitrum One address on each of the 548 days from 1 April 2025 to 30 September 2026: no day is taken from a second key of the service, filled from the previous day or corrected as a one-day error. Against the Chainlink feeds read on

Table 8.3: The five tokens of the analysis, with their Chainlink feed (market-risk category), the positive latest allowances between contracts at T_{obs} and those that stay positive after the balance cap, the days of the price series filled from the second DefiLlama key, filled from the previous day and replaced as one-day errors (of the 548 days from 1 April 2025 to 30 September 2026 that the windows cover), and the median and largest absolute difference, in percent, between the DefiLlama price and the Chainlink feed read on the chain on the 20 check dates among those days. The five-token events of the typing window number 1,240,275,042; 603,475,784 of them have a contract at both ends.

Token	Address	Dec.	Chainlink feed	Allowances	After the cap	2nd key	Filled	Errors	CL diff. (%)
USDC	`0xaf88…5831`	6	USDC / USD (low)	255,339	108,916	0	0	0	0.01 / 0.02
USDT0	`0xfd08…cbb9`	6	USDT / USD (low)	36,468	5,648	0	0	0	0.01 / 0.15
WETH	`0x82af…bab1`	18	ETH / USD (low)	30,826	1,395	0	0	0	0.04 / 0.14
WBTC	`0x2f2a…5b0f`	8	WBTC / USD (low)	11,841	386	0	0	0	0.05 / 0.24
ARB	`0x912c…6548`	18	ARB / USD (low)	4,808	378	0	0	0	0.06 / 0.24

the chain on 20 of these days per token, the DefiLlama prices differ by a median of 0.01% to 0.06% per token and by at most 0.24%.

Unlimited allowances are common. Of the $339{,}282$ positive latest allowances between contracts at T_{obs}, $223{,}892$ (66.0%) are of 1.15×10^{77} base units or more, close to $2^{256} - 1$, and $230{,}313$, the unlimited ones included, are worth at least the cap U before the balance cap (Table 8.5 gives the unlimited allowances per token). In the registered analysis, which kept every token in raw base units, 99.52% of the ego transfers and 99.78% of the $28{,}069{,}418$ latest allowances at T_{obs} were of ten base units or more, where V with $b = 1$ is within 10^{-4} of one, and $1{,}532{,}301$ of the $13{,}029{,}306$ owners with a positive latest allowance held both an unlimited and a smaller allowance.

8.4.3 Contract graph: account types, smart accounts and balances

Table 8.4 gives the account types behind the contract graph; Section 3.5.1 states the rule and Appendix 8.1.3 the reads.

The reads alone miss ERC-4337 accounts whose proxy hides the `validateUserOp` selector. Two lists of code families, drawn from the code alone without reading a label, a cohort membership or a score of the addresses they move, type $192{,}656$ more addresses as smart accounts. Counts of the structure pilot's graph with and without the candidate families were known when the lists were drawn; in each of them the rules of the protocol set the same reveal mode, schedule and hot-set

Table 8.4: Account types of the contract graph. The five-token events of the typing window have 25,537,280 distinct addresses other than the zero address; 17,477,779 of them sent a transaction and are user accounts by that fact. The table types the 6,230,270 addresses that were read with `eth_getCode` at block 513,216,506: every address that sent no transaction and has a counterparty that is neither a transaction sender nor the zero address, the members of the cohorts and two audit samples. An address outside this set that sent no transaction has only user accounts as counterparties and cannot end an edge between contracts. Smart accounts are user accounts: a Safe answers its threshold and owners, and an ERC-4337 account answers a canonical EntryPoint and carries a `validateUserOp` selector and no paymaster selector; 192,656 further accounts whose proxy hides that selector were moved by listed code families (37,337 by code hash and 155,319 by code hash and implementation). Of the addresses read, 524,454 are ERC-1967 proxies, 503,516 ERC-1167 clones and 45,362 Safe proxies. With the transaction senders, 21,877,279 addresses are typed user accounts and 1,829,719 contracts.

Code read at the block	Class	Addresses
Externally owned account without code	user account	2,909,509
Externally owned account with an EIP-7702 delegation	user account	1,165,025
Safe (threshold and owners answered)	user account	46,204
ERC-4337 account, by the rule of the reads	user account	87,157
ERC-4337 account, by the listed code families	user account	192,656
Any other code	contract (graph node)	1,829,719
Addresses read		6,230,270

share, and the stop flag stayed (Section 3.7). The first holds 20 code hashes of small proxies whose implementation the reads did not find and whose representative answers a canonical EntryPoint (37,337 addresses). The second holds seven pairs of a code hash and an implementation of ERC-1967 proxies whose implementation carries the selector and whose getter `entrypoint()` answers a canonical EntryPoint where `entryPoint()` answers none (155,319 addresses).

Kinds of smart account that the rule does not find stay contracts and graph nodes. After the two lists, 115,084 typed contracts still carry a `validateUserOp` selector and no paymaster selector in the code examined: 104,187 of them share one code hash whose getter `entrypoint()` reverts or does not exist, and 8,793 run one implementation whose EntryPoint is not a canonical one. Contracts whose code suggests a smart account of another kind, neither a Safe nor an ERC-4337 account, stay as well, as do the contracts that services create for each payment, order or user, which no rule picks out.

The balance of an owner in a token at a freeze is the net flow of the token's `Transfer` logs to the

owner from the first block of the chain up to the freeze, set to zero when it is negative or below 10^{-6} token (Section 3.5.2). It was computed for the $6{,}033{,}263$ pairs of a token and an owner with an approval of that token from 1 April 2025 to T_{obs}. A check compared $1{,}500$ of these pairs, for each token 150 with a zero and 150 with a positive reconstructed balance at T_{obs}, with `balanceOf` at block $479{,}089{,}705$, the last block of 30 June 2026: $1{,}493$ agree (99.5%, against a threshold of 98%). For the other seven, four of WETH and three of USDT0, each with a reconstructed balance that counts as zero, the call returned no value, and the check counts them as disagreements; no pair with an answer differs.

Table 8.5 gives the effect of the balance cap on the latest allowances between contracts, per token. Most of the allowances that stay positive are counted at a balance below the allowed amount ($114{,}454$ of $116{,}723$ at T_{obs}), and the two freezes differ in USDC, of which $108{,}916$ allowances stay positive at T_{obs} and $11{,}381$ at t_1. After the cap 150 allowances at T_{obs} and 204 at t_1 are worth U or more, so the cap U of the allowance layer of C-PR binds on few of them. The labels, unlike the graphs, count every account, and Table 8.6 splits each label of the two label windows by the class of the account it counts (Section 4.7).

8.4.4 Discovery-order test: universes, coverage, sensitivity and diagnostics

This section holds the tables of the discovery-order test that Section 4.6 reads; Section 3.7 defines their terms. The test was defined after the registered results were known, and it decides no rule of the analysis plan or the registration.

Table 8.7 gives the sizes of the contract graph, of its bow-tie and of the two universes at both freezes, with the sizes of the hot sets. Table 8.8 shows how much of the two importances each universe holds: at T_{obs} the universe of the `both` mode holds 96.5% of I_T and 93.0% of I_A by value, but 64.2% and 38.1% of the contracts that make up the top 1% by each.

Table 8.9 changes one setting of the deciding run at a time. In no row does the interval of D_2 lie above zero; the interval of D_1 on $H_q(I_A)$ lies below zero in two rows and contains zero in two others. Table 8.10 gives the share of picks that each ordering decides by a tie, the share of early

Table 8.5: Latest allowances between contracts and the balance cap, per token and freeze. An allowance is the latest `Approval` of a (token, owner, spender) triple in the score window, kept when its value is positive. It is counted up to the owner's balance of the token at the freeze: unlimited allowances are those of at least 1.1×10^{77} base units, an allowance is capped to zero when the owner holds no balance of the token at the freeze, and capped below the allowance when the balance is positive and smaller than the allowance. Only the allowances that stay positive enter the allowance layer. The last column is the sum of the capped values in US dollars. At T_{obs} 65.6% of the allowances are capped to zero and 66.0% are unlimited (at t_1: 93.6% and 67.4%). The median positive capped value at t_1 is \$0.071, the constant m_A.

Token	Positive allowances	Unlimited	Capped to zero	Capped below	Positive after the cap	Capped value (USD)
At T_{obs}						
USDC	255,339	167,068	146,423	106,988	108,916	3.3×10^7
USDT0	36,468	29,609	30,820	5,522	5,648	6.5×10^6
WETH	30,826	17,608	29,431	1,270	1,395	2.0×10^7
WBTC	11,841	5,592	11,455	341	386	7.8×10^6
ARB	4,808	4,015	4,430	333	378	8.4×10^5
All five tokens	339,282	223,892	222,559	114,454	116,723	6.8×10^7
At t_1						
USDC	223,685	146,928	212,304	9,640	11,381	3.3×10^7
WETH	36,916	24,198	35,474	1,294	1,442	3.2×10^7
USDT0	33,673	27,950	27,666	5,851	6,007	7.6×10^6
WBTC	9,853	5,409	9,413	390	440	1.1×10^7
ARB	4,795	3,760	4,442	305	353	6.1×10^5
All five tokens	308,922	208,245	289,299	17,480	19,623	8.4×10^7

Table 8.6: The labels of the two label windows by the class of the account they count: for each label, its total over the spender cohort, the parts counted on user accounts and on contracts, the share of the contracts, and the spenders that have a positive part of each kind. An approval pair is counted by the class of its owner and a sender by its own class; a handful of senders are neither (the zero address of mints), so the two parts can fall short of the total by them. The split is a description outside every rule: the labels are those of the twelve-month analysis, unchanged, and most of what they count are user accounts, which no score of the contract graph reads.

Label	Total	From user accounts	From contracts	Contracts' share	Spenders with user accounts	Spenders with contracts
W0 ($n = 7{,}864$ spender contracts)						
New approval pairs	748,258	676,525	71,733	9.6%	1,322	876
New transfer senders	651,474	562,007	89,458	13.7%	1,044	883
Revocations	61,869	60,111	1,758	2.8%	947	287
Drained owners	570,724	547,427	23,297	4.1%	1,271	926
W1 ($n = 7{,}915$ spender contracts)						
New approval pairs	702,275	617,088	85,187	12.1%	1,220	641
New transfer senders	618,461	509,194	109,258	17.7%	970	855
Revocations	43,625	42,270	1,355	3.1%	761	279
Drained owners	528,557	503,788	24,769	4.7%	1,139	847

Table 8.7: Sizes of the contract graph and of the two universes of the discovery-order test at both freezes. The union graph holds the distinct ordered pairs of different contracts with an edge in either layer; the bow-tie is that of the union graph. SCC ∪ OUT holds fewer than half the contracts of the weakly connected component of the giant SCC (WCC) at t_1, so by the rule of the protocol the `both` mode, whose universe is the WCC, is the primary mode at both freezes and the `out` mode the sensitivity. A hot set $H_q(I)$ holds the contracts of the universe with the largest positive importance I, $q = 1/100$ of the universe with ties included, so it is smaller when fewer contracts have a positive importance.

Quantity	T_{obs} (primary)	t_1
Contracts of G (one numbering for both freezes)	1,403,959	1,403,959
Capped allowance pairs	115,006	17,940
Transfer pairs	2,580,709	2,269,073
Union graph: contracts with a pair	1,299,438	1,111,554
Union graph: pairs	2,575,632	2,269,421
Giant strongly connected component (SCC)	265,804	222,592
IN (reaches the SCC)	174,239	169,237
OUT (reached from the SCC)	242,833	109,804
Other contracts with a pair	616,562	609,921
|SCC ∪ OUT|, universe of the `out` mode	508,637	332,396
Union pairs inside it	1,546,906	1,283,946
|WCC|, universe of the `both` mode (primary)	1,169,044	1,015,867
Union pairs inside it	2,378,418	2,121,895
Capped allowance pairs inside the WCC	16,776	17,741
$\lvert H_q(I_T)\rvert$, `both` mode	13,141	12,697
$\lvert H_q(I_A)\rvert$, `both` mode	11,662	8,136
$\lvert H_q(I_T)\rvert$, `out` mode	5,087	3,324
$\lvert H_q(I_A)\rvert$, `out` mode	5,088	3,324

visits that go to contracts without user-account demand and the visit at which each ordering reaches the hubs: the median hub is visit 950 under AWP′ and $4{,}088$ under C-PR′, against $25{,}565$ under IB'_U.

Figure 8.1 shows the curves of the `out` mode at T_{obs}, which carries no verdict. In that mode C-PR′ reaches the hot contracts sooner than the breadth-first crawl and than IB'_U on both importances (Table 4.12), but it does not lead the walks on the transfer layer alone: on $H_q(I_T)$ it lies below AWP′ (-0.0617, $[-0.0709, -0.0526]$) and below its own transfer layer, C-PR′ ($\lambda = 0$) (-0.0305, $[-0.0343, -0.0264]$), and on $H_q(I_A)$ it is level with AWP′ (-0.0016, $[-0.0091, +0.0063]$) and above C-PR′ ($\lambda = 0$) ($+0.0153$, $[+0.0108, +0.0206]$).

Table 8.8: Coverage of the two importances by the contract graph and by the universes of the discovery-order test. I_T is the US-dollar value that user accounts transferred to a contract in the score window and I_A the value they had authorized it to spend at the freeze, each counted up to U_{user} per account and, for I_A, up to the account's balance. At T_{obs} 809,296 typed contracts have a positive importance, 112,210 of them outside G (at t_1: 715,480 and 109,094). The parts of the bow-tie are those of the union graph and sum to 100%. User accounts also transferred $1.68 billion at T_{obs} and $1.37 billion at t_1 to addresses that were not typed because every counterparty of theirs is a user account; these addresses are never graph nodes, and the sum bounds the inflow that the coverage cannot place.

	T_{obs}		t_1	
Share of the importance	I_T	I_A	I_T	I_A
Sum over all typed contracts	$16.61 billion	$309.5 million	$16.62 billion	$343.8 million
Giant SCC	71.2%	50.2%	74.1%	49.1%
IN	9.7%	1.7%	9.0%	2.6%
OUT	10.1%	40.4%	6.6%	38.2%
Other contracts with a pair	8.1%	3.4%	9.5%	6.3%
Contracts of G without a union pair	0.4%	3.6%	0.5%	3.3%
Contracts outside G	0.4%	0.7%	0.3%	0.5%
Held by the WCC (`both` mode)	96.5%	93.0%	97.0%	93.8%
Held by SCC ∪ OUT (`out` mode)	81.3%	90.6%	80.7%	87.3%
Top 1% of the contracts by the importance: share in the WCC	64.2%	38.1%	64.8%	34.6%
Top 1%: share in SCC ∪ OUT	29.6%	26.2%	32.5%	23.7%

8.5 EndorseRank and AWP Side by Side

Chapter 4 reports every score of the thesis together and reads each PageRank against the raw degree it smooths. This appendix collects the direct comparisons between the two single-layer scores, EndorseRank and AWP: their family means and paired differences in the same window, their per-proxy coefficients, and their paired differences out of window. None of these comparisons belongs to rule A or to rule B, and none carries a claim of the thesis. The two scores walk on different edges and are built for different constructs, so on each check of a whole cohort the score ahead is the one whose edges the check counts. Their coverage differs as well: 5,193 of the matched contracts lie outside the allowance graph and 2,384 outside the transfer graph, and each score is zero there (Table 4.6). Out of window EndorseRank ranks later new approval pairs below the in-approve degree, with uniform and with activity restarts, while AWP ranks later new senders slightly above the transfer in-degree on the whole cohorts and below it among the W0 spenders that another

Table 8.9: Discovery-order test at T_{obs} in the primary mode: sensitivity of the two decision contrasts, D_1 (C-PR′ minus BFS) and D_2 (C-PR′ minus IB′$_U$), each over its 95% percentile intervals of a paired bootstrap over the hot contracts (400 resamples, seed 42). The rows change one setting at a time: the share x of the universe over which the area is taken, the share q of the hot set, the hot set without the hubs (the contracts with the highest union in-degree, removed from the hot set but kept in the graph), the importance measured by the number of user accounts with at least one dollar, a bootstrap over the seeds in place of the hot contracts, and a schedule that recomputes the ordering four times as often, on the first seeds. Only 11,662 contracts of the universe have a positive I_A, so the hot set of I_A is the same at both shares q. The other mode and the other freeze are in Table 4.12. None of these rows carries a verdict. The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

	On $H_q(I_T)$		On $H_q(I_A)$	
Setting	D_1	D_2	D_1	D_2
Decision ($x = 10\%$, $q = 1\%$)	+0.0520 [+0.0485, +0.0558]	-0.0051 [-0.0087, -0.0010]	+0.0061 [+0.0014, +0.0111]	-0.0571 [-0.0621, -0.0519]
$x = 5\%$	+0.0673 [+0.0635, +0.0714]	+0.0026 [-0.0010, +0.0064]	+0.0472 [+0.0429, +0.0521]	-0.0132 [-0.0178, -0.0084]
$x = 20\%$	+0.0199 [+0.0169, +0.0230]	-0.0063 [-0.0091, -0.0034]	-0.0518 [-0.0575, -0.0460]	-0.0966 [-0.1023, -0.0905]
$q = 5\%$	+0.0132 [+0.0121, +0.0144]	-0.0067 [-0.0078, -0.0057]	+0.0061 [+0.0014, +0.0111]	-0.0571 [-0.0621, -0.0519]
Without the 1,170 hubs	+0.0306 [+0.0274, +0.0341]	-0.0126 [-0.0159, -0.0088]	-0.0019 [-0.0065, +0.0026]	-0.0607 [-0.0654, -0.0553]
Importance by counts (N_T, N_A)	+0.0188 [+0.0163, +0.0211]	-0.0282 [-0.0308, -0.0254]	+0.0191 [+0.0129, +0.0262]	-0.0473 [-0.0537, -0.0408]
Bootstrap over seeds	+0.0520 [+0.0320, +0.0717]	-0.0051 [-0.0249, +0.0179]	+0.0061 [-0.0315, +0.0398]	-0.0571 [-0.0873, -0.0221]
$\rho/4$ (recomputed at each growth of 1/200; 5 seeds)	+0.0381 [+0.0347, +0.0418]	-0.0129 [-0.0167, -0.0087]	-0.0134 [-0.0181, -0.0082]	-0.0713 [-0.0765, -0.0658]

contract had paid before the freeze (Section 4.7; computed after the labels were known), so a lead of EndorseRank over AWP does not show that propagation over approvals adds anything to the count of approving owners.

8.5.1 Same window

Table 4.5 gives the family means of the two scores with their intervals, and the last row of Table 4.6 their agreement. The two patterns cross: AWP tracks the transfer family most closely and the allowance family least, while EndorseRank tracks the allowance family most closely. Table 8.11 tests five paired differences on the same contract resamples (Section 3.11), numbered i to v in the order of its rows.

Table 8.10: Discovery-order test at T_{obs}: diagnostics of the crawls. Ties: the share of picks whose group of tied scores held more than one contract, in the primary mode (`both`) and in the `out` mode; a tie is broken by the earlier discovery and then by the lower address. Wasted visits: the share of the first 1% and of the first 10% of visits that went to contracts with $I_T = I_A = 0$ (primary mode). Hub visit: over the 1,170 hubs of the primary universe, the median of each hub's mean visit index across seeds; the largest hub has a union in-degree of 441,612. The count orderings and the walks on the allowance layer alone decide nearly every pick by a tie.

	Picks decided by a tie		Wasted visits		Median hub
Ordering	`both`	`out`	first 1%	first 10%	visit index
Breadth-first (BFS)	0.000	0.000	0.891	0.845	73,219
Random	0.000	0.000	0.841	0.682	272,806
IB'_A (allowance backlinks)	1.000	1.000	0.874	0.862	59,089
IB'_T (transfer backlinks)	1.000	0.999	0.678	0.888	25,566
IB'_U (contract backlinks)	1.000	0.999	0.675	0.888	25,565
EndorseRank$'$	0.999	0.998	0.874	0.863	60,494
EndorseRank-activity$'$	0.991	0.997	0.692	0.857	66,699
AWP$'$	0.252	0.005	0.522	0.835	950
C-PR$'$ ($\lambda = 1$)	1.000	0.999	0.874	0.863	60,840
C-PR$'$ ($\lambda = 0$)	0.531	0.019	0.747	0.911	5,156
C-PR$'$ ($\lambda = 0.25$)	0.534	0.019	0.727	0.908	4,185
C-PR$'$ ($\lambda = 0.5$)	0.535	0.023	0.719	0.900	4,088
C-PR$'$ ($\lambda = 0.75$)	0.537	0.025	0.719	0.889	4,203
S-PR$'$	0.531	0.019	0.743	0.910	4,885
δ' (normalized in-strength)	0.634	0.283	0.497	0.650	1,404

Table 8.11: Paired bootstrap tests of differences between Kendall τ coefficients of EndorseRank and AWP in the same window (matched cohort, $n = 6{,}443$ contracts). Family entries use the family-mean τ; the same contract resamples are used for a and b, so the interval is on the paired difference. An interval excluding 0 indicates a difference not attributable to sampling variation at the 5% level. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Contrast (a minus b)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank allowance minus EndorseRank transfer	0.839	0.297	+0.542	[0.520, 0.563]	1.000
AWP transfer minus EndorseRank transfer	0.755	0.297	+0.458	[0.438, 0.476]	1.000
EndorseRank allowance minus AWP allowance	0.839	0.227	+0.613	[0.591, 0.632]	1.000
AWP sybil-stability minus EndorseRank sybil-stability	0.643	0.220	+0.422	[0.404, 0.438]	1.000
EndorseRank in-approve degree minus EndorseRank in-degree	0.847	0.305	+0.542	[0.521, 0.564]	1.000

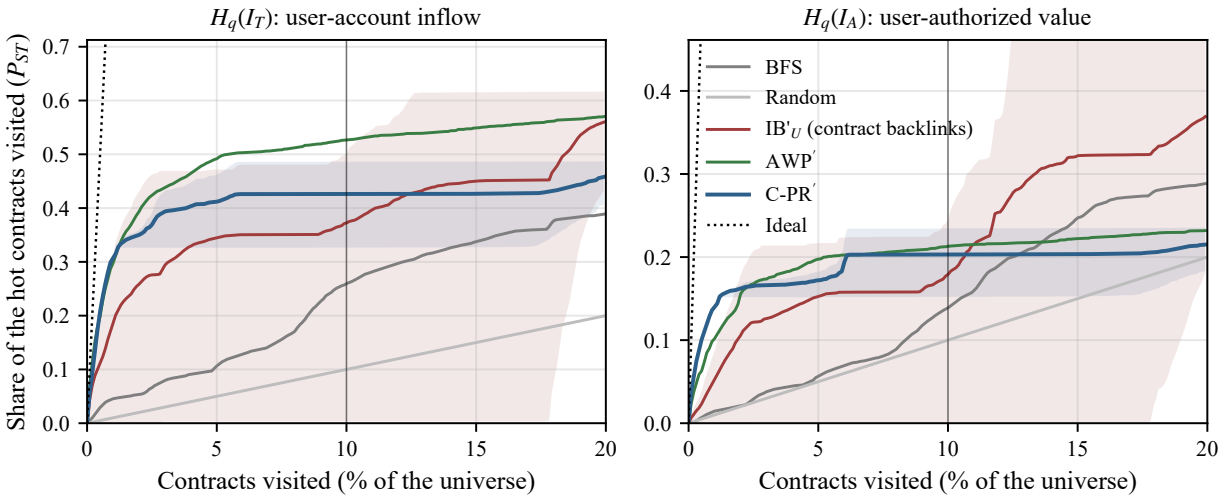


Figure 8.1: Discovery-order test at T_{obs} in the `out` mode, which carries no verdict: the share of the hot contracts visited, P_{ST}, against the contracts visited as a percentage of the universe, for the hot sets of user-account inflow I_T (left) and of user-authorized value I_A (right). In this mode a visit reveals out-edges only, and the universe is the giant strongly connected component with the contracts reachable from it. The curves are means over the 20 seeds for the breadth-first and random orderings, the backlink count IB'_U, AWP′ and C-PR′; the shaded bands give the minimum and maximum over the seeds for IB'_U and C-PR′, the dotted line is the ideal order, and the vertical line marks 10% of the universe, the range of $\text{AUC}_{10\%}$. The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

All five intervals lie above zero. AWP's transfer alignment is higher than EndorseRank's (contrast ii, $+0.458$, $[0.438, 0.476]$), EndorseRank's allowance alignment is higher than AWP's (contrast iii, $+0.613$, $[0.591, 0.632]$), and AWP leads EndorseRank on the Sybil-stability family (contrast iv, $+0.422$, $[0.404, 0.438]$). The remaining two compare EndorseRank with itself: its allowance alignment is higher than its transfer alignment (contrast i, $+0.542$, $[0.520, 0.563]$), and its in-approve-degree alignment is higher than its in-degree alignment (contrast v, $+0.542$, $[0.521, 0.564]$). Ties weigh on these coefficients: EndorseRank ties 65.4% of the contract pairs and AWP 13.7%, mostly among the contracts outside each graph (Section 4.4; Appendix 8.3.3). With those contracts kept as isolated nodes and not scored zero, EndorseRank's allowance mean rises, its transfer and Sybil-stability means fall and AWP's means stay as they are, so the gaps of contrasts i to iv widen in their point estimates (Table 4.6).

The opposite directions of contrasts ii and iii follow from the constructs. Each score is ahead on

the family built from its own edges, and the main proxies of each family include the in-degree that the corresponding PageRank smooths, so the agreement is partly built in and checks the intended construct of each score. Neither difference says which score is the better measure of reputation.

8.5.2 Proxy detail

Each family mean condenses several per-proxy coefficients. Tables 8.12 to 8.14 report them one proxy at a time for EndorseRank and AWP, as Spearman ρ, Kendall τ and the 95% interval of τ; the transfer proxies are the ones used in the evaluation of AWP by its authors (T. Do et al., 2023). Every proxy is computed from the events between contracts, and a contract outside a layer has zero on the proxies of that layer.

Table 8.12: Domain alignment of EndorseRank and AWP with the transfer proxies (in-degree and in-value of each contract), same window (matched cohort, $n = 6{,}443$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
In-degree	0.345	0.305	[0.287, 0.326]
In-value (sum inbound)	0.338	0.289	[0.269, 0.309]
AWP			
In-degree	0.873	0.764	[0.757, 0.772]
In-value (sum inbound)	0.872	0.746	[0.738, 0.754]

On the transfer family (Table 8.12) AWP agrees with the in-value almost as closely as with the in-degree, for the reason given in Section 4.3. EndorseRank's transfer coefficients have intervals that overlap neither zero nor AWP's intervals. Part of them is coverage: the $2{,}173$ contracts outside both graphs lie at the bottom of both orderings, and among the $4{,}270$ contracts inside the union graph the two coefficients are 0.174 ($[0.146, 0.200]$) and 0.153 ($[0.127, 0.177]$).

On the allowance family (Table 8.13) EndorseRank follows the in-approve degree and the in-approve value about equally, for the reason given in Section 4.3. AWP's coefficients lie above zero, also among the contracts inside the union graph (0.122, $[0.091, 0.151]$, and 0.116, $[0.087, 0.145]$): among the cohort's contracts, those with capped allowances from more owner contracts also tend

Table 8.13: Domain alignment of EndorseRank and AWP with the allowance proxies (in-approve degree and value of each spender contract), same window (matched cohort, $n = 6{,}443$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
In-approve degree	0.883	0.847	[0.835, 0.857]
In-approve value (sum)	0.882	0.831	[0.820, 0.841]
AWP			
In-approve degree	0.271	0.229	[0.207, 0.248]
In-approve value (sum)	0.269	0.225	[0.204, 0.243]

to be central in the transfer graph.

Table 8.14: Domain alignment of EndorseRank and AWP with the Sybil-adjusted stability proxies from transfer activity, same window (matched cohort, $n = 6{,}443$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
Inbound counterparty ratio	0.177	0.148	[0.130, 0.166]
Transfer tenure (days)	0.317	0.271	[0.253, 0.289]
Active months	0.281	0.243	[0.224, 0.262]
AWP			
Inbound counterparty ratio	0.648	0.428	[0.414, 0.443]
Transfer tenure (days)	0.862	0.711	[0.702, 0.719]
Active months	0.917	0.789	[0.781, 0.797]

On the Sybil-adjusted stability family (Table 8.14) AWP is ahead of EndorseRank on all three proxies, as one would expect from a transfer score that rewards activity spread over the window; the paired difference on the family mean excludes zero (Table 8.11, contrast iv). Coverage shapes these coefficients. The $2{,}384$ contracts outside the transfer graph have zero on all three proxies and score zero under AWP, and $2{,}173$ of them score zero under EndorseRank as well. Among the $4{,}270$ contracts inside the union graph AWP's family mean is 0.380 ($[0.367, 0.393]$) and EndorseRank's 0.028 ($[0.009, 0.046]$); on the inbound counterparty ratio AWP's coefficient does not exclude zero there (0.018, $[-0.007, 0.042]$), and EndorseRank's is negative (-0.091, $[-0.112, -0.068]$). The

ratio falls when the same senders return, so inside the graph the contracts that EndorseRank ranks high are, on this measure, contracts whose senders return.

8.5.3 Out of window

Table 8.15 collects the paired differences between the two scores on the spender cohorts of both label windows. In every row a is EndorseRank and b AWP. The rows were computed after the labels were known.

Table 8.15: Paired bootstrap differences between EndorseRank and AWP out of window (W0 spenders $n = 7{,}864$; W1 spenders $n = 7{,}915$). None of them enters rule A or the two applied criteria of rule B: the rows were computed after the labels were known, the W1 rows from the saved W1 scores with the resamples of Table 4.20. Within each block a and b share the contract resamples (400; seed 42).

Contrast (a minus b)	Label	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
W0, spenders ($n = 7{,}864$)						
EndorseRank minus AWP	new approval pairs	0.168	0.098	+0.070	[0.042, 0.096]	1.000
EndorseRank minus AWP	new transfer senders	0.084	0.330	-0.246	[-0.271, -0.220]	0.000
W1, spenders ($n = 7{,}915$)						
EndorseRank minus AWP	new approval pairs	0.147	0.064	+0.083	[0.059, 0.110]	1.000
EndorseRank minus AWP	new transfer senders	0.064	0.314	-0.250	[-0.273, -0.225]	0.000

On the spenders of W0, EndorseRank ranks new approval pairs above AWP, a paired difference of $+0.070$ (interval $[0.042, 0.096]$). On new transfer senders the order reverses, with a difference of -0.246 ($[-0.271, -0.220]$). Every label falls after all scored events. The first gap rests on coverage. AWP scores zero for the $2{,}421$ spenders without a transfer pair before the freeze, nearly a third of the cohort, which only allowance edges can order, and EndorseRank for the $5{,}966$ without an allowance edge of positive weight. On the $4{,}569$ spenders that another contract had paid before the freeze, AWP's coefficient is the higher on both labels (Table 4.17), without a paired test.

In July–September the pattern repeats: EndorseRank leads on new approval pairs ($+0.083$, $[0.059, 0.110]$) and AWP on new senders (-0.250, $[-0.273, -0.225]$).

8.6 The Registered Analysis

The registered analysis is the one that the analysis plan and the registration of W1 fixed before the study extracted the data it judges (Appendix 8.1.1): amounts in raw token base units, every ERC-20 token, the value transform with $b = 1$, scores from every event since 1 October 2023, graphs that hold every account of the cohort's events, and the plan's cohort of $27{,}844$ contracts. After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. The chapters report this analysis (Section 3.5.3), in which rules A and B are applied to the scores of the contract graph as a re-analysis. Table 8.16 sets the decision contrasts and verdicts of the registered analysis beside the same contrasts on those scores; its last four rows, degree contrasts computed after the labels were known, carry no verdict. Of rule B the thesis applies two of its registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds. The registration is the file `docs/registration_w1.md` of the public repository (Section 3.14). The two columns reach the same verdicts: neither part of rule A is met, F1 holds and F2 fails, also in the two registered sensitivity analyses, so rule B is not met, and F6 fails. Only the verdicts of the registered column are outcomes of tests fixed before the study extracted the data they judge. The complete registered results, with every coefficient, interval and verdict, are published unchanged in `data/3-published-results-for-thesis/` with their digest, `results_digest.md`.

Table 8.16: Decision contrasts and verdicts of the registered analysis (amounts in raw base units, every ERC-20 token, $b = 1$, scores from every event since 1 October 2023, graphs that hold every account of the cohort's events), exactly as obtained under the analysis plan and the W1 registration, beside the same contrasts on the scores of the contract graph (amounts in USD, the five tokens, scores from the twelve months before each freeze, nodes that are contracts other than smart accounts, each allowance counted up to its owner's balance). The contract-graph analysis was defined after the registered results were known; its column applies the registered rules and thresholds to its scores and is not a registered test. Under the registration rule B holds only if each of its criteria holds; the table applies two of them, F1 and F2, in both columns. The last block holds degree contrasts that no rule of either analysis decides; they are reported in both. $\Delta\tau$ with 95% paired bootstrap interval (400 resamples, seed 42); the rules and their thresholds are in Sections 3.8 and 3.8.1.

Quantity	Registered analysis	Contract-graph analysis
Matched cohort (contracts)	27,844	6,443
W0 spender cohort	33,101	7,864
W1 spender cohort	34,587	7,915
EndorseRank vs. AWP, same window (τ)	0.217 [0.209, 0.225]	0.275 [0.258, 0.294]
A, W0: C-PR minus C-PR ($\lambda = 1$), new approval pairs	$+0.008$ [0.003, 0.013]	-0.025 [-0.045, -0.004]
A, W0: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.030 [-0.034, -0.026]	-0.014 [-0.019, -0.008]
Rule A, primary part	not met	not met
Rule A, secondary part	not met	not met
B, F1: C-PR minus C-PR ($\lambda = 0$), new approval pairs	$+0.101$ [0.092, 0.111]	$+0.074$ [0.060, 0.087]
B, F2: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.025 [-0.029, -0.021]	-0.016 [-0.022, -0.012]
F1 and F2, two of rule B's registered criteria	not met (F2 fails)	not met (F2 fails)
F1 and F2, sensitivity 1 (AWP as comparator)	not met (F2 fails)	not met (F2 fails)
F1 and F2, sensitivity 2 (isolated nodes)	not met (F2 fails)	not met (F2 fails)
F6 (rule A, primary part, on W1)	fails	fails
W0: EndorseRank minus in-approve degree, new pairs	-0.011 [-0.016, -0.006]	-0.036 [-0.046, -0.027]
W0: AWP minus transfer in-degree, new senders	$+0.038$ [0.035, 0.042]	$+0.012$ [0.003, 0.023]
W1: EndorseRank minus in-approve degree, new pairs	-0.029 [-0.032, -0.025]	-0.034 [-0.043, -0.025]
W1: AWP minus transfer in-degree, new senders	$+0.020$ [0.016, 0.023]	$+0.011$ [0.001, 0.022]

8.7 Ethics Clearance Certificate

8.7.1 Certificate

This study uses public blockchain data only and involves no human participants (Section 3.17). The ethics clearance certificate issued by the Unisa School of Computing / College of Agriculture and Environmental Sciences research ethics review committee is reproduced below.

Ethics clearance certificate

Reference: [Unisa SoC/CAES ethics clearance ref: TBD]

Date of issue: [TBD]

(Scanned certificate to be inserted here.)